

DETECTING ECCENTRICITY: WHAT IS THE CURRENT EVIDENCE?

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GW Eccentricity Corfu, Greece September 2026



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WHAT IS DISCOVERY: WHAT IS BAYES?

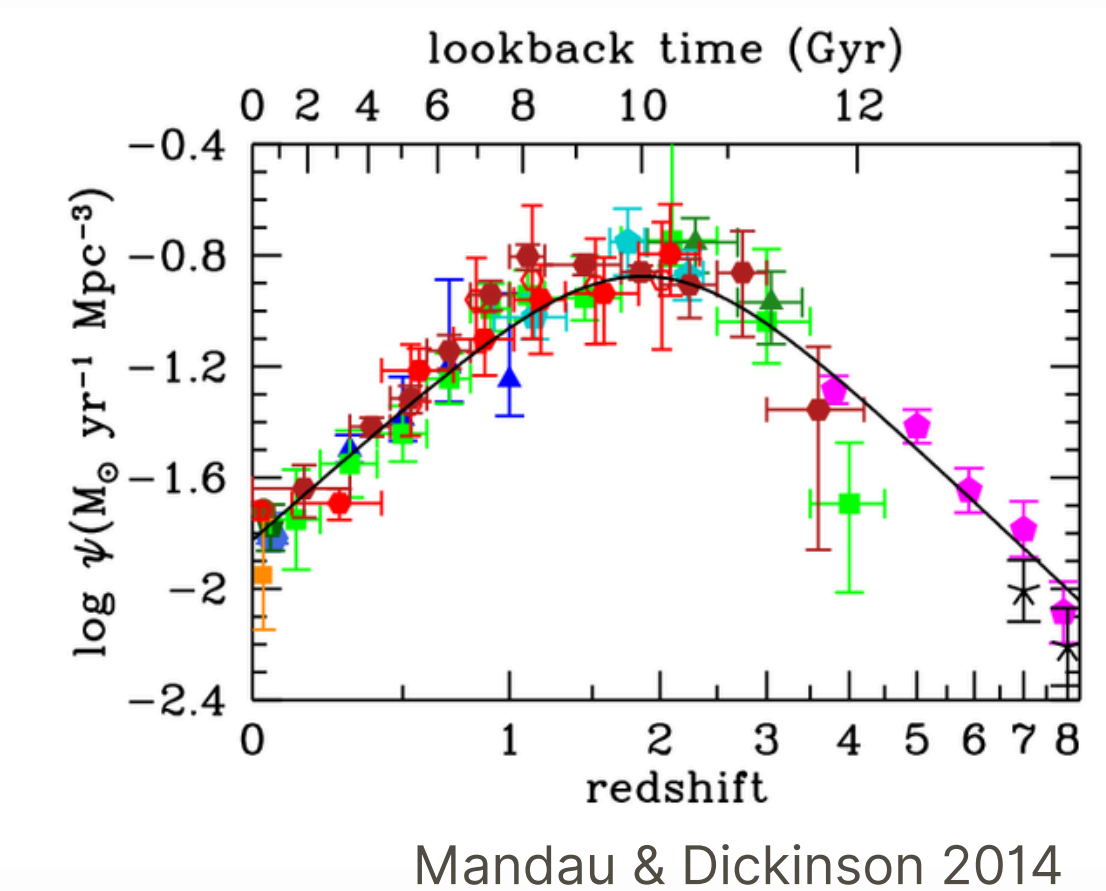
Bayes/Laplace told us that

Parameters

$$p(\theta | d, M) = \frac{p(d|\theta, M)p(\theta|M)}{p(d|M)}$$

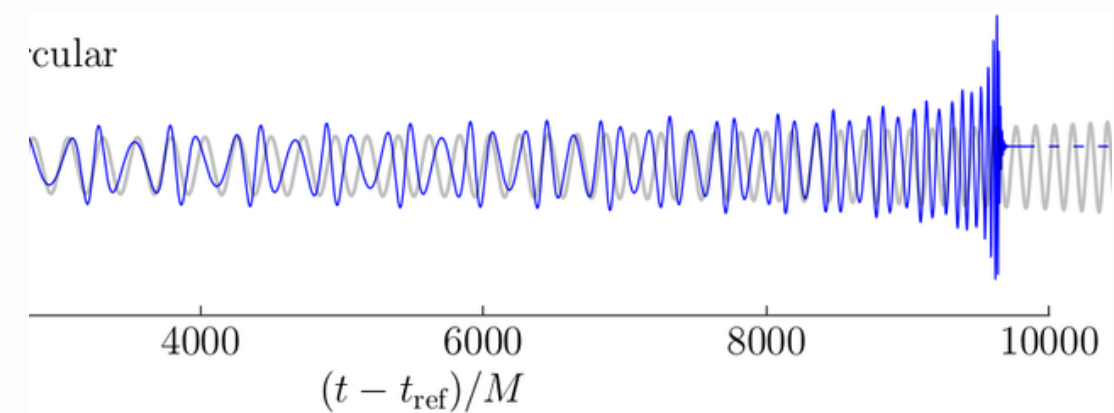
Data Model

The prior is **model dependent** and encodes our belief about the parameters before seeing the data



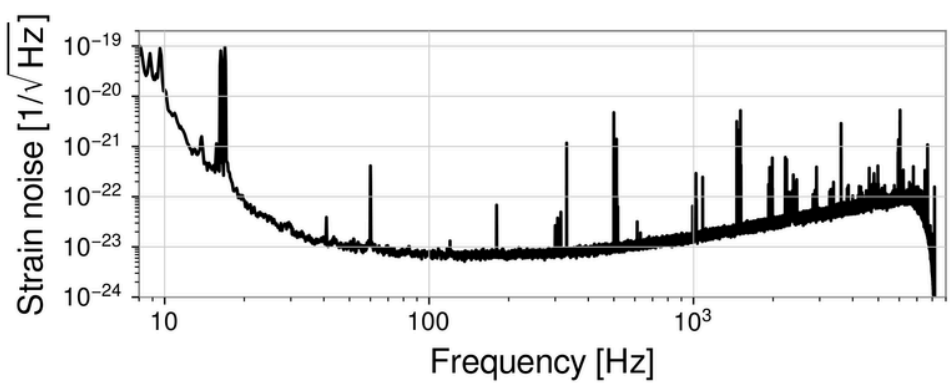
The **likelihood** is **model dependent** and encodes our belief about the **data generating process**

Signal model



SXS Collaboration 2025

Noise Model



LVK Collaboration 2020

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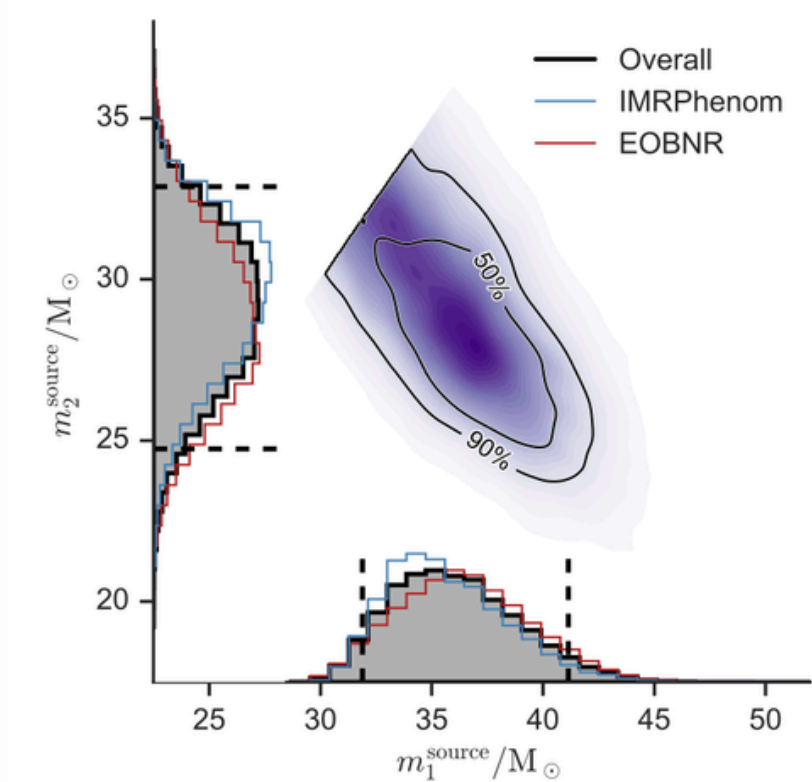
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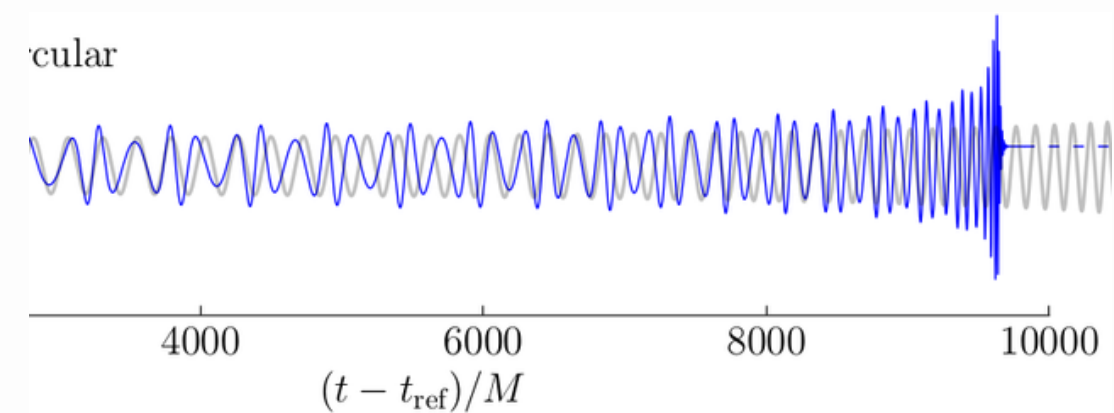
The **posterior** is **model dependent** and encodes our belief about the parameters after seeing the data



LVK Collaboration 2015

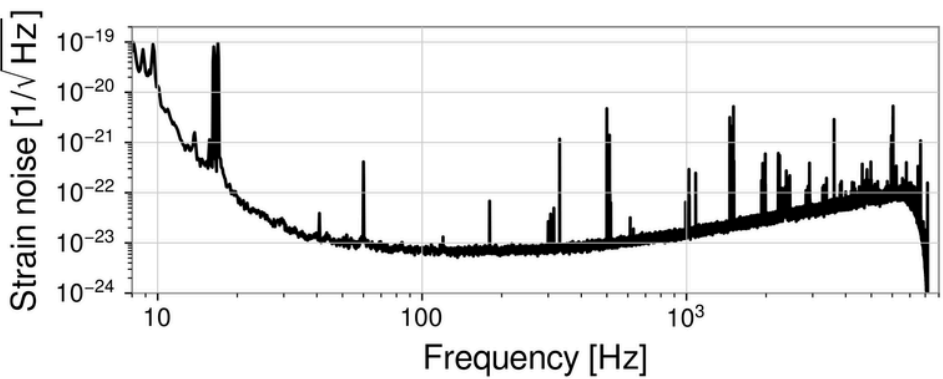
The **likelihood** is **model dependent** and encodes our belief about the **data generating process**

Signal model



SXS Collaboration 2025

Noise Model



LVK Collaboration 2020

WHAT IS DISCOVERY: WHAT IS SIGNIFICANCE?

One problem with just looking at the posterior is that it doesn't allow us to compare models. We can instead look at the Bayes factors

$$\text{BF}_{12} = \frac{p(d|M_1)}{p(d|M_2)}$$

Kass and Raftery table

Look elsewhere effect

Prior odds

Systematics

LETS BUILD A PRIOR TOGETHER

One question we can ask ourselves is, what statistical significance is needed (Lyons+ 2013)

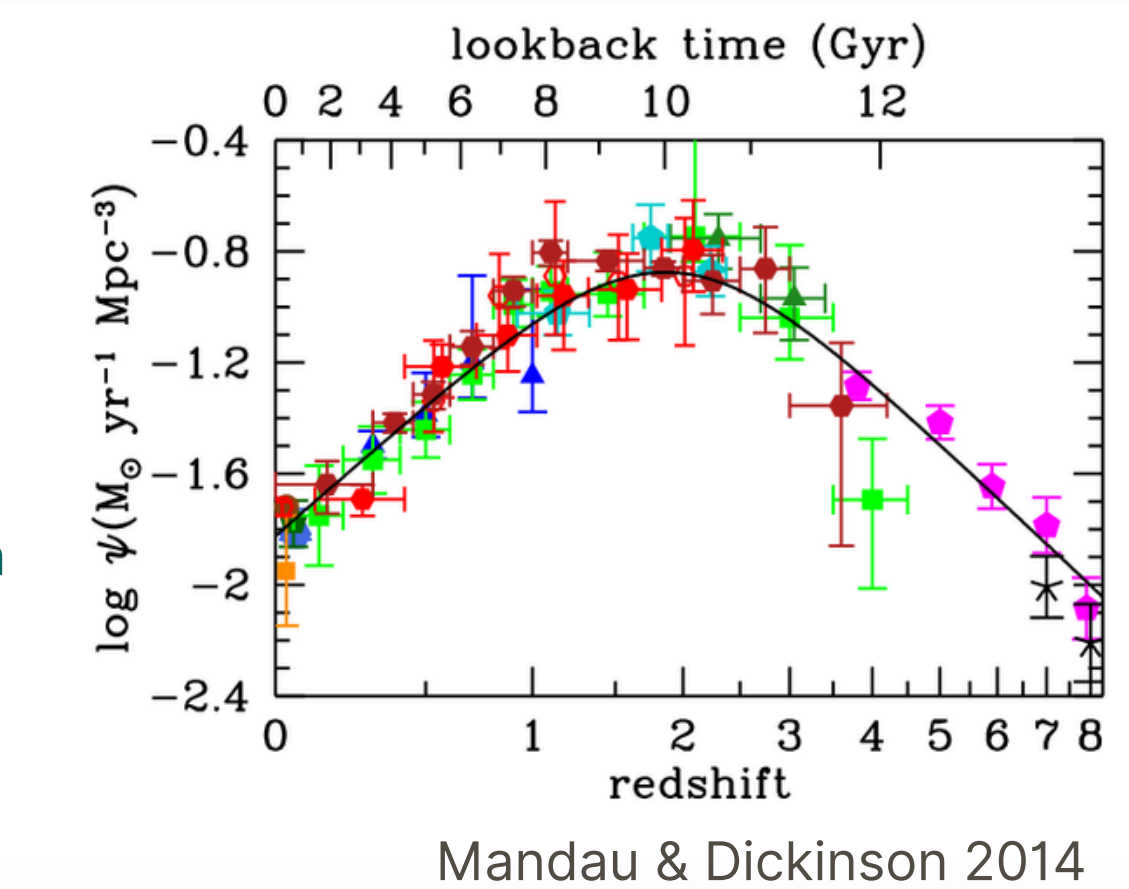
Look elsewhere effect

Prior odds

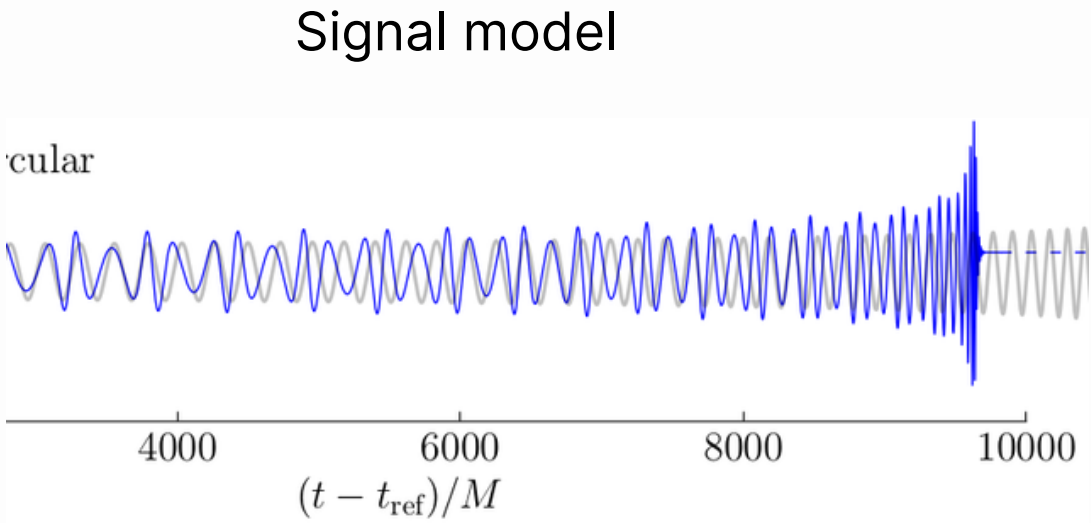
Systematics

WHAT IS DISCOVERY: WHAT IS BAYES?

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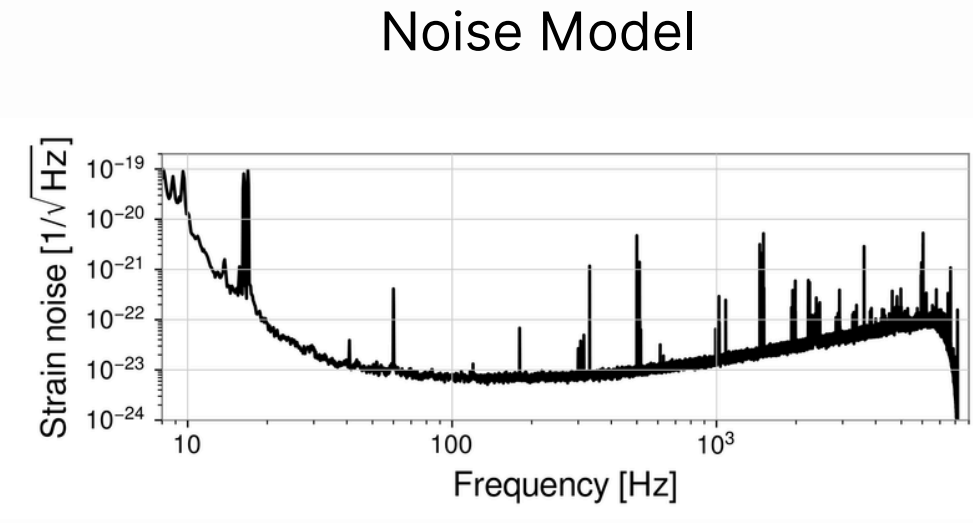


The **likelihood** is **model dependent** and encodes our belief about the data generating process



SXS Collaboration 2025

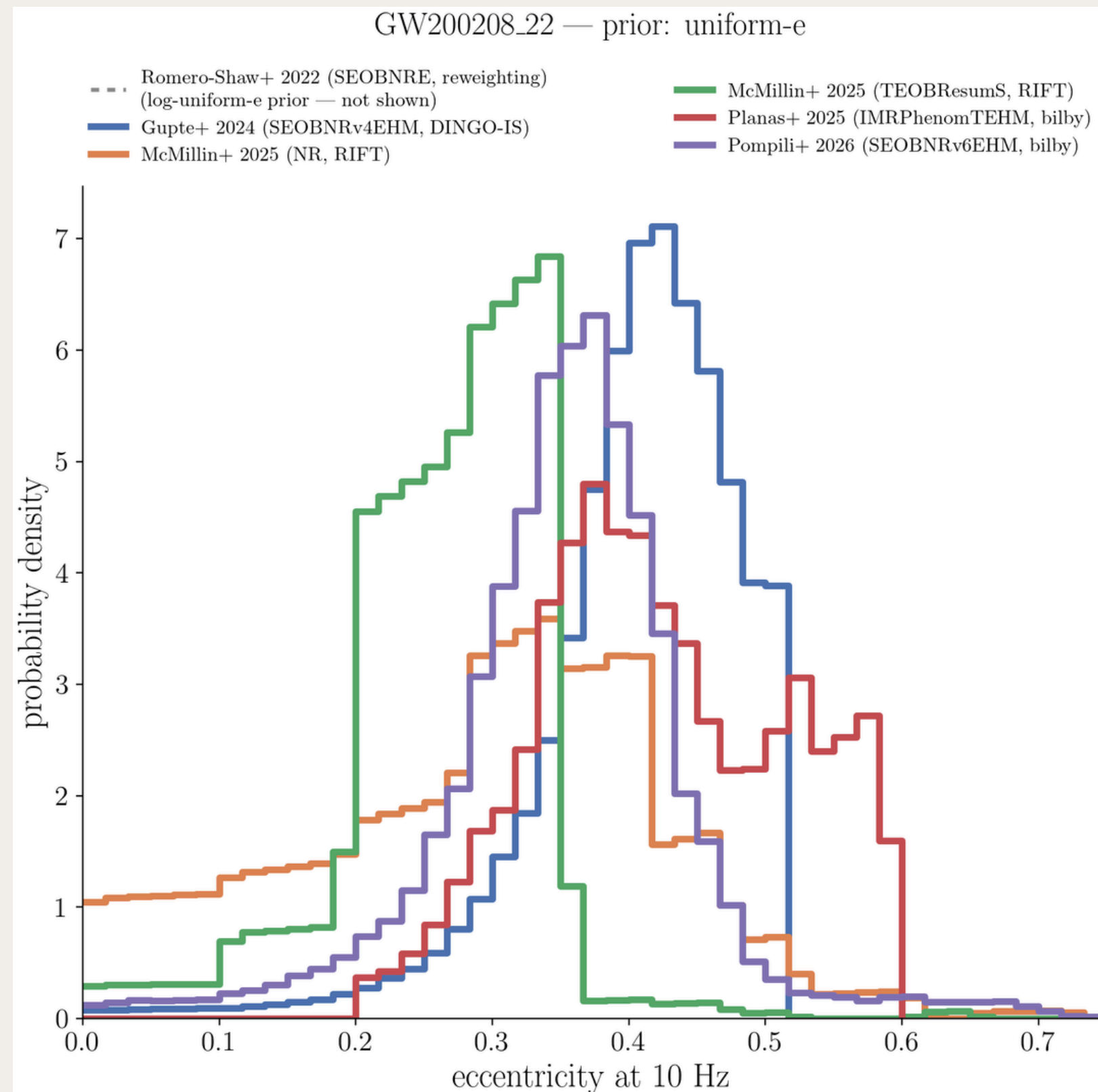
+



LVK Collaboration 2020

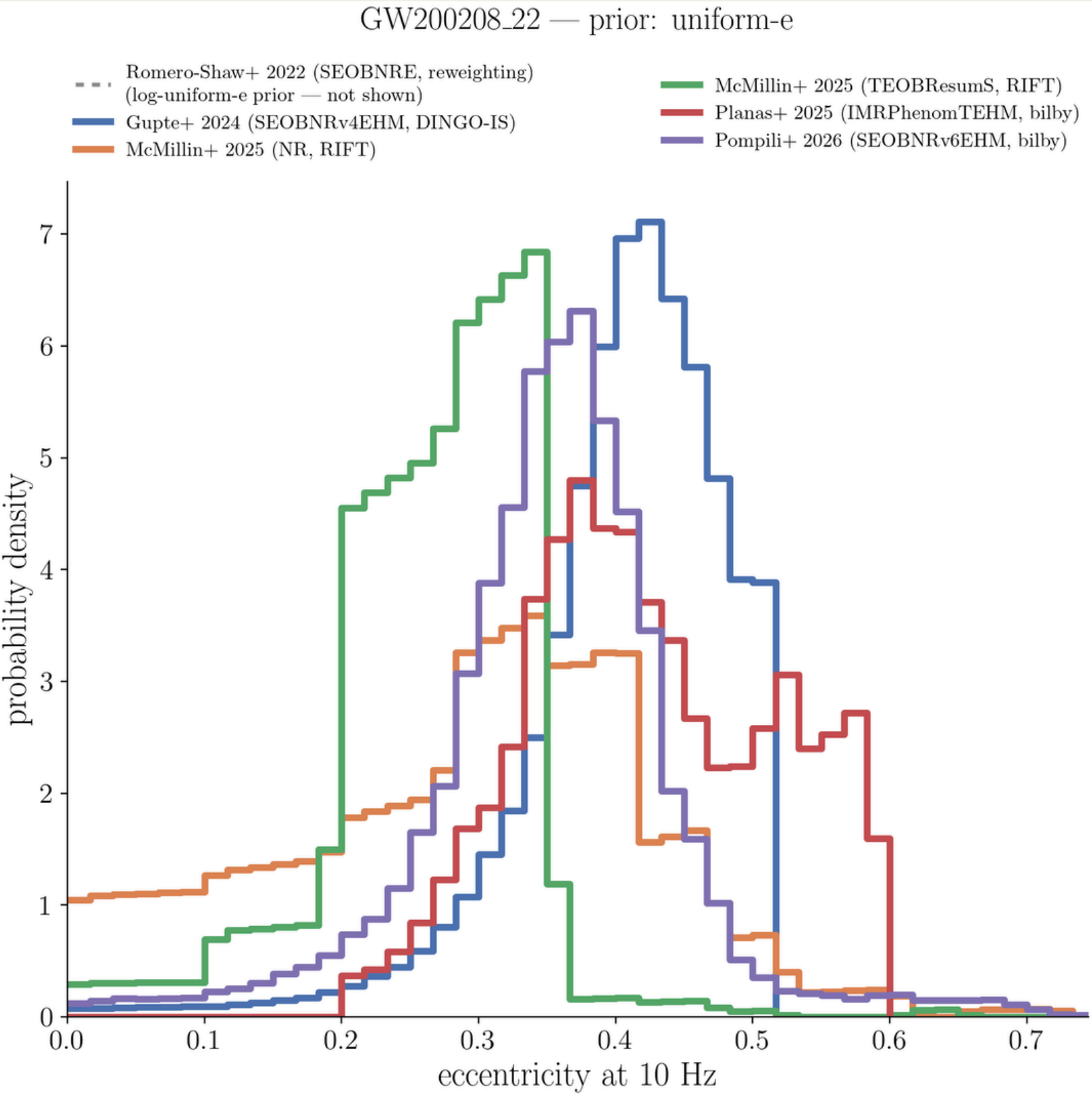
GW200208_22

This is a ~66 solar mass BBH with 14 cycles in band.
Bayes factor ~ 3, 17, _ (uniform, log-uni, expert prior)



GW200208_22

This is a ~20-20 solar mass BBH with 14 cycles in band.
Bayes factor ~ 3, 17, — (uniform, log-uni, expert prior)

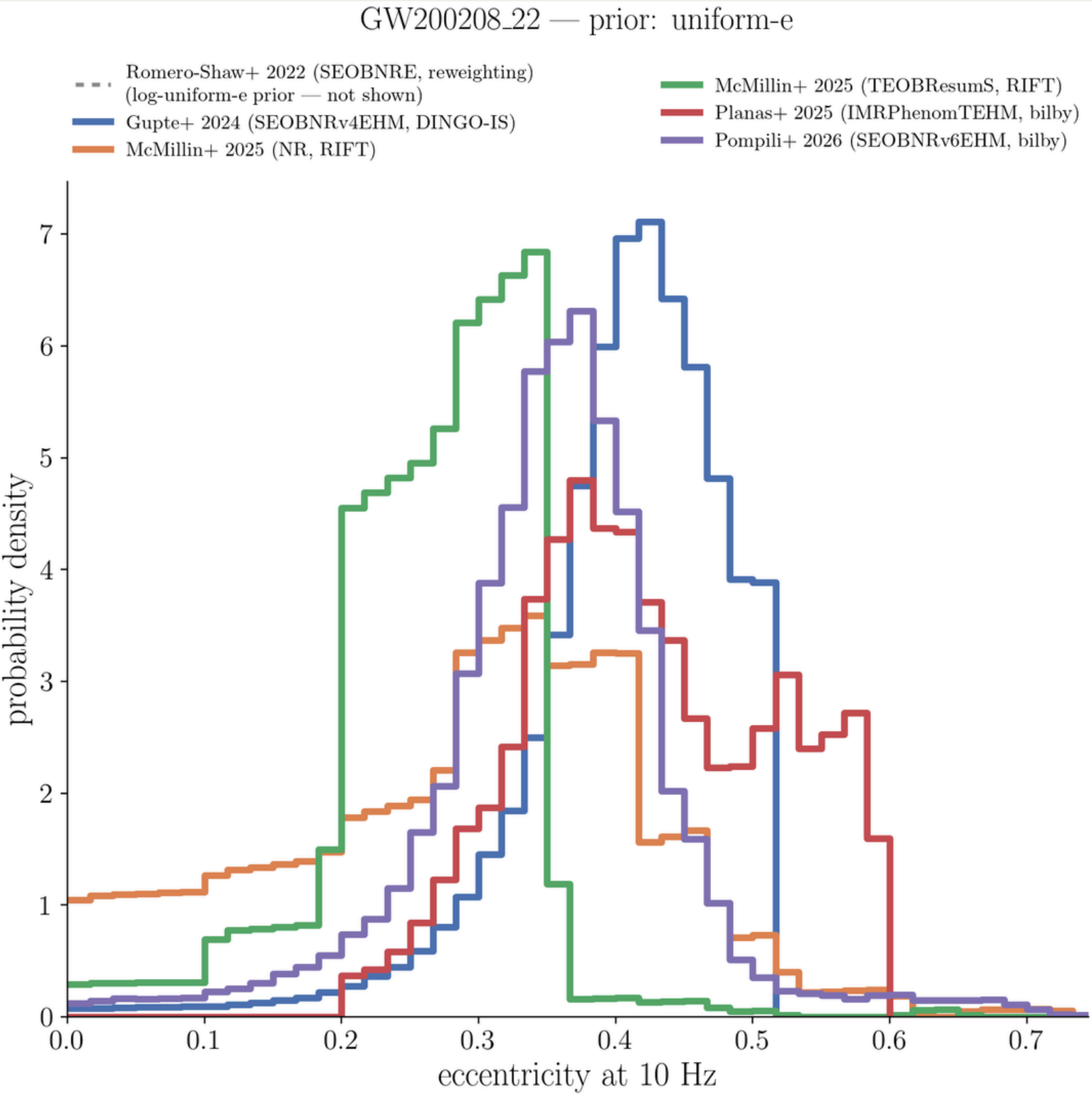


BUT IS IT AN EVENT?

LVK 2021 found that the FAR is 5 per year. The p-astro is 70%.

GW200208_22

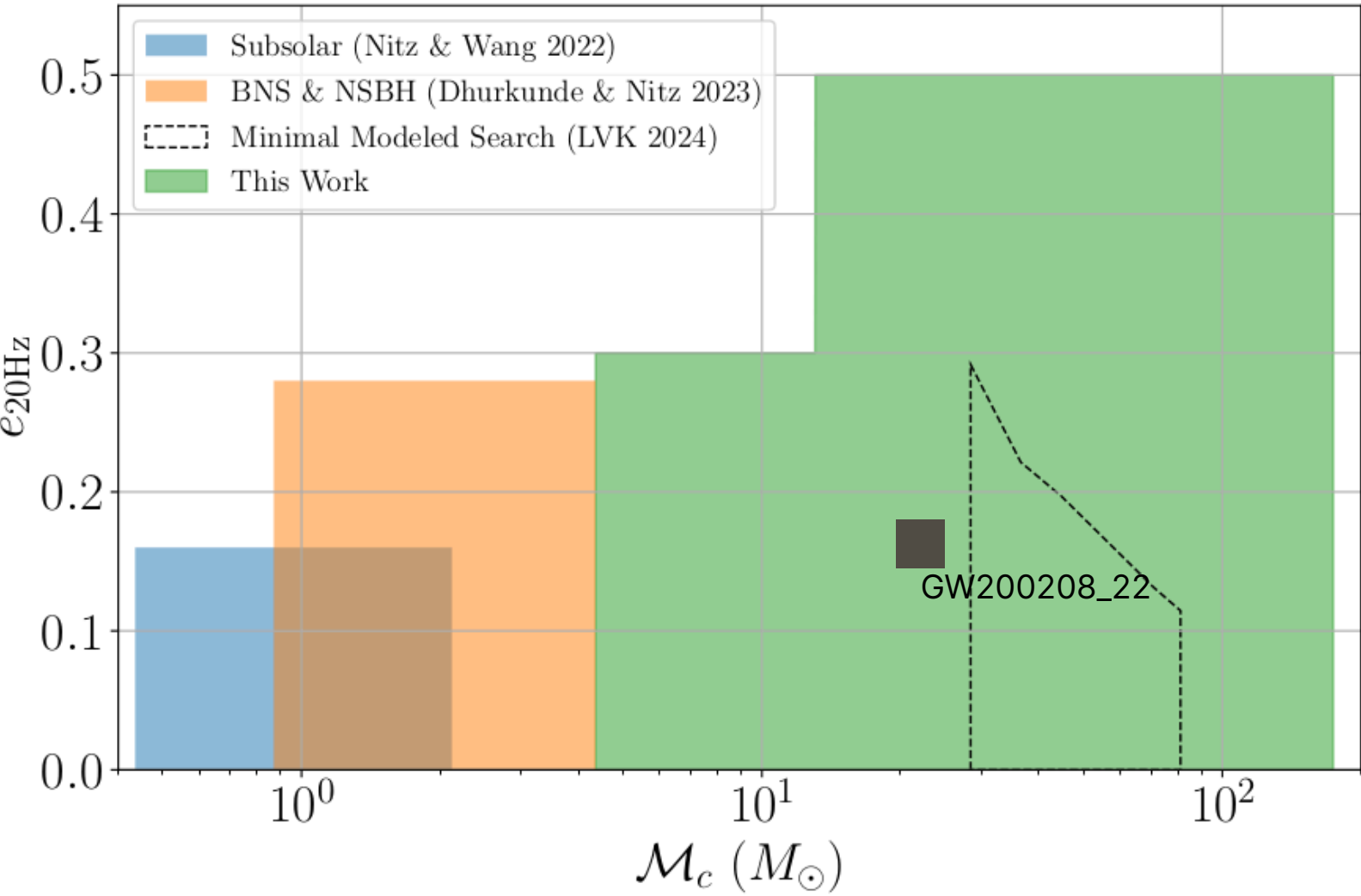
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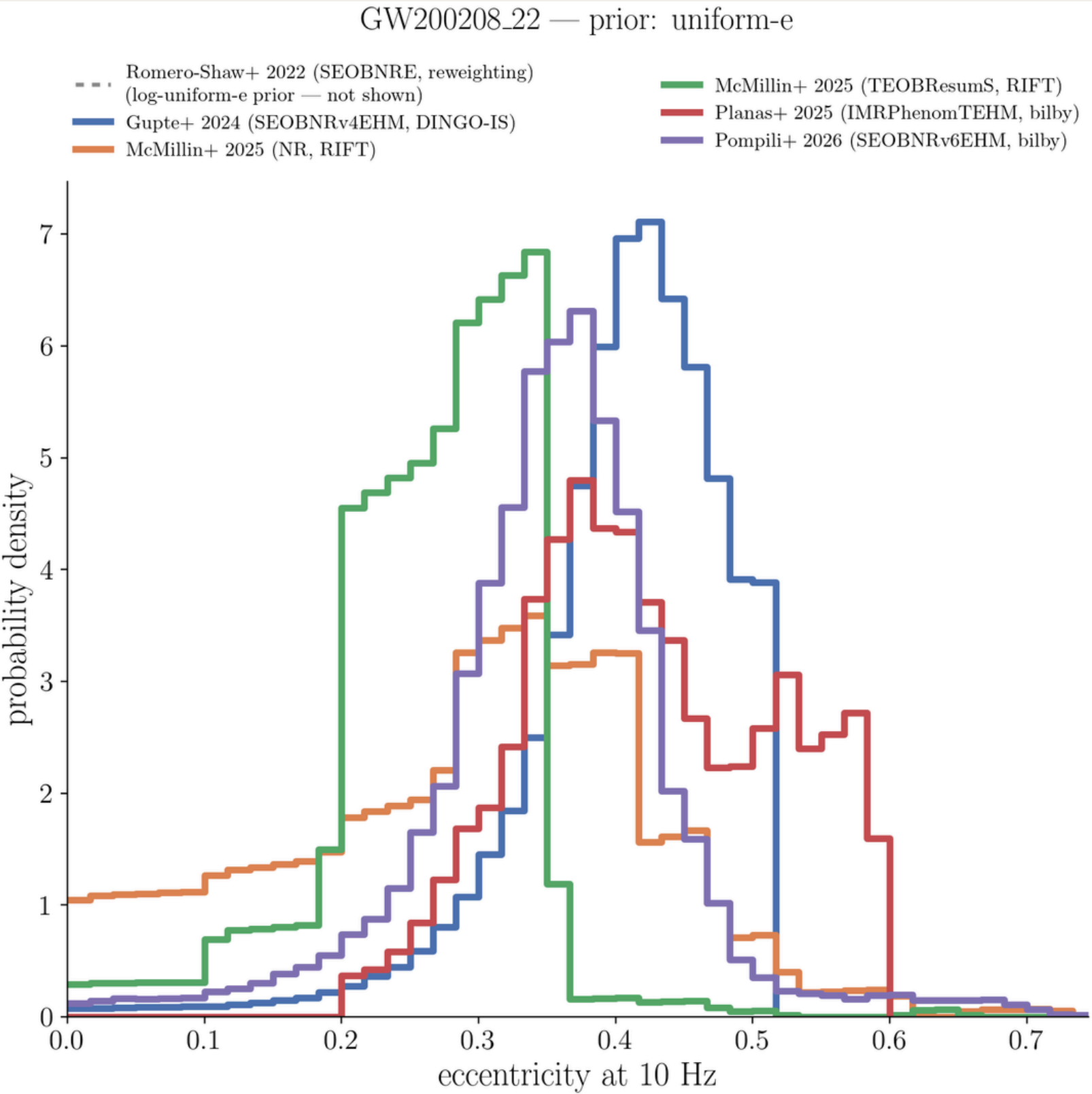
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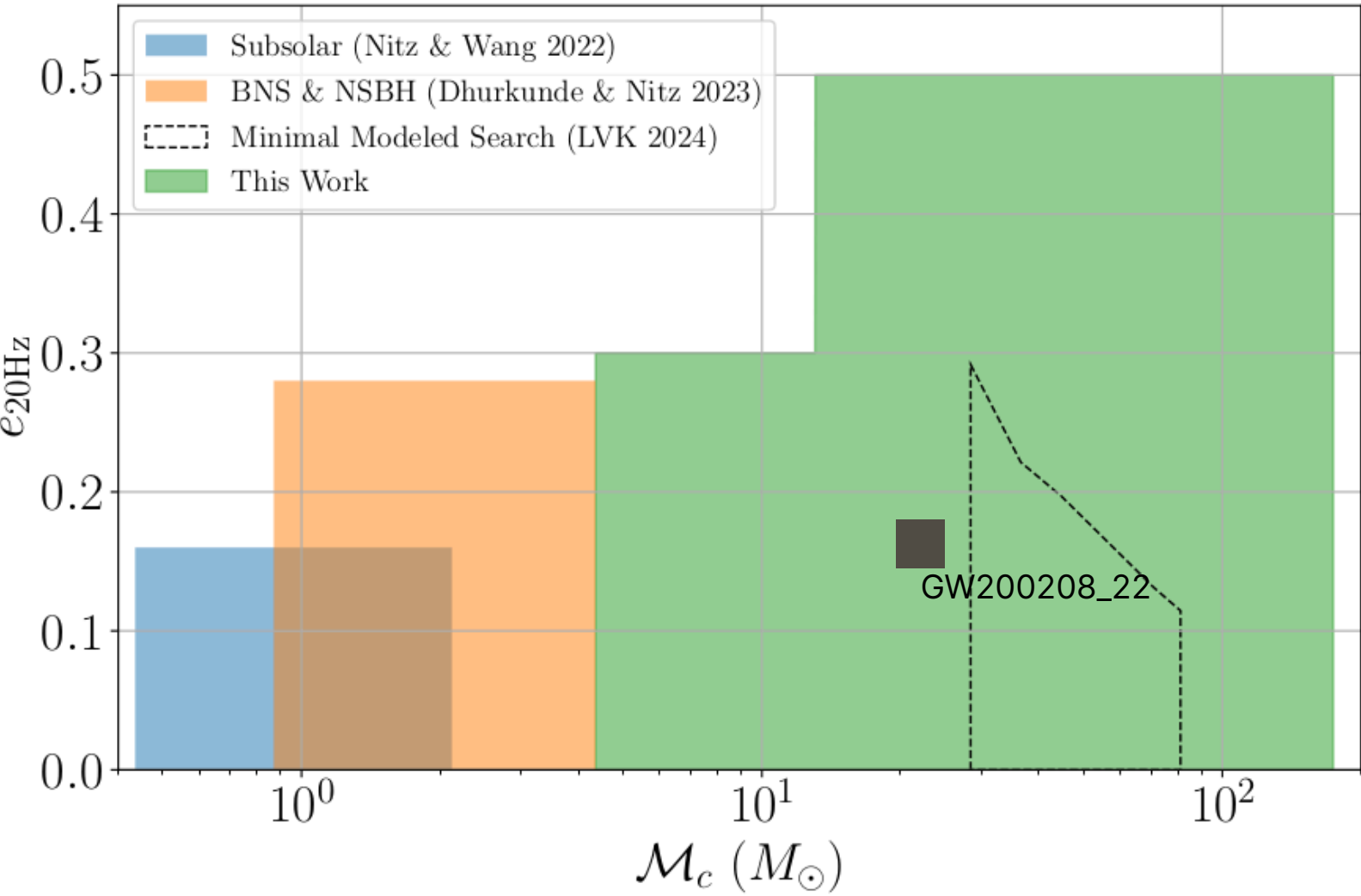
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BUT IS IT AN EVENT? PROBABLY NOT.

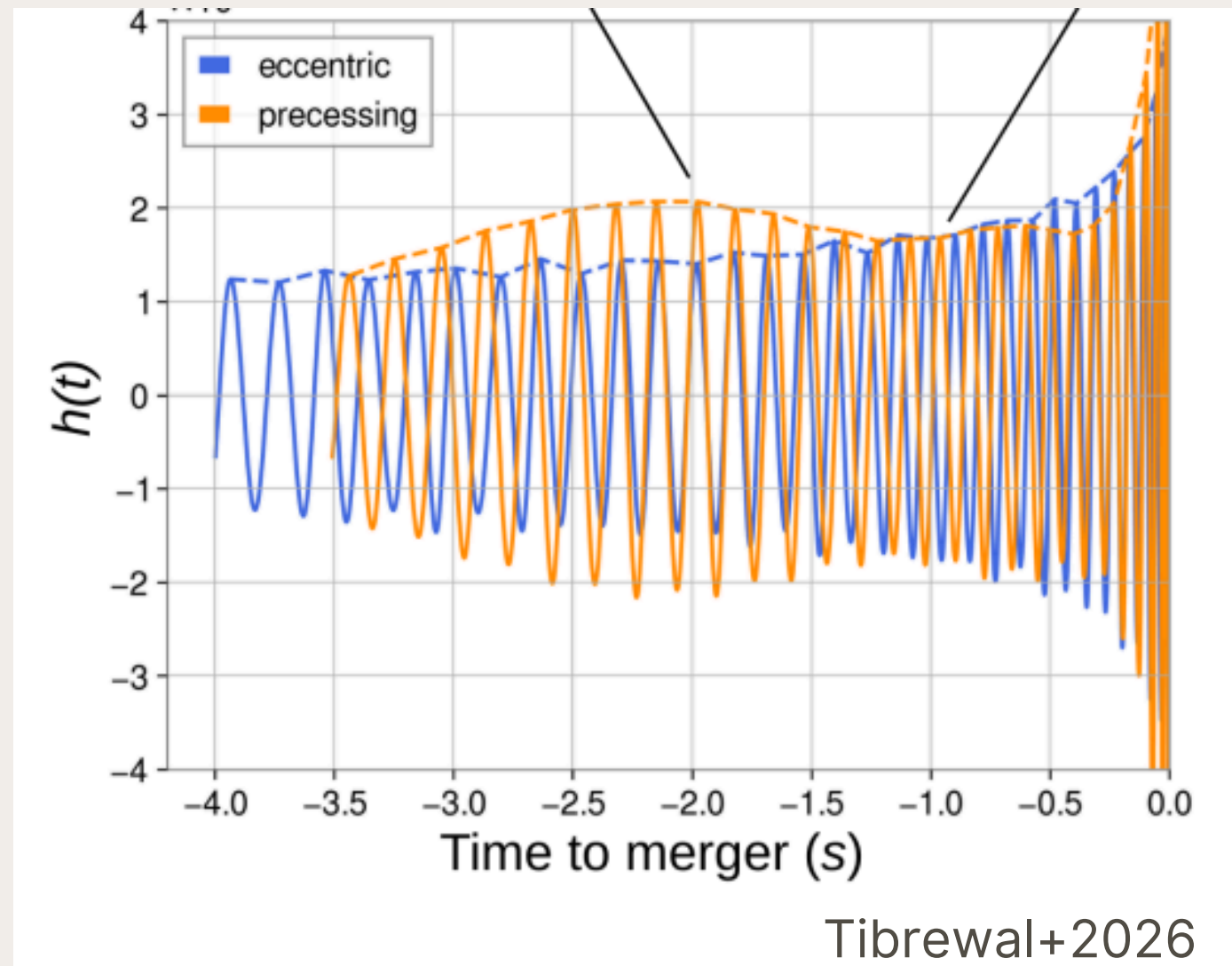
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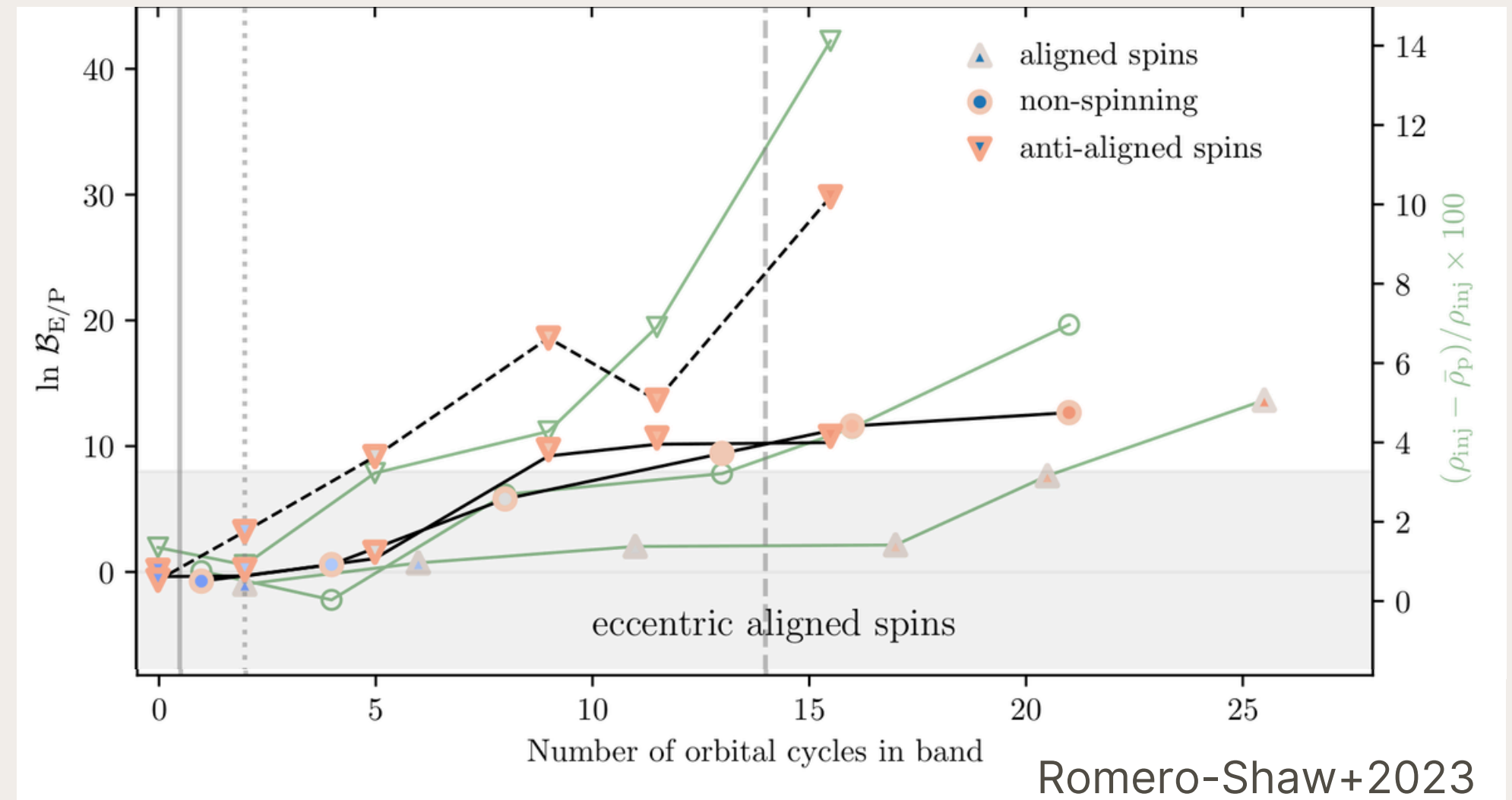
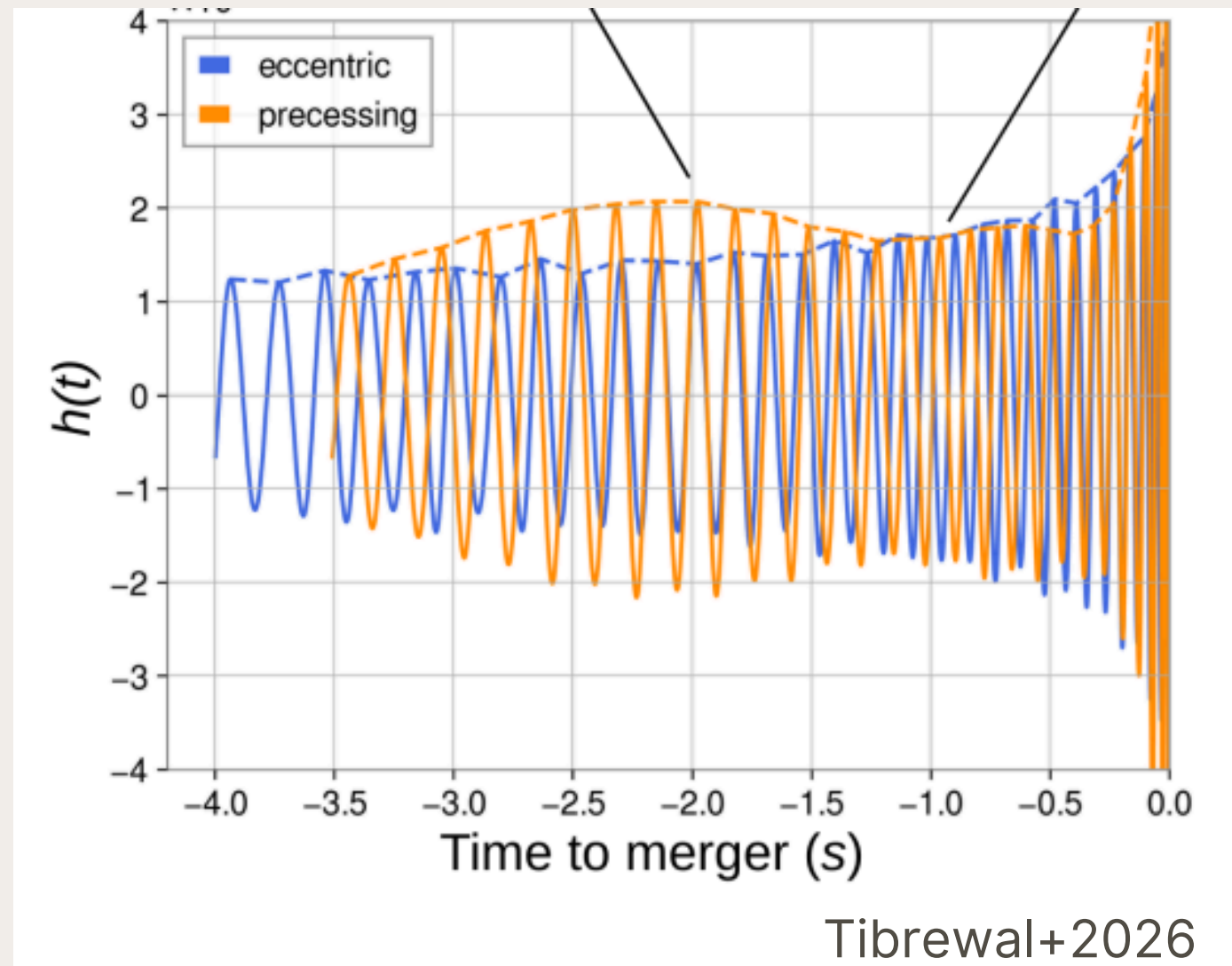
AN ASIDE ON DEGENERACY & WAVEFORM LENGTH

The Bayes factors on the previous slides compared the eccentric aligned-spin vs the quasi-circular spin-precessing hypothesis. This is because eccentricity and spin precession can be degenerate (Bustillo+ 2021, Romero-Shaw+2023, Divyajyoti+ 2025, Tibrewal+ 2026).



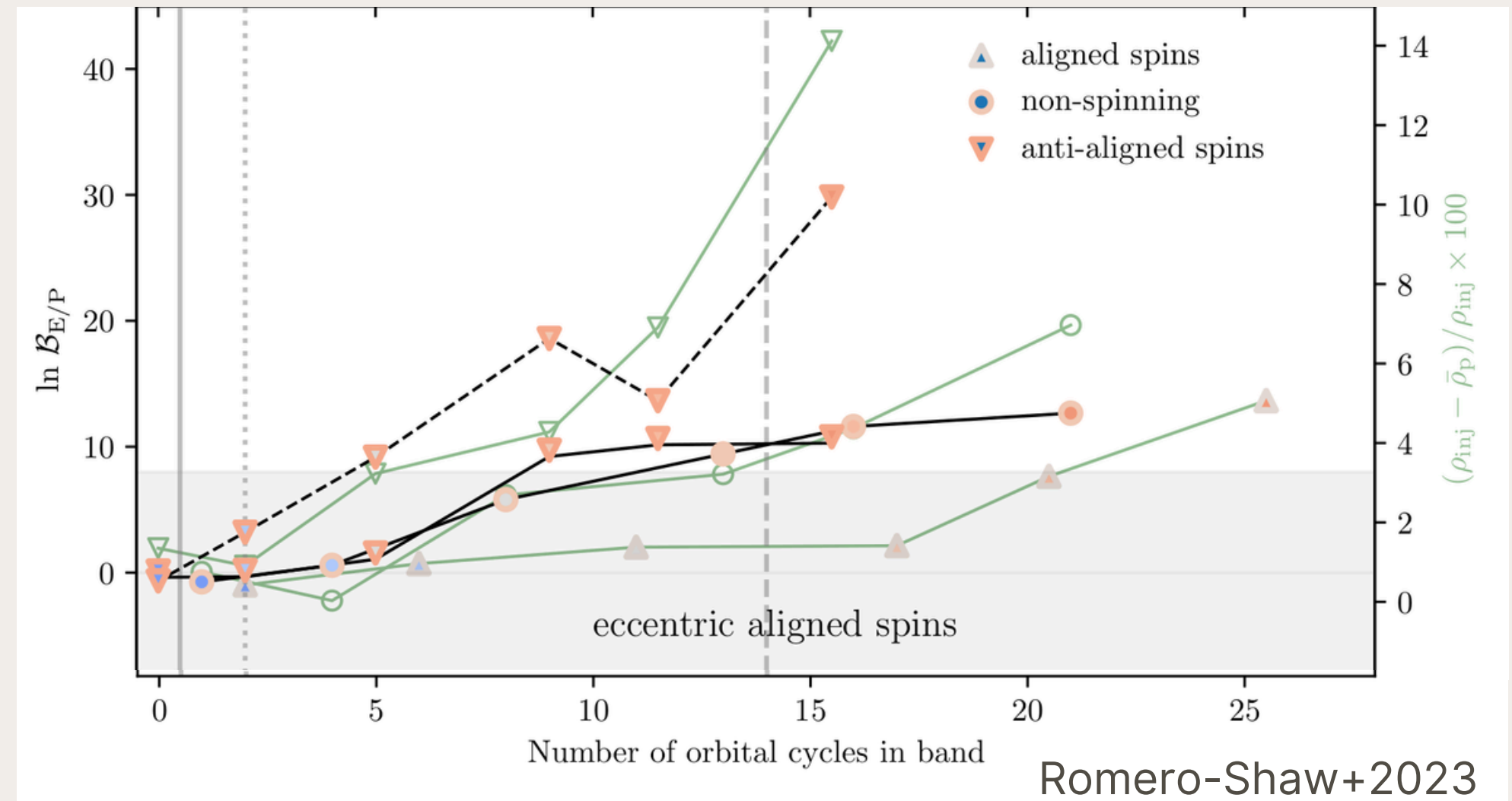
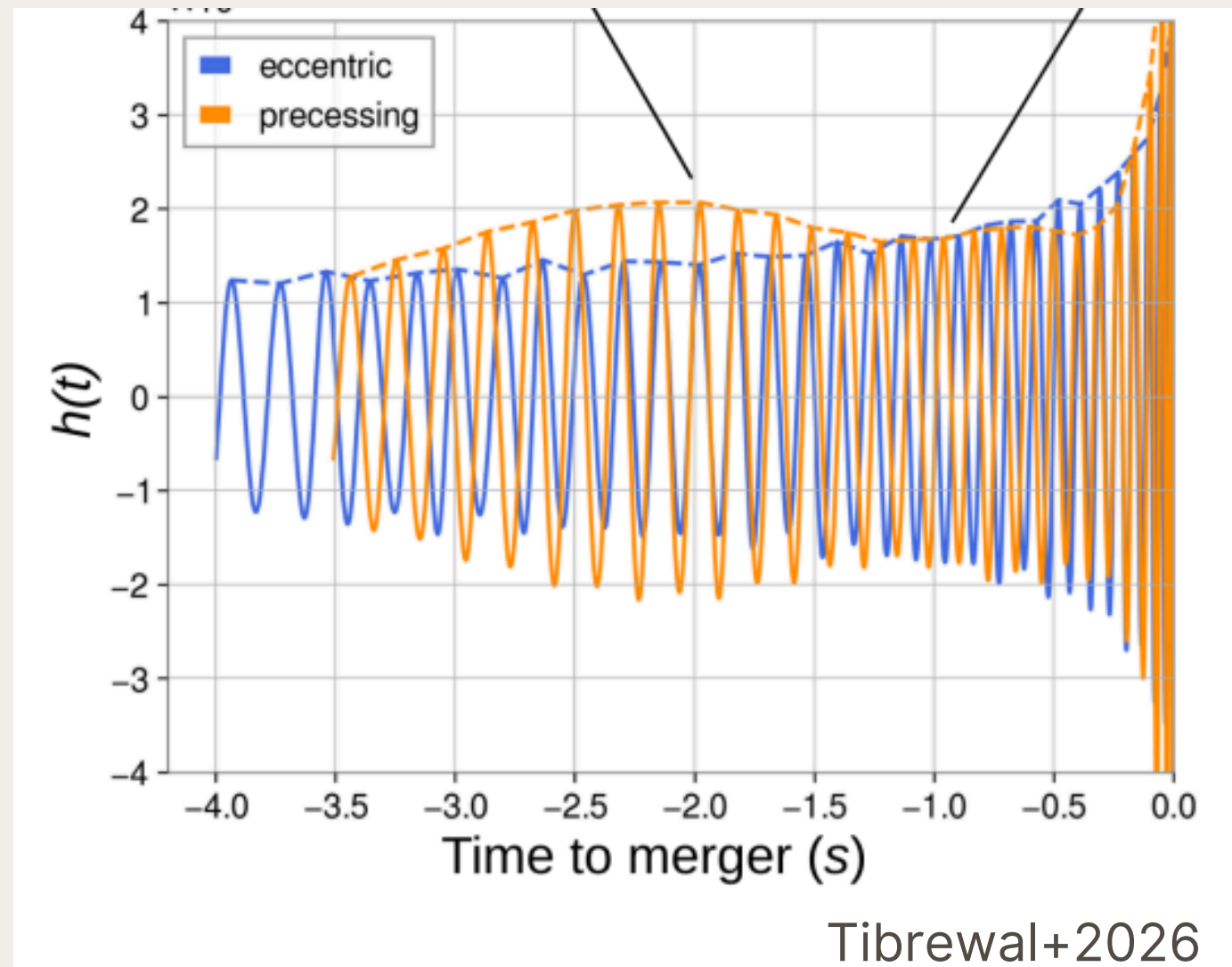
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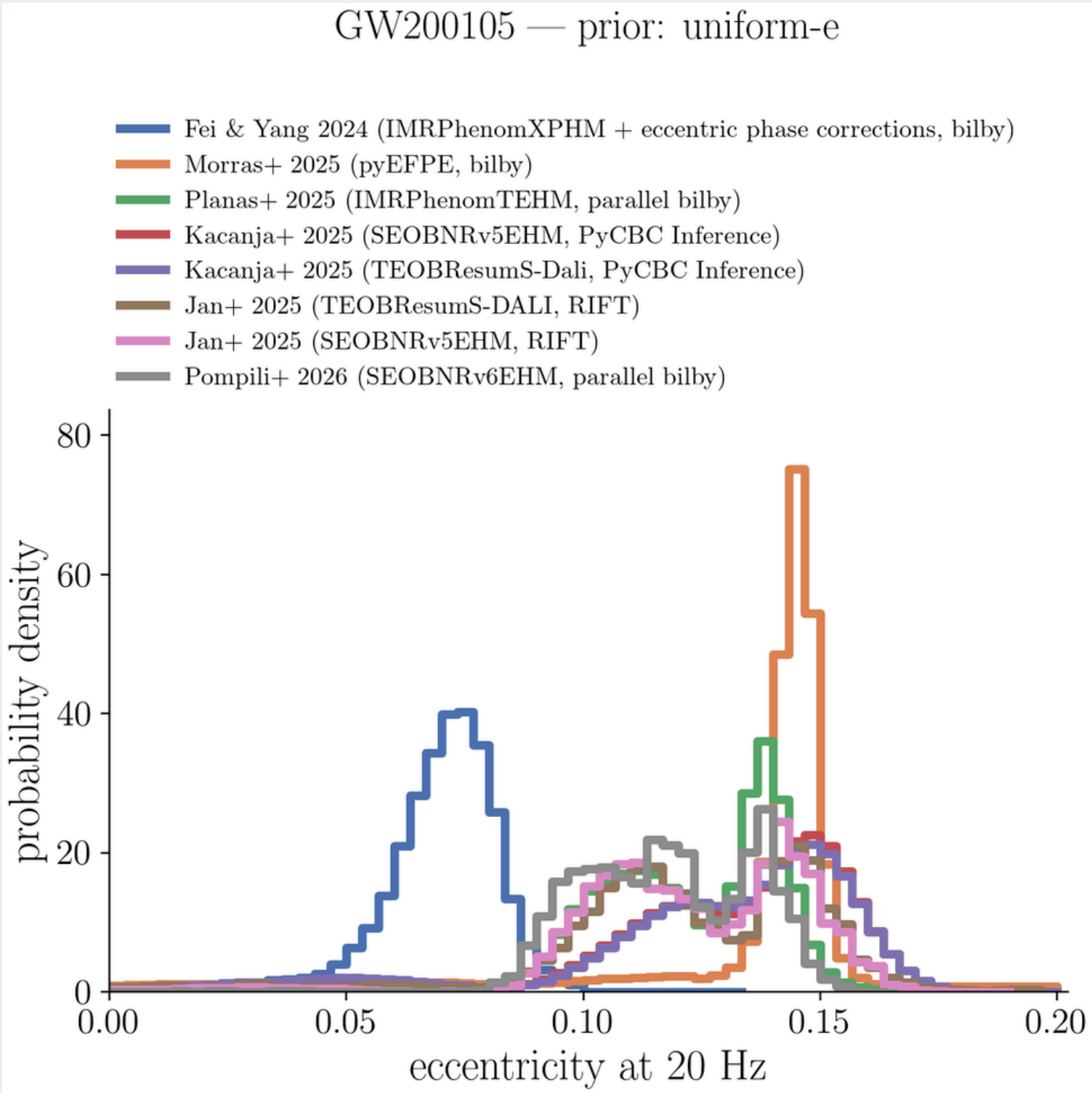


This is a statement on $p(\text{confusion} \mid \text{number of cycles})$. It is **not** a statement about $p(\text{confusion} \mid \text{number of cycles, data})$.

Consequence: if two observations have the same Bayes factor but one is shorter it does not mean that the longer one is more trustworthy.

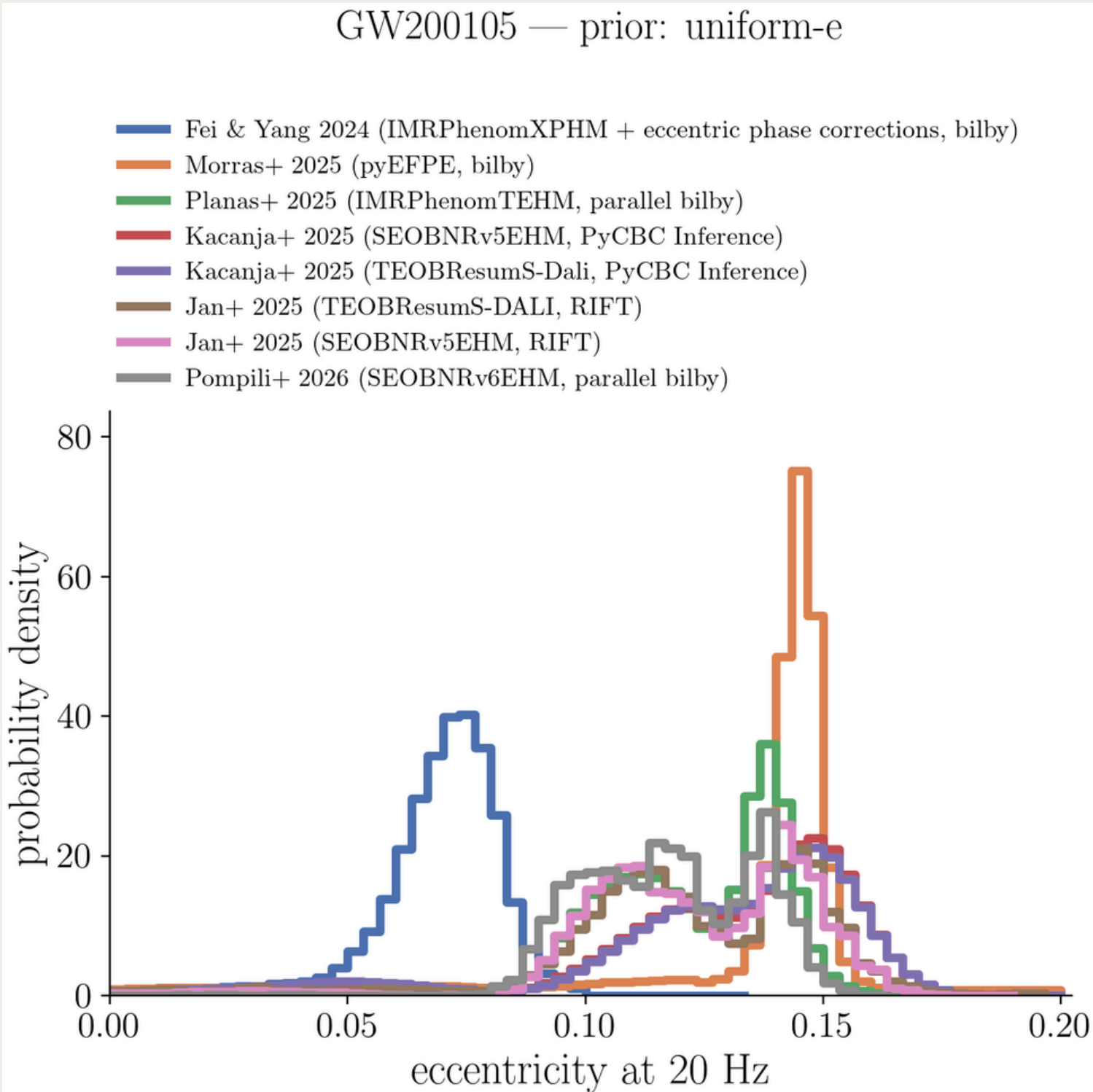
GW200105 – AN NSBH?

It was reported that there were signatures of eccentricity in a NSBH event GW200105. Note pyEFPE (Morras+ 2025) is inspiral only but **includes eccentricity and spin-precession!**



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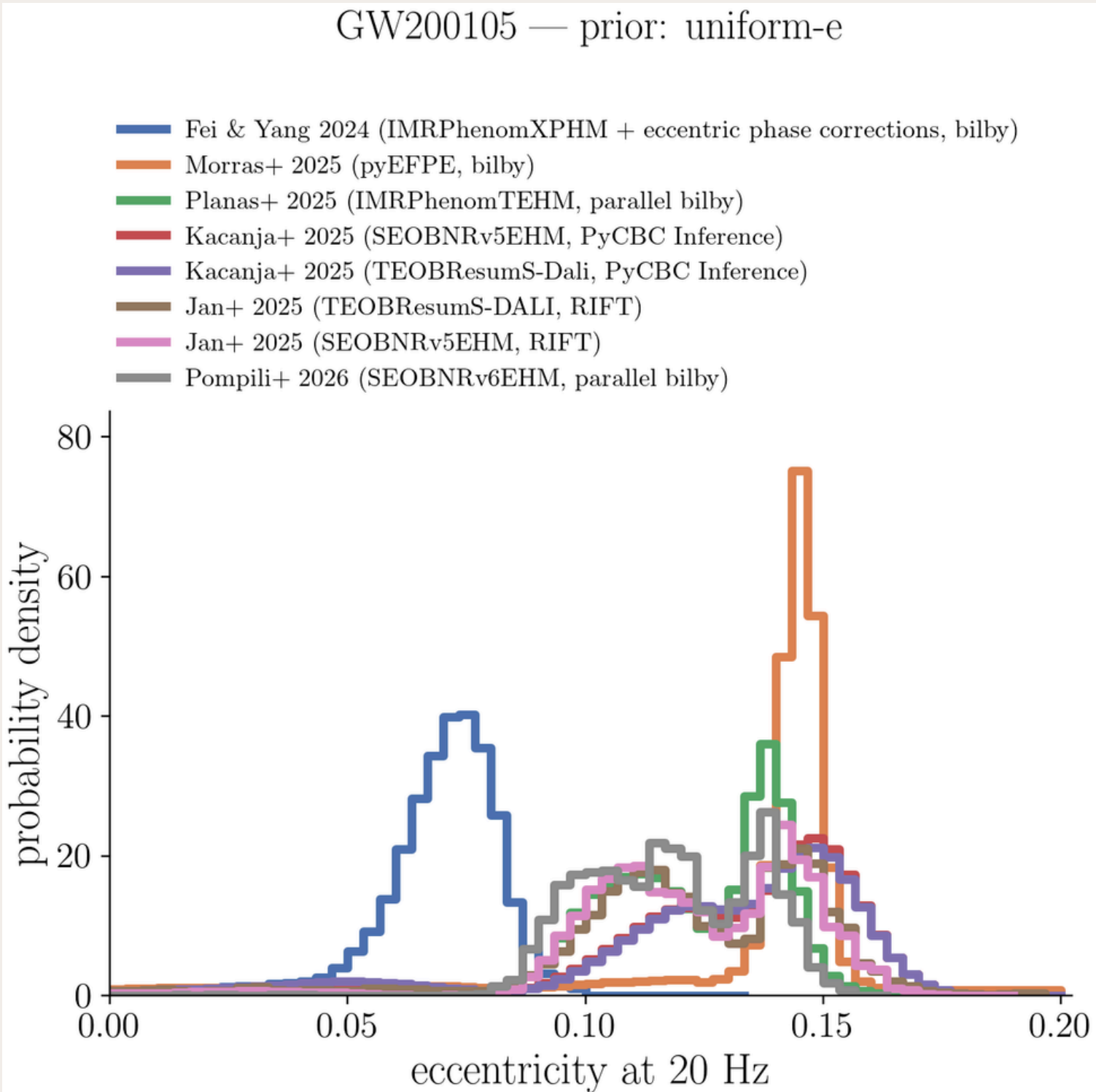
Criticism of this event was that GW200105 was reported as a low-significance single detector trigger (**FAR of 1 per 2.8 years**, (LVK 2021).

Davies & Harry 2022 showed with updated methods the **FAR is 1 per 50,000 yrs**. Phukon+ 2025 further showed that eccentric templates also decrease the FAR. The glitch also has negligible impact (Morras+ 2025)

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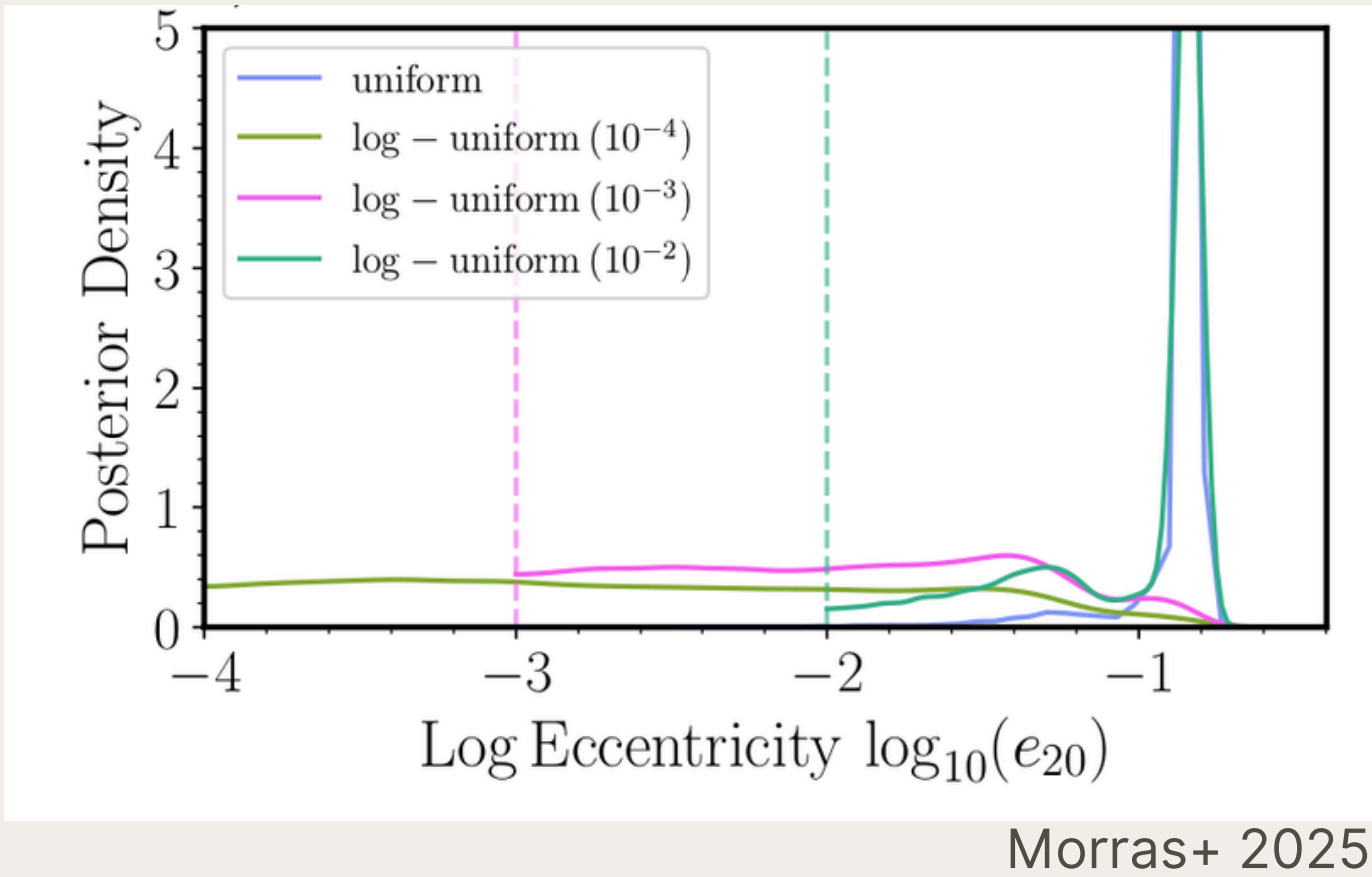
Bayes factor ~ 17, 1.3, __ (uniform, log-uni, expert prior)



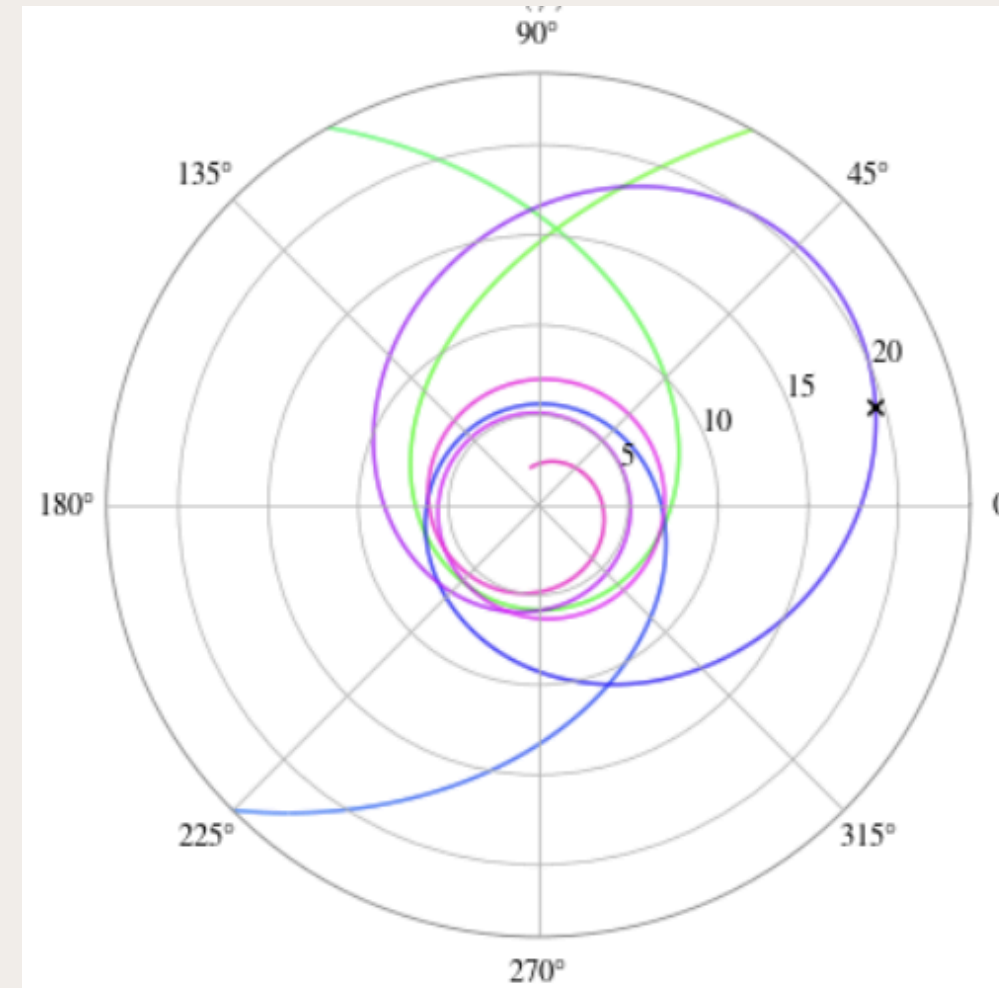
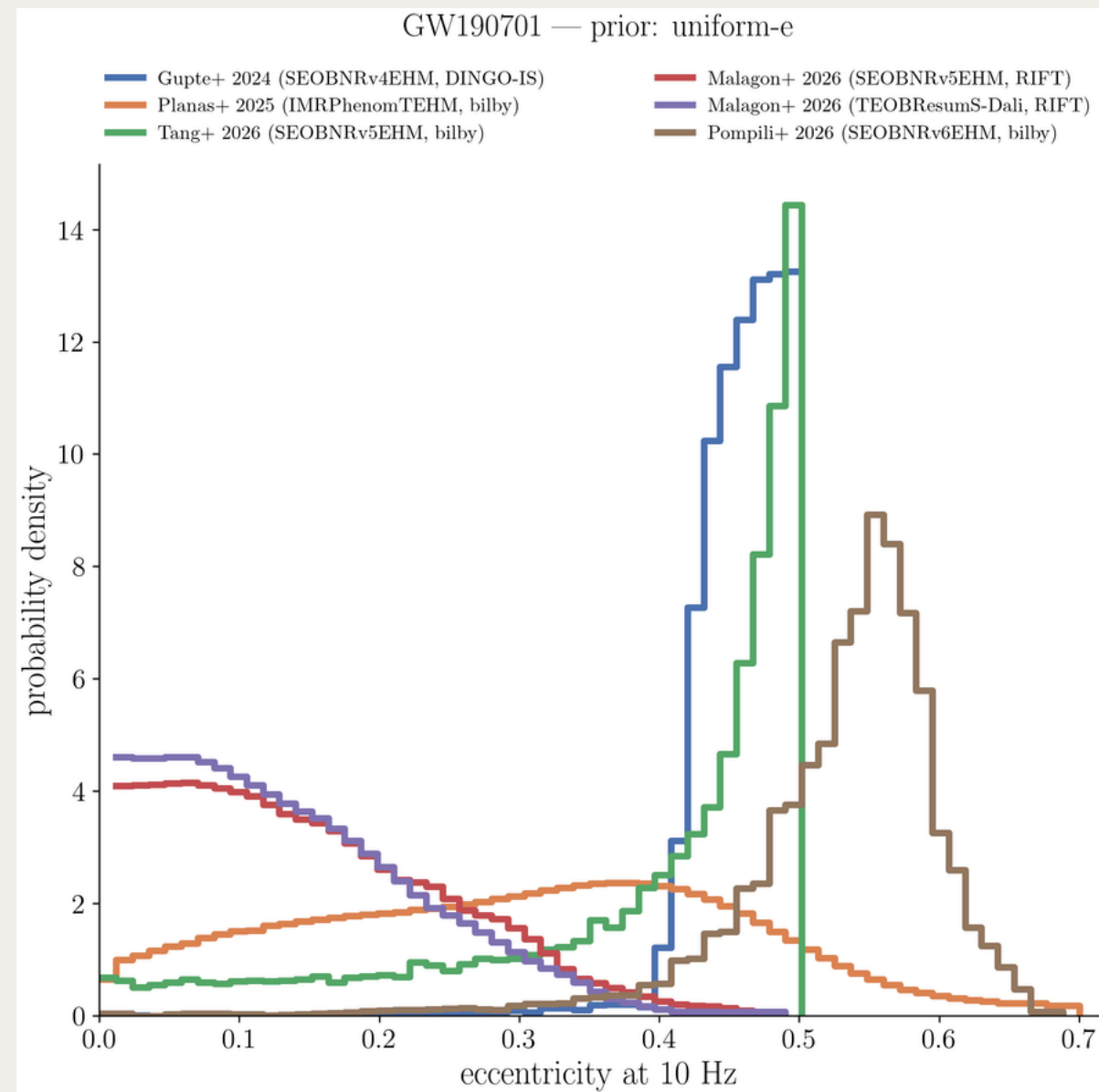
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GW190701



Gupte+ 2024 data release

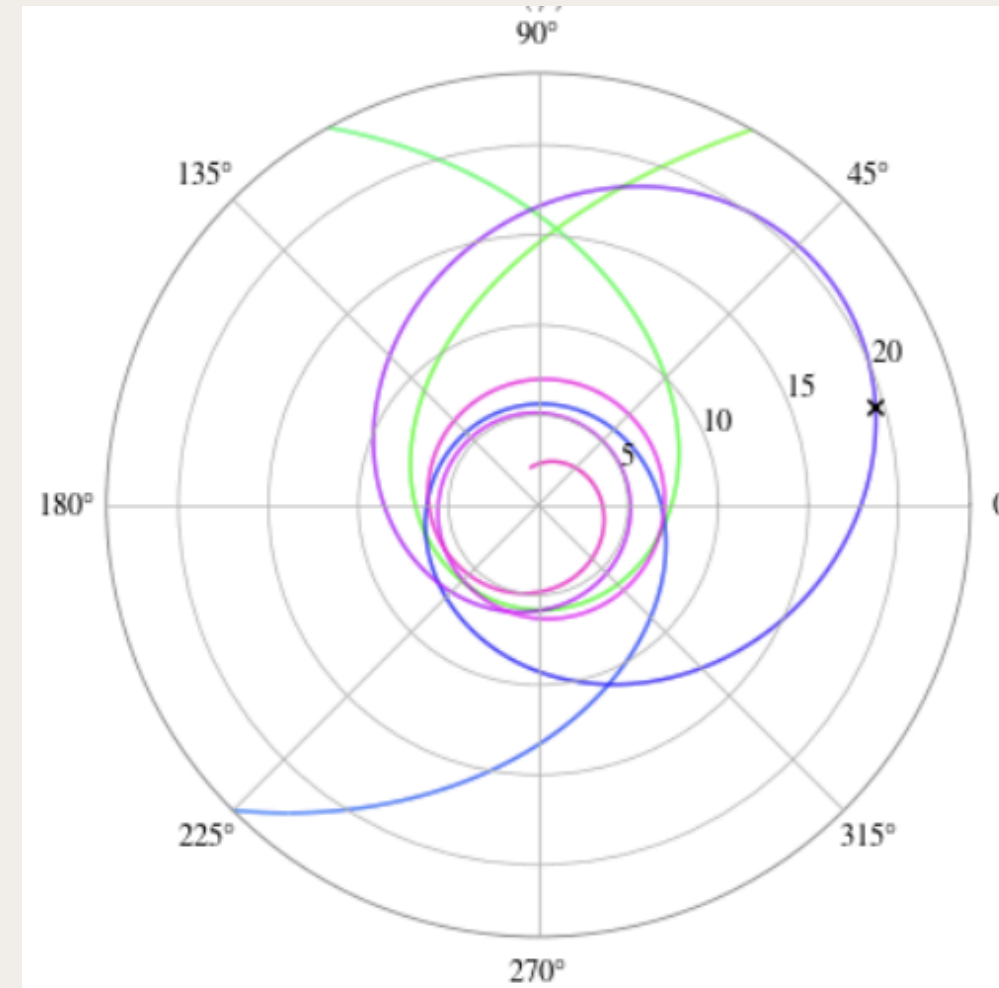
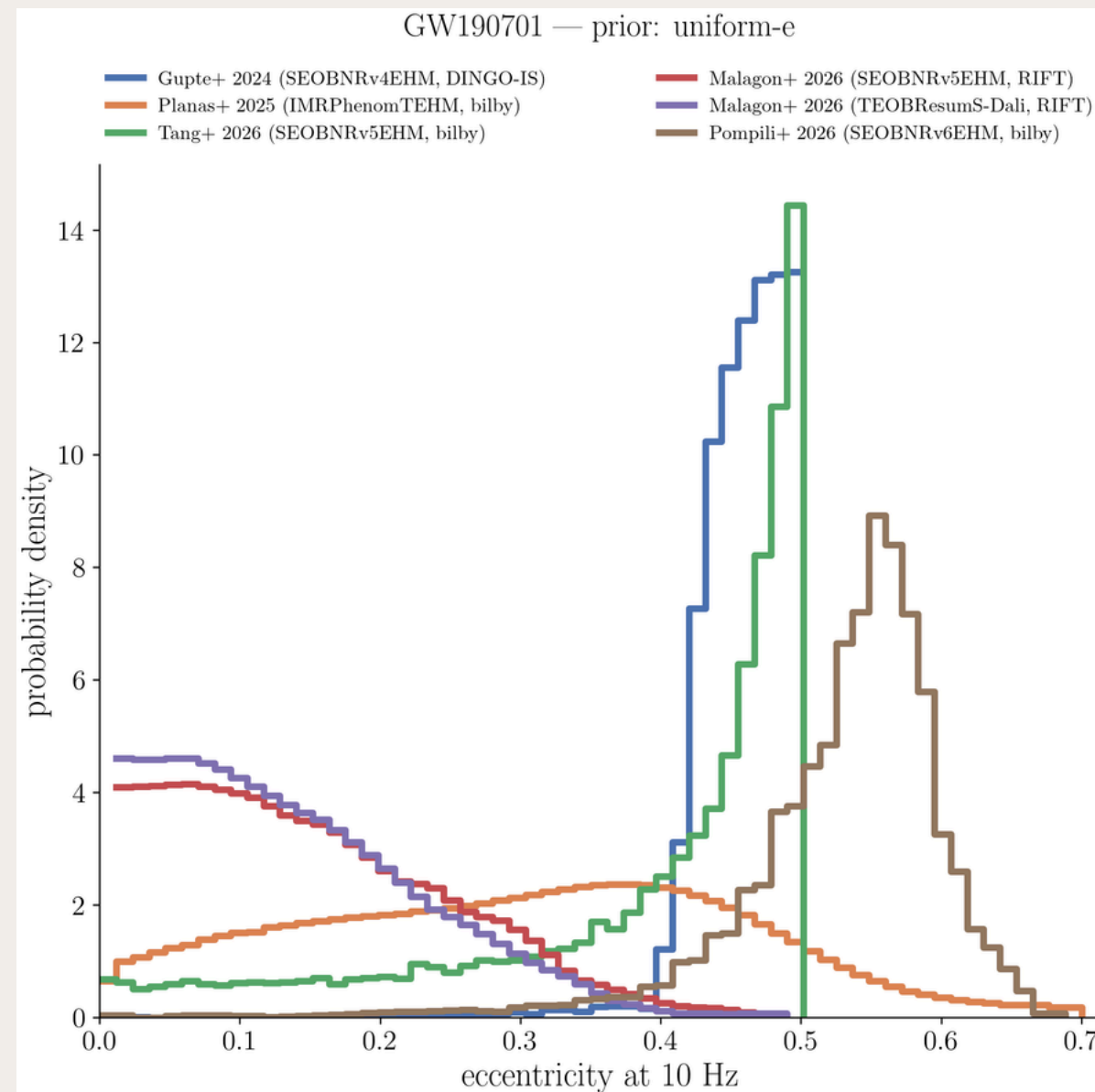
> This is a heavy BBH with a source frame mass of 140

> There are waveform systematics and sampler systematics. IMRPhenomTEHM find less support than SEOBNR family

> RIFT and Bilby disagreement?

> This event had a high Bayes factor 407, 51, — (uniform, log-uni, expert prior)

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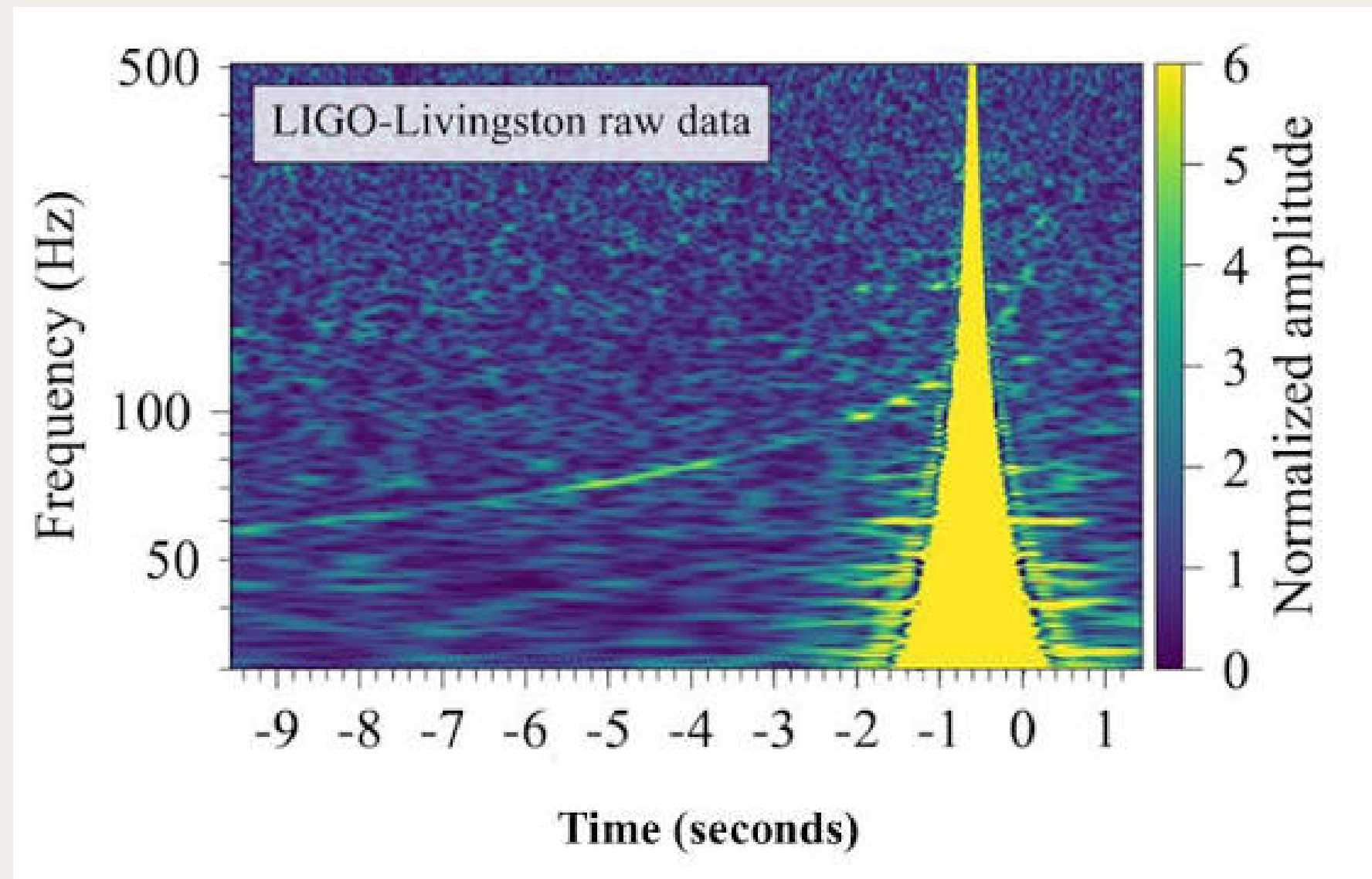
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There was a scattered light glitch in LIGO Livingston during the detection. Therefore, the “BayesWave” (Cornish+ 2015) glitch mitigation algorithm was employed to subtract the glitch.

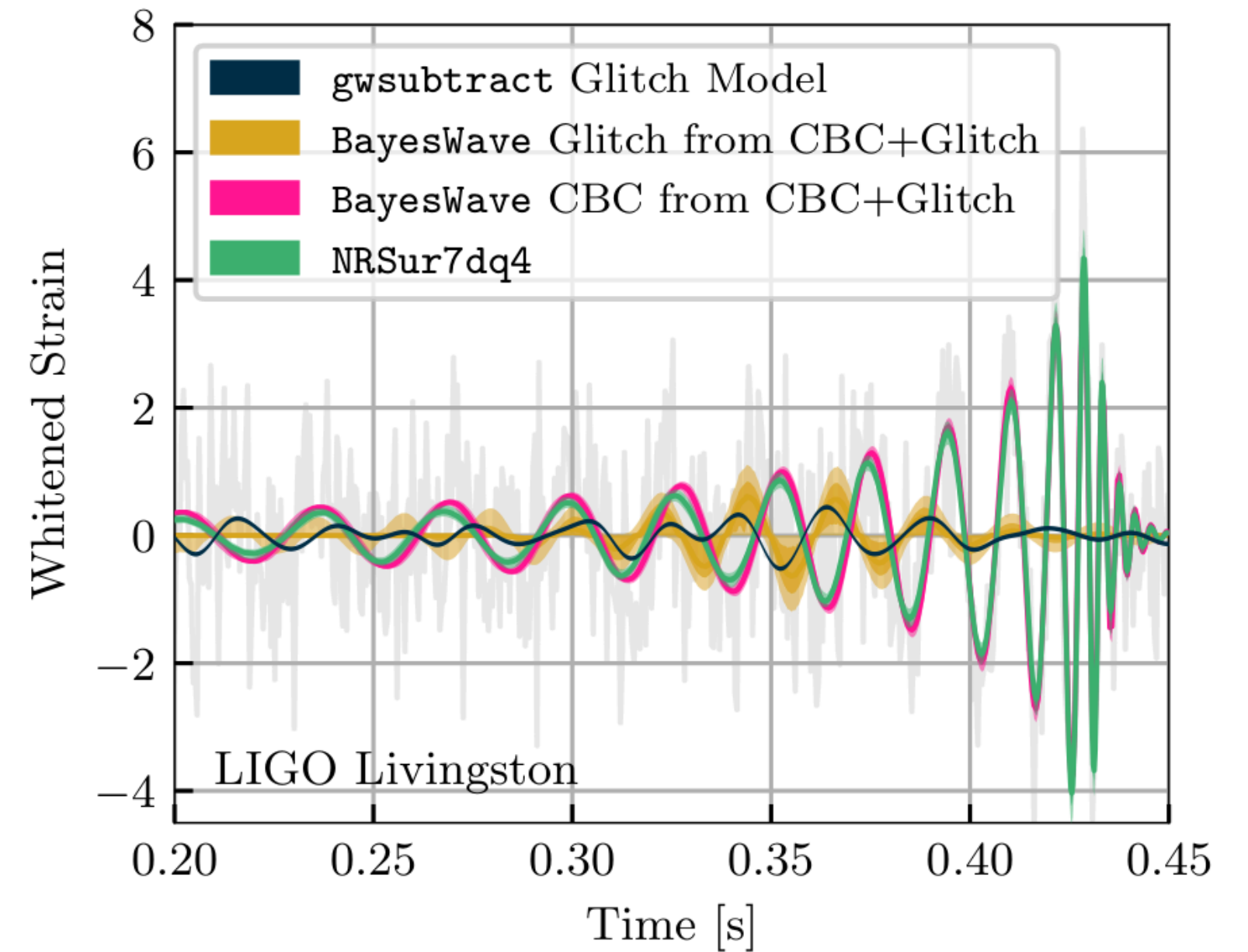
HOW DOES GLITCH MITIGATION WORK?



LVK Collaboration 2017

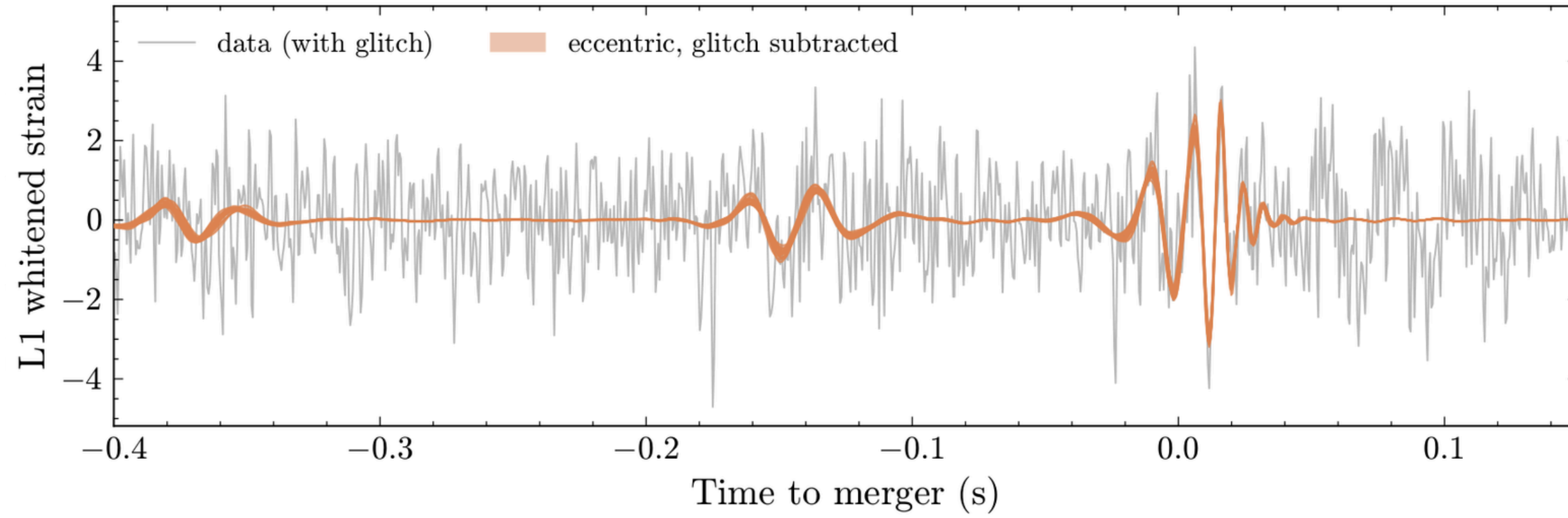
HOW DOES GLITCH MITIGATION WORK?

- > How does “BayesWave” work?
- > Fit the signal coherently with a waveform model
- > Fit the glitch incoherently with sine-Gaussian wavelets
- > **Note this is not waveform model agnostic.**



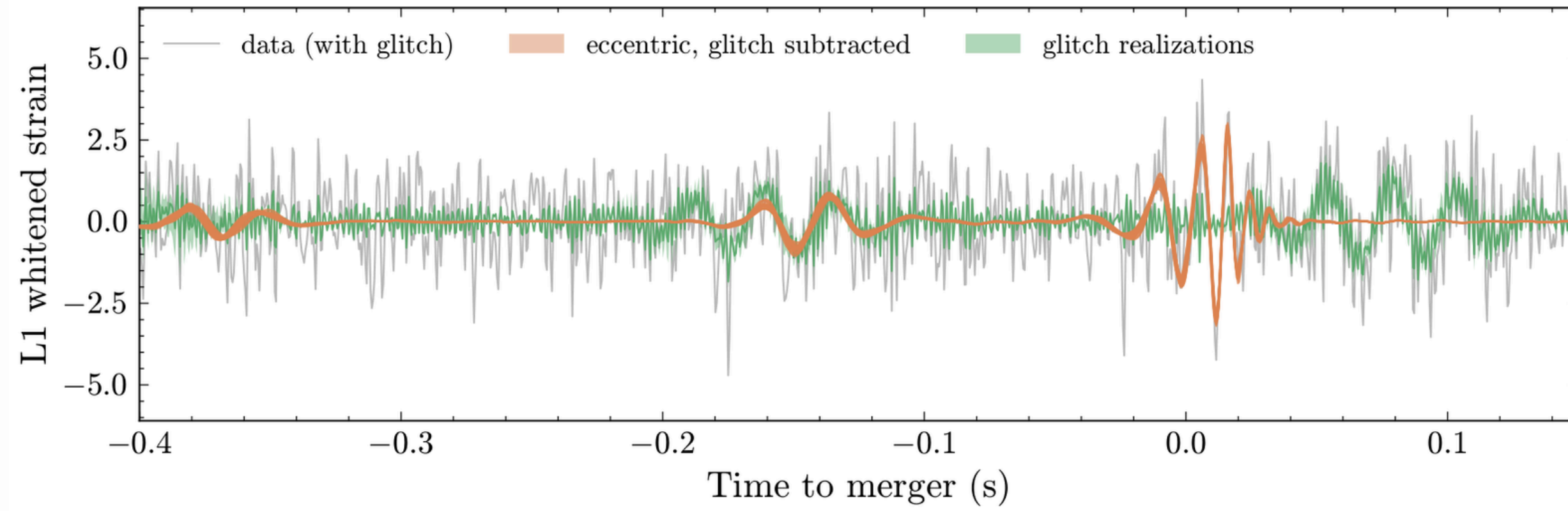
Payne+ 2022

GW190701

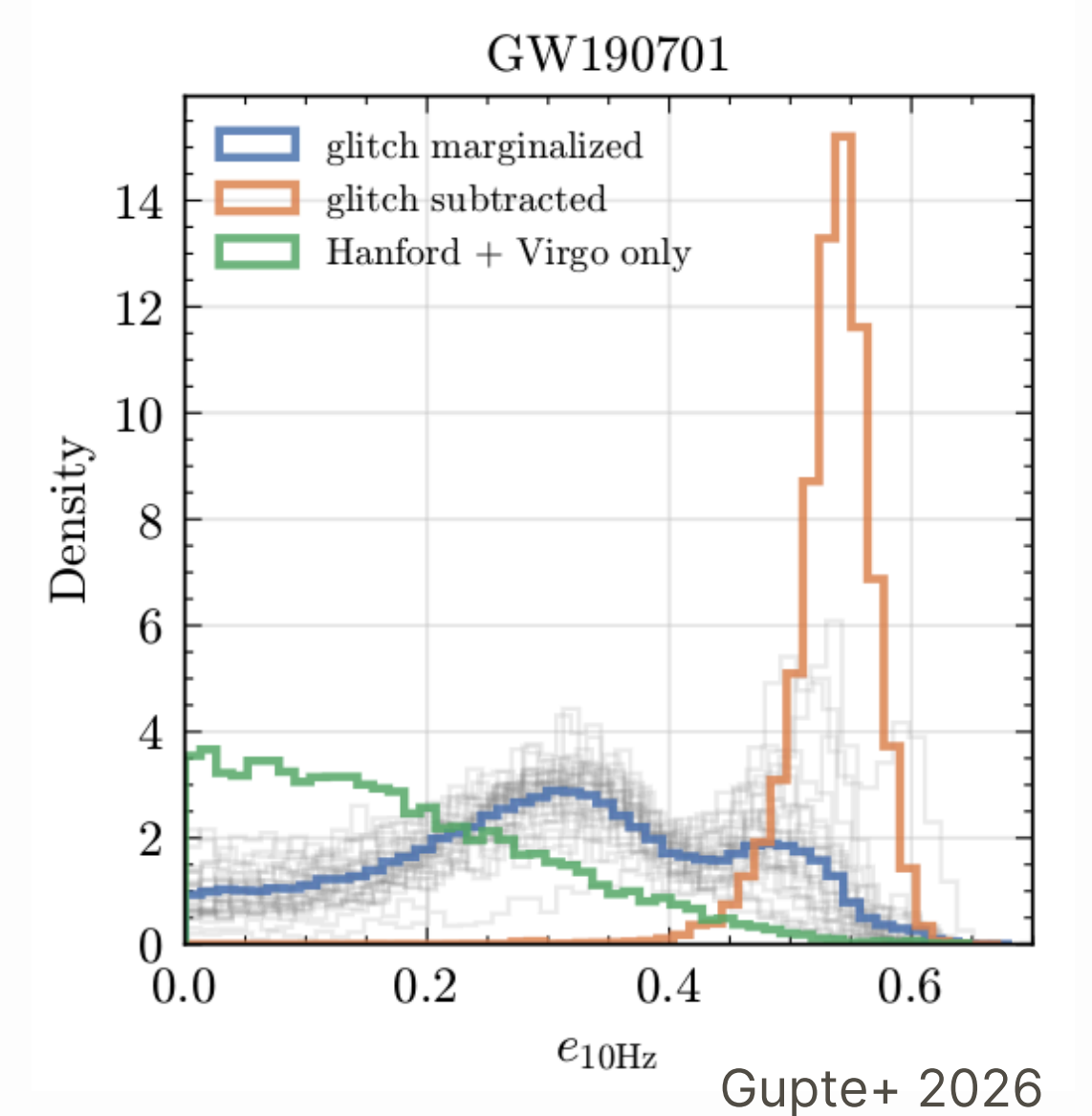
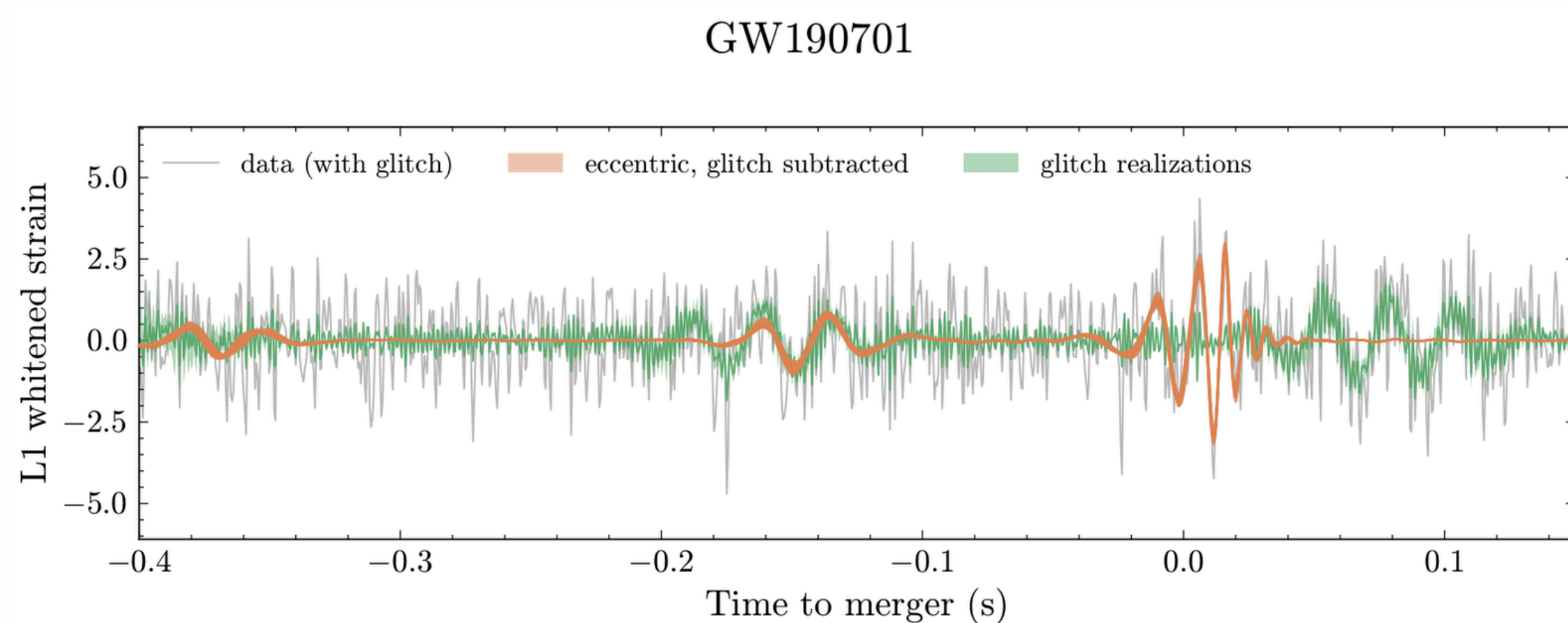


GW190701 - GLITCH

GW190701



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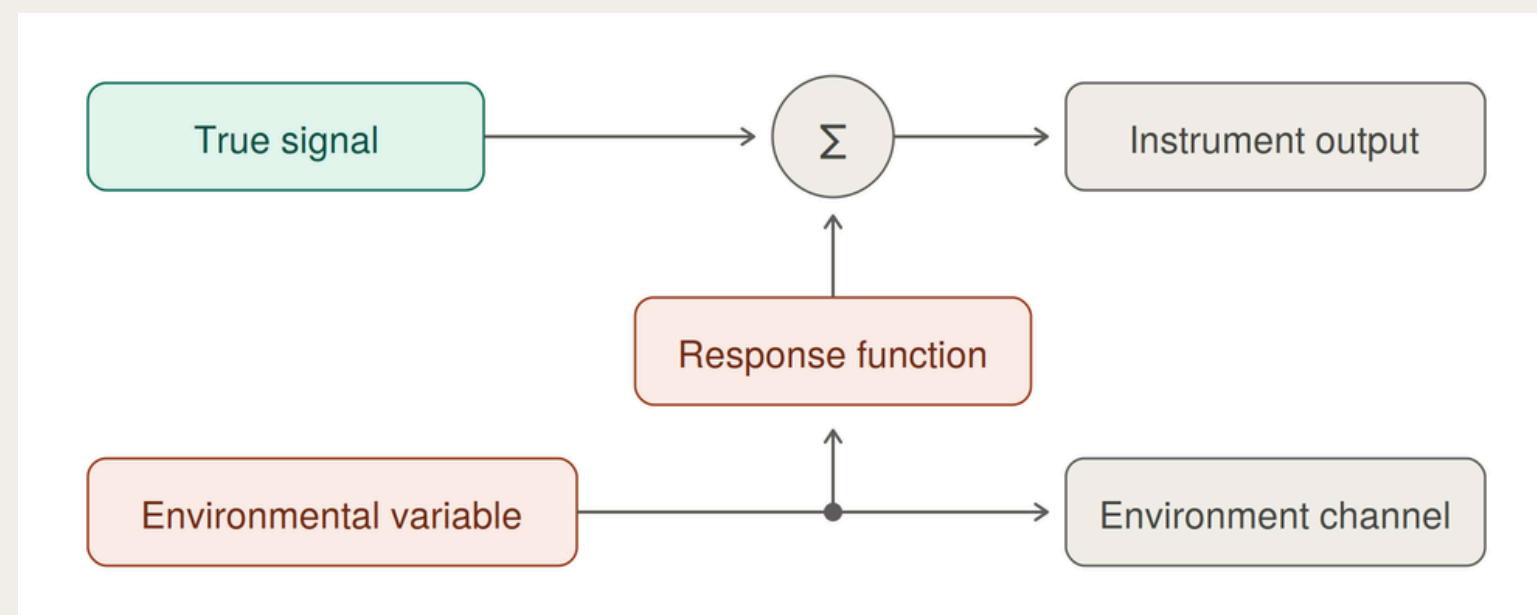


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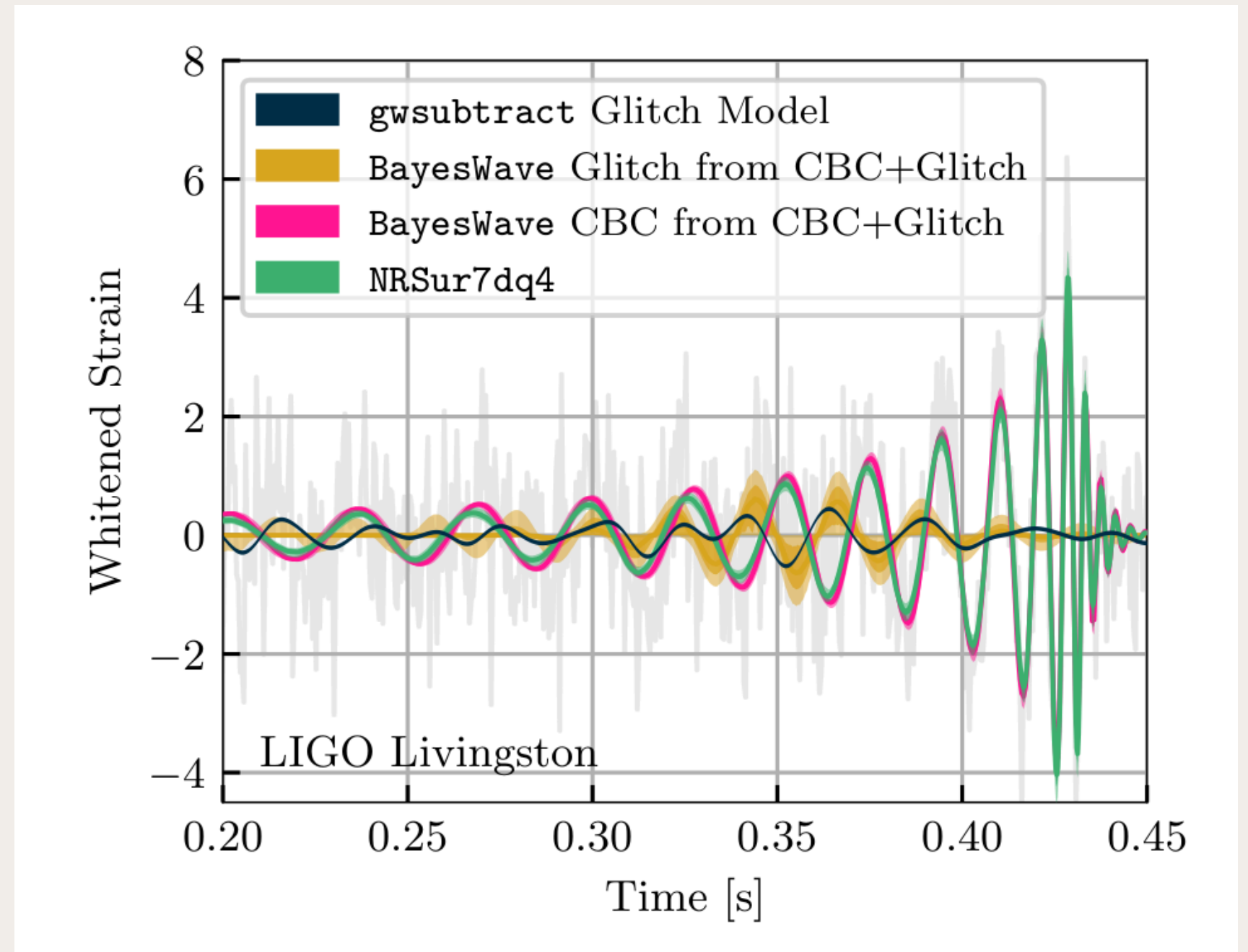
- > Just using one draw of the BayesWave glitch mitigation, can you massively differ results compared to marginalizing over the glitch uncertainty
- > The majority of the eccentric evidence comes from L1.
- > The glitch-marginalized run still has some support for high eccentricities. To what extent is the glitch fitting the eccentricity and the eccentricity fitting the glitch?

HOW DOES GLITCH MITIGATION WORK?

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Adapated from Allen+1998



Payne+ 2022

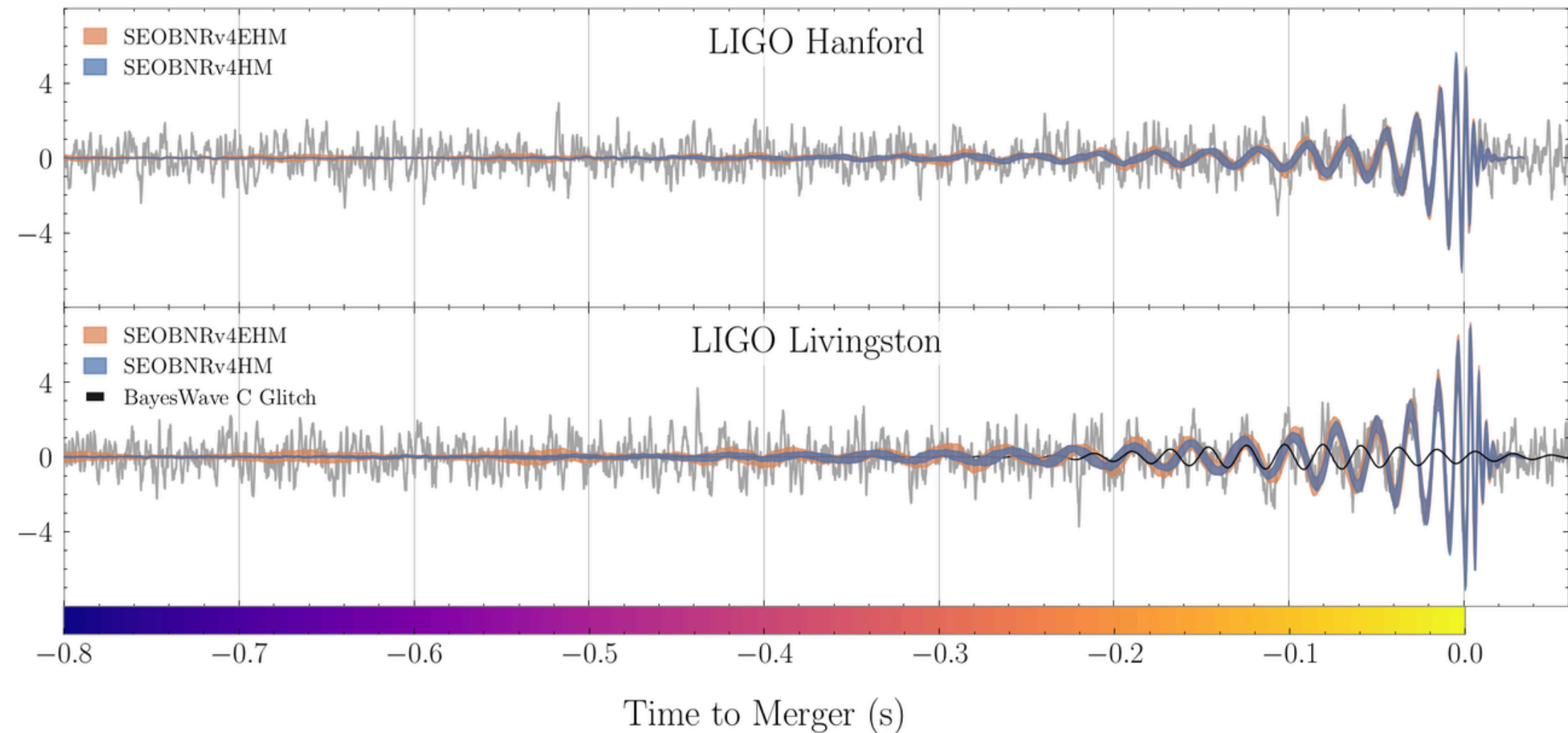
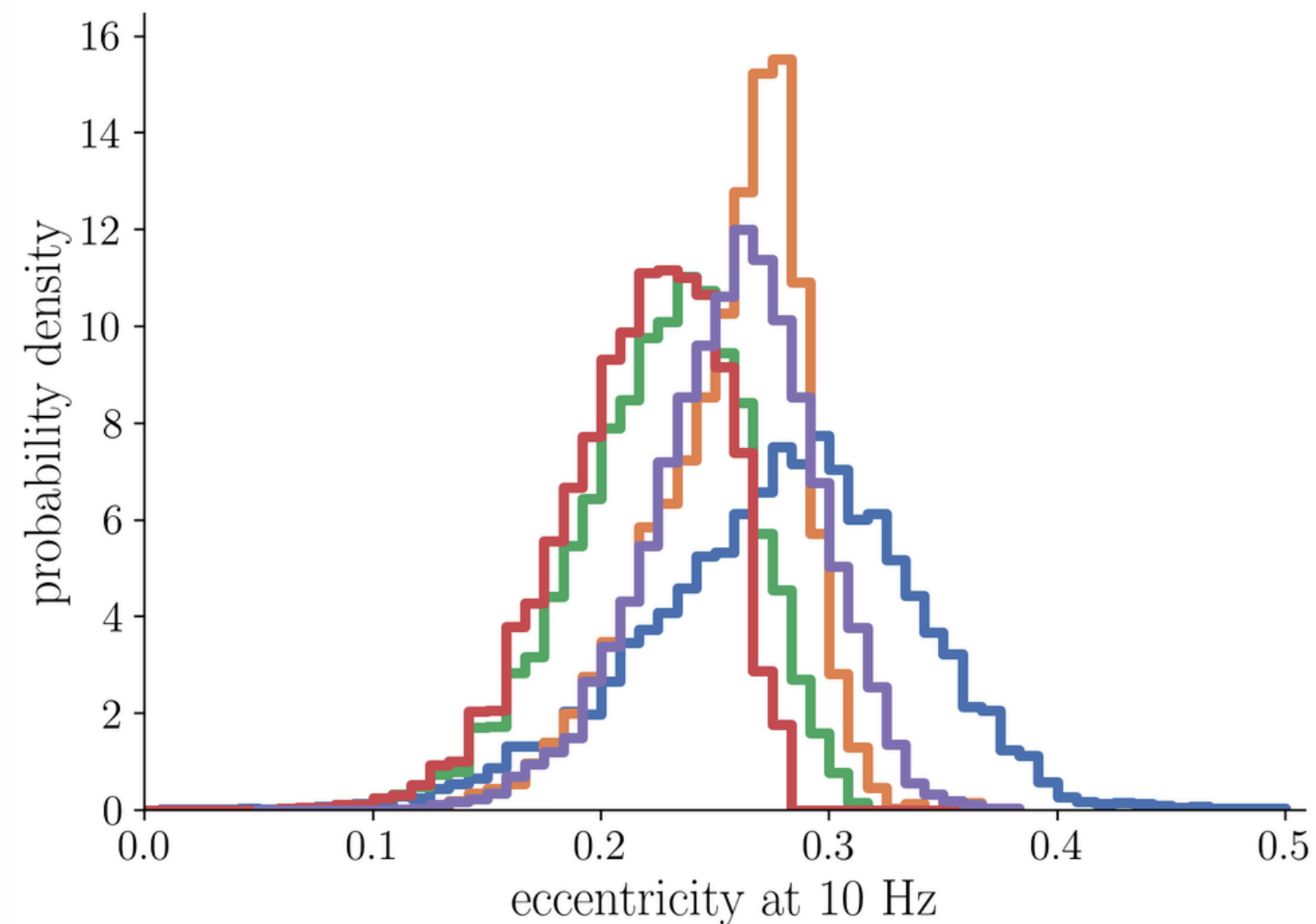
- > Use auxillary channels to deglitch (Allen+ 1999, Davis+ 2022, Macas+ 2024)?
- > **Note, this is waveform model agnostic.** It uses more information, but it doesn't output uncertainty.

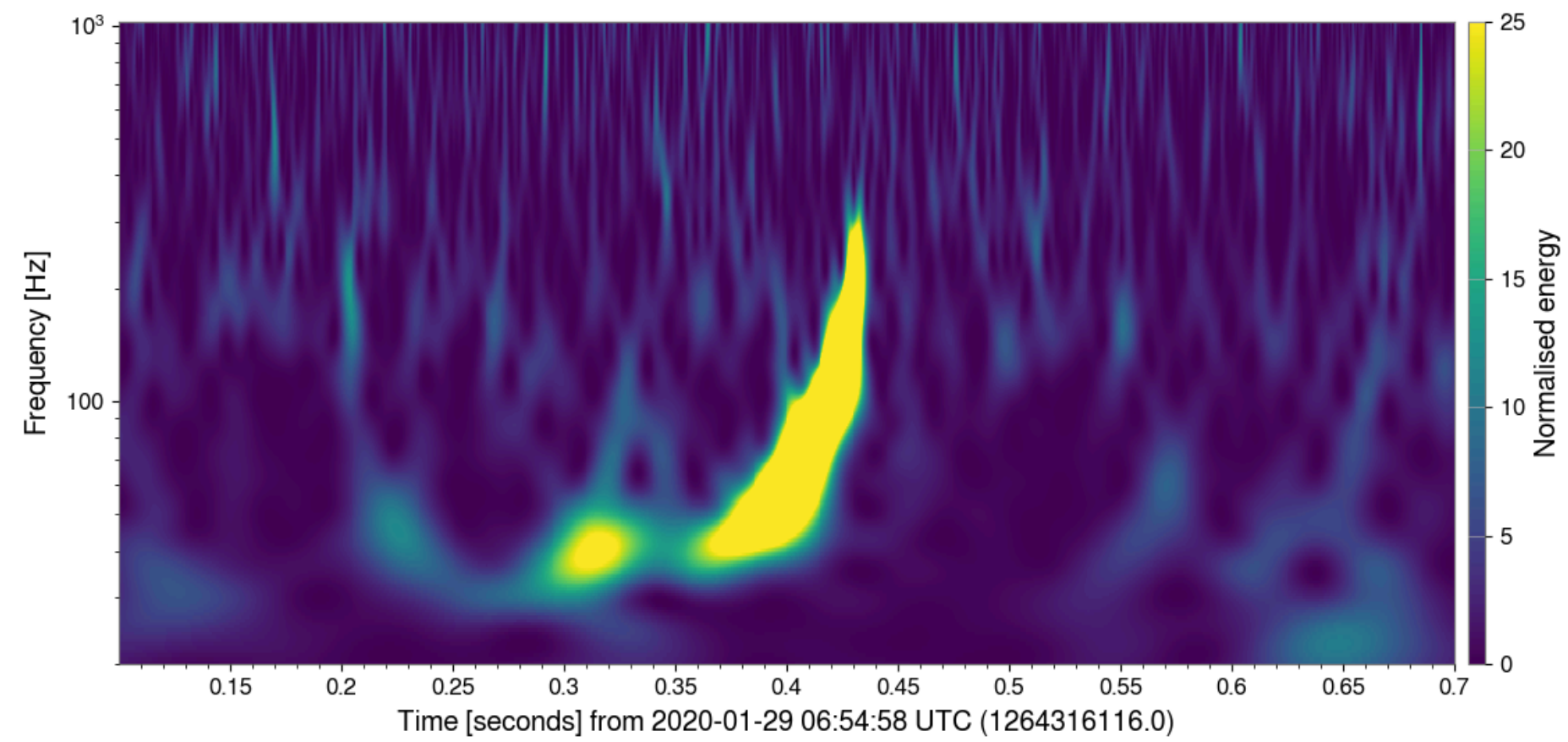
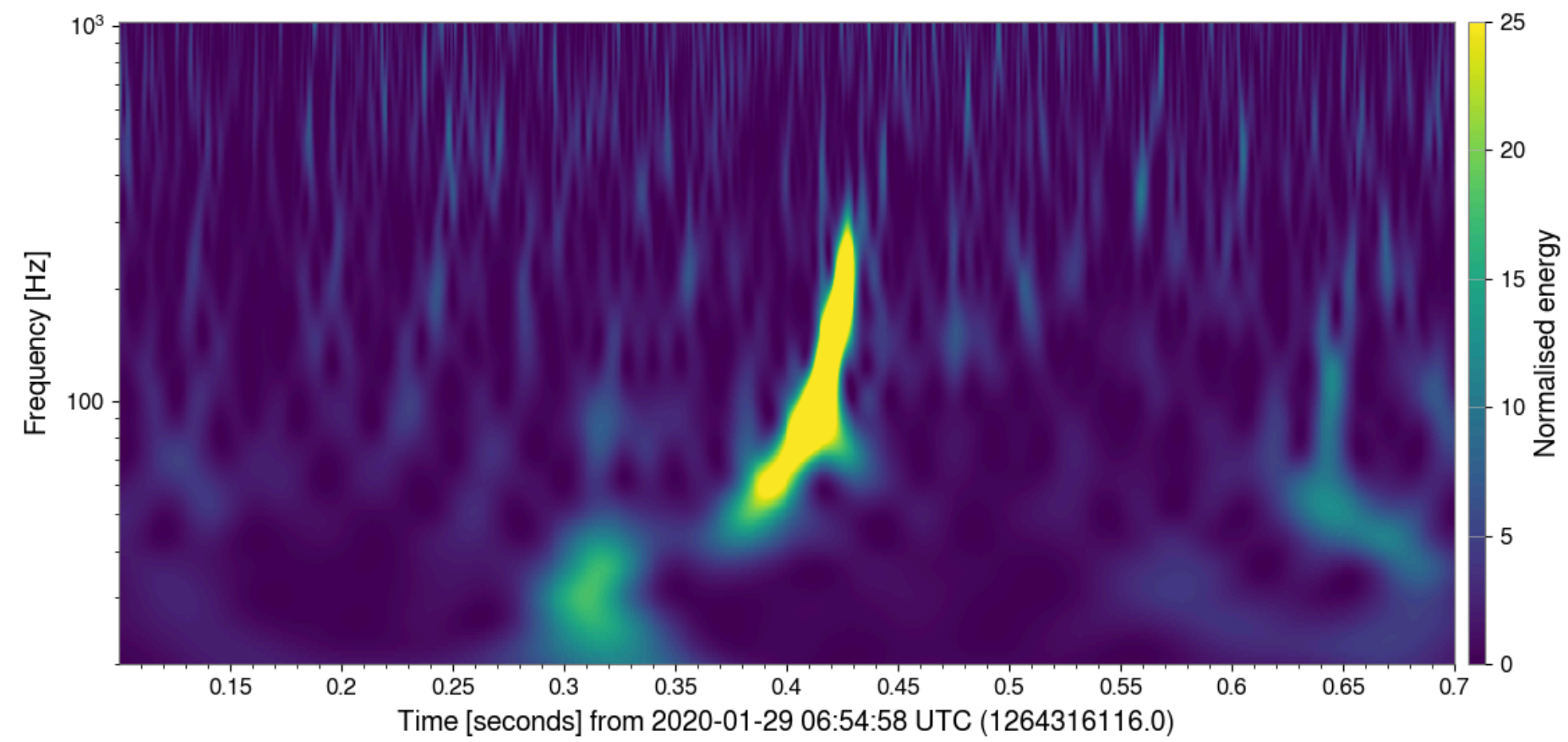
GW200129

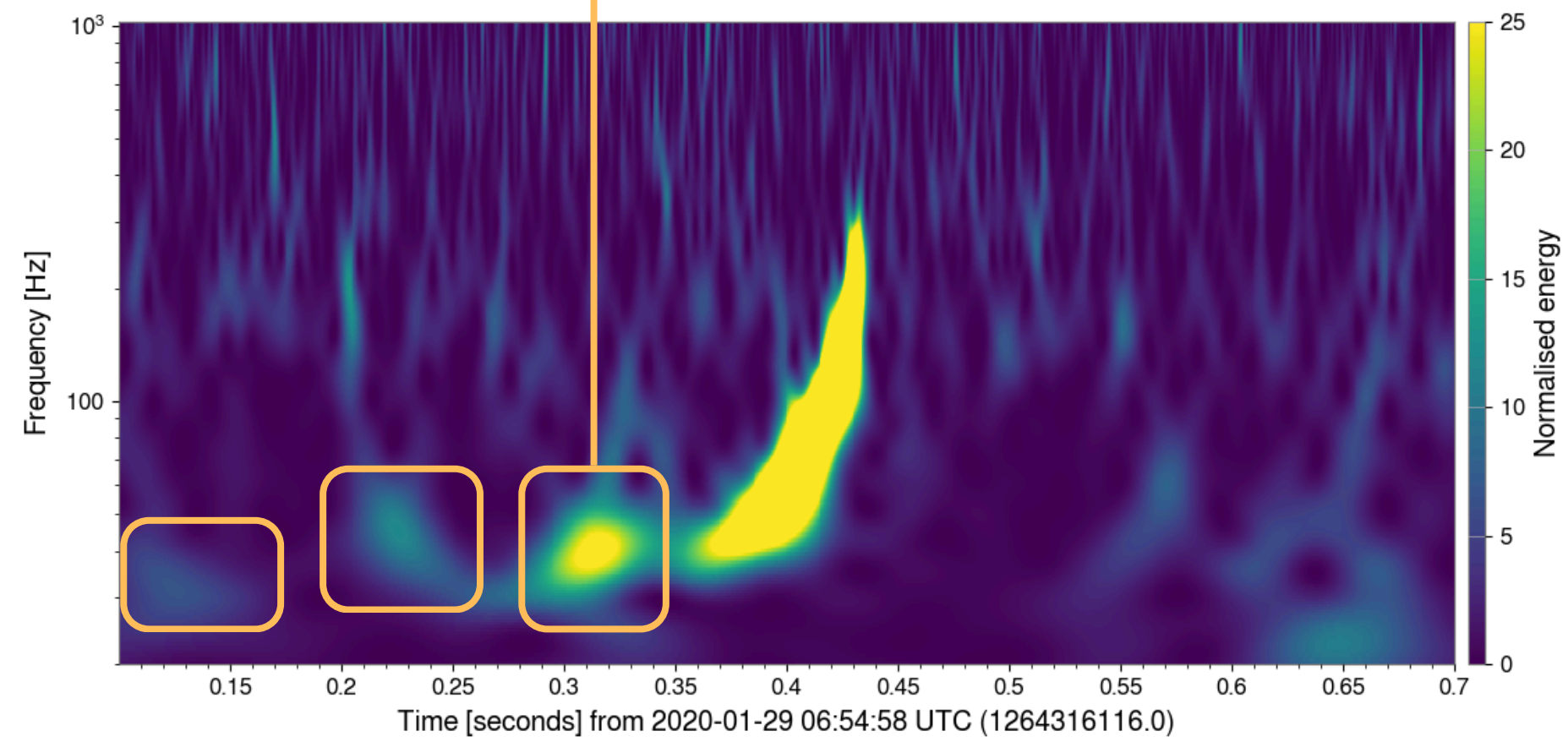
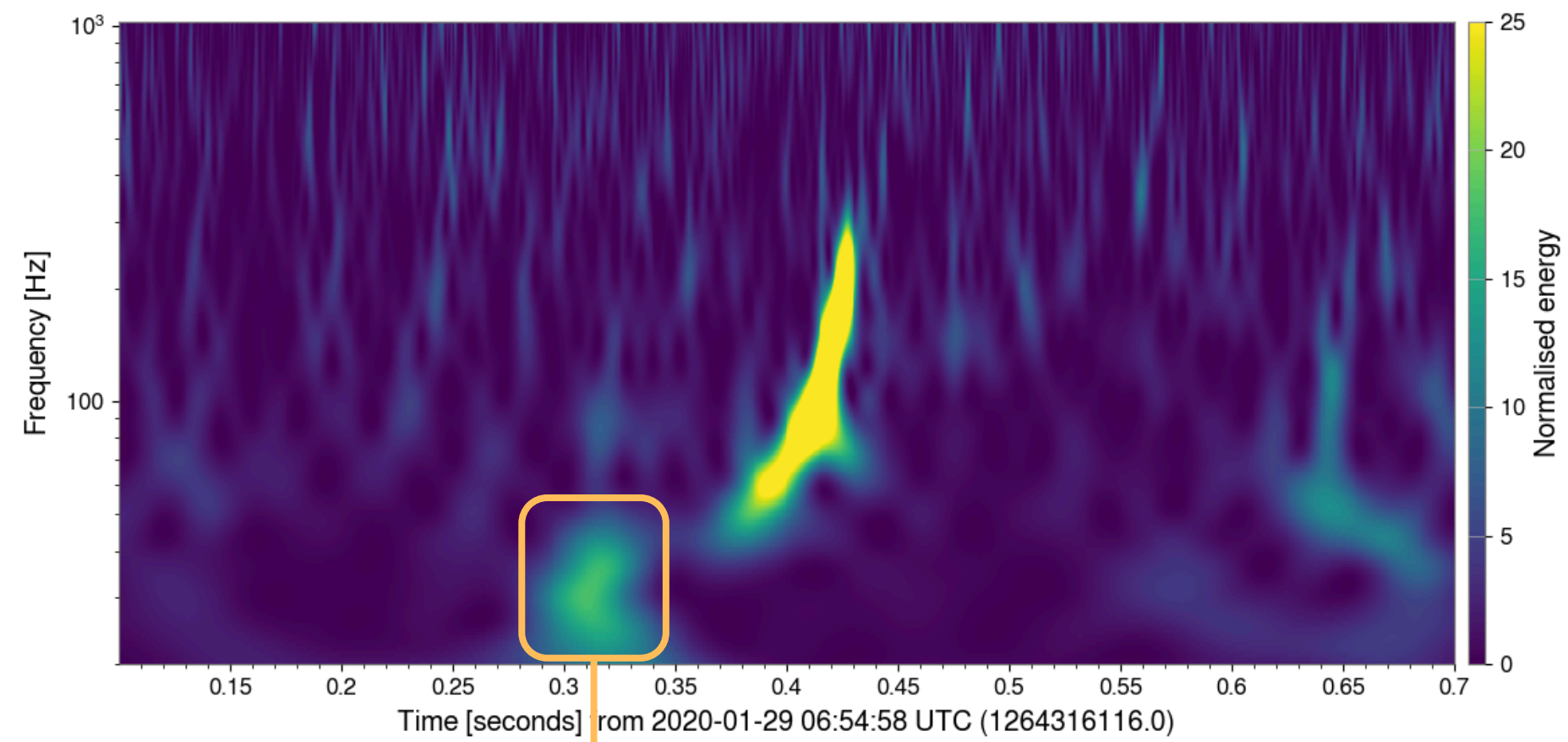
> This is a 70 solar mass BBH. The Bayes factors are 10,000, 1,698 (uniform prior, log uniform prior, expert prior)

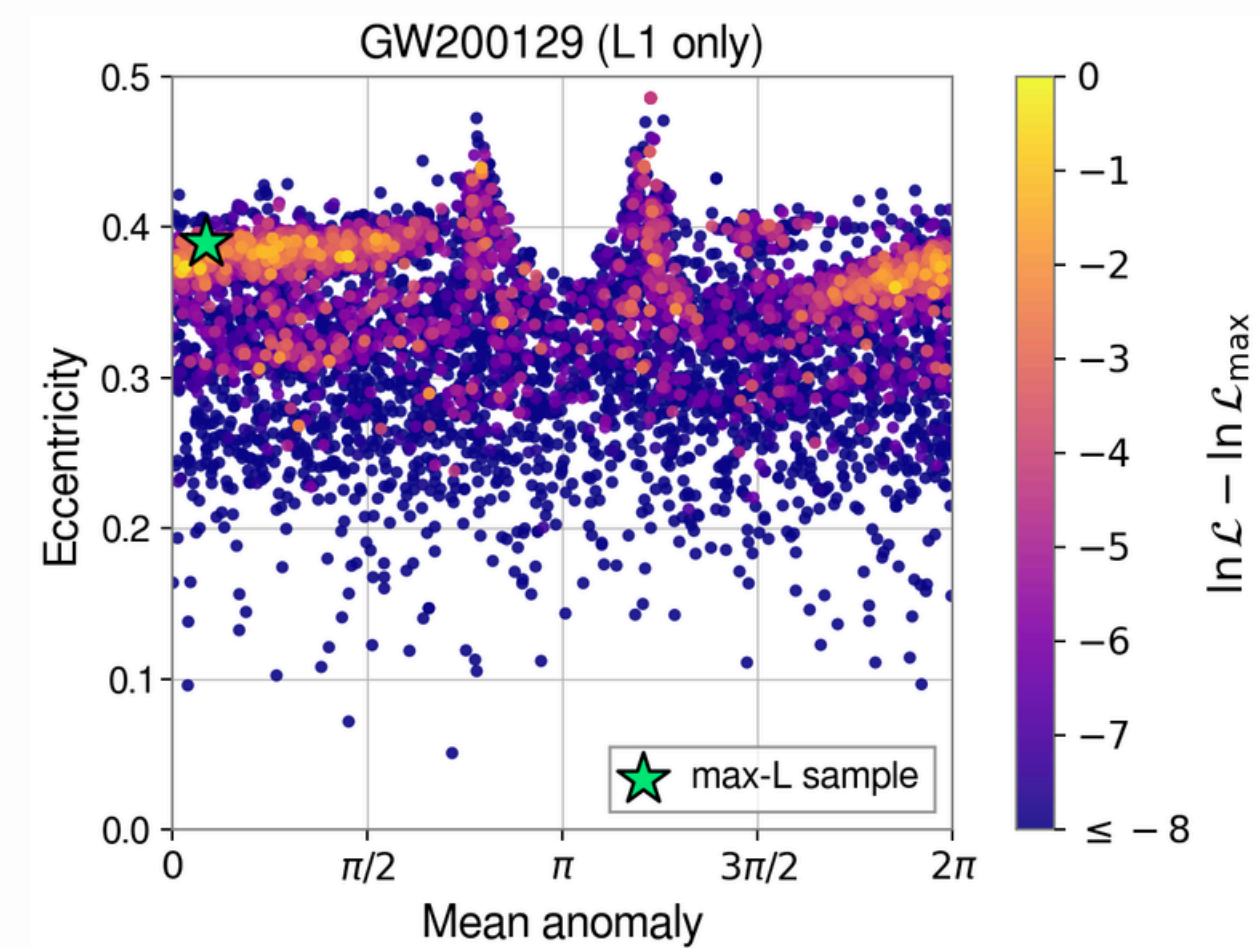
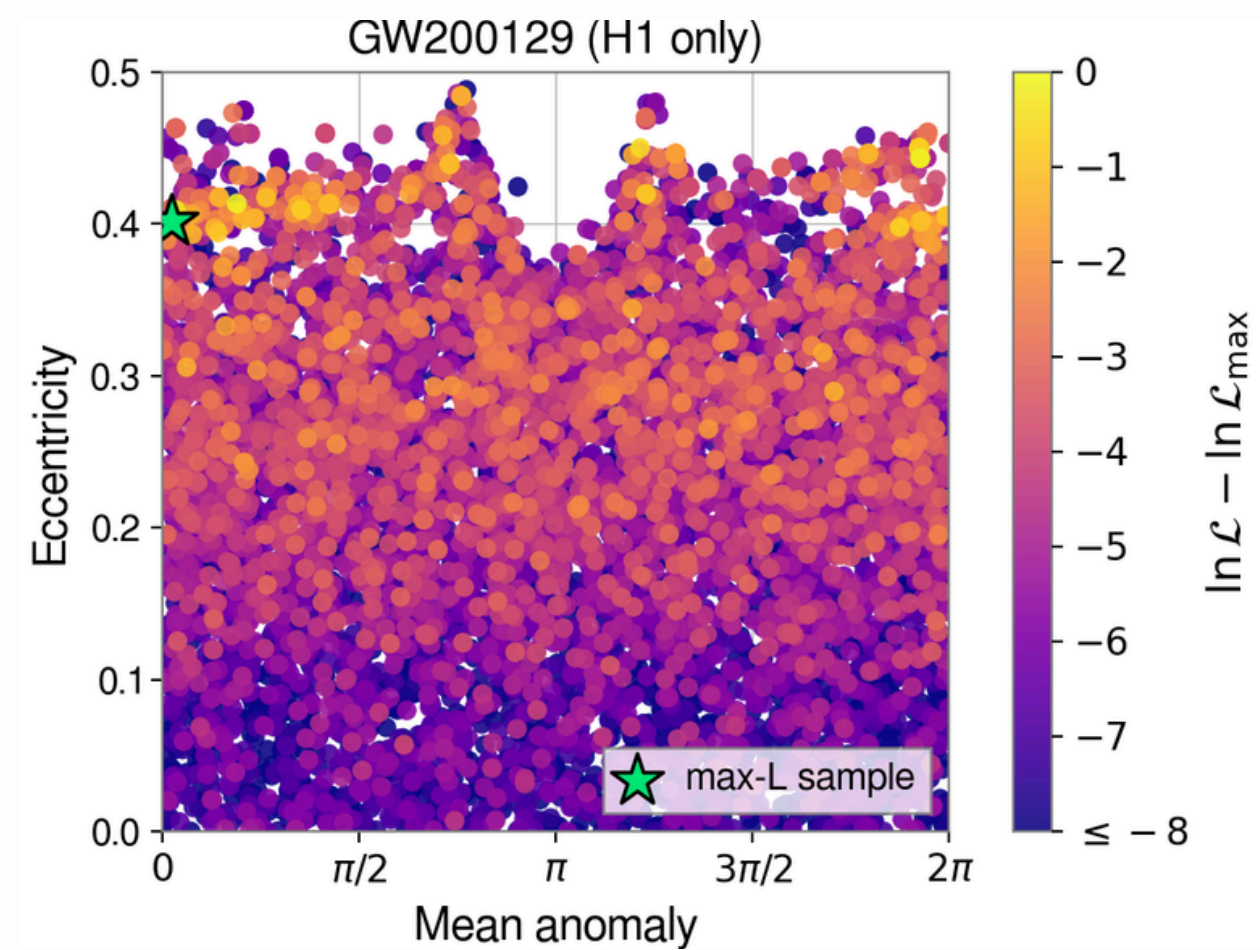
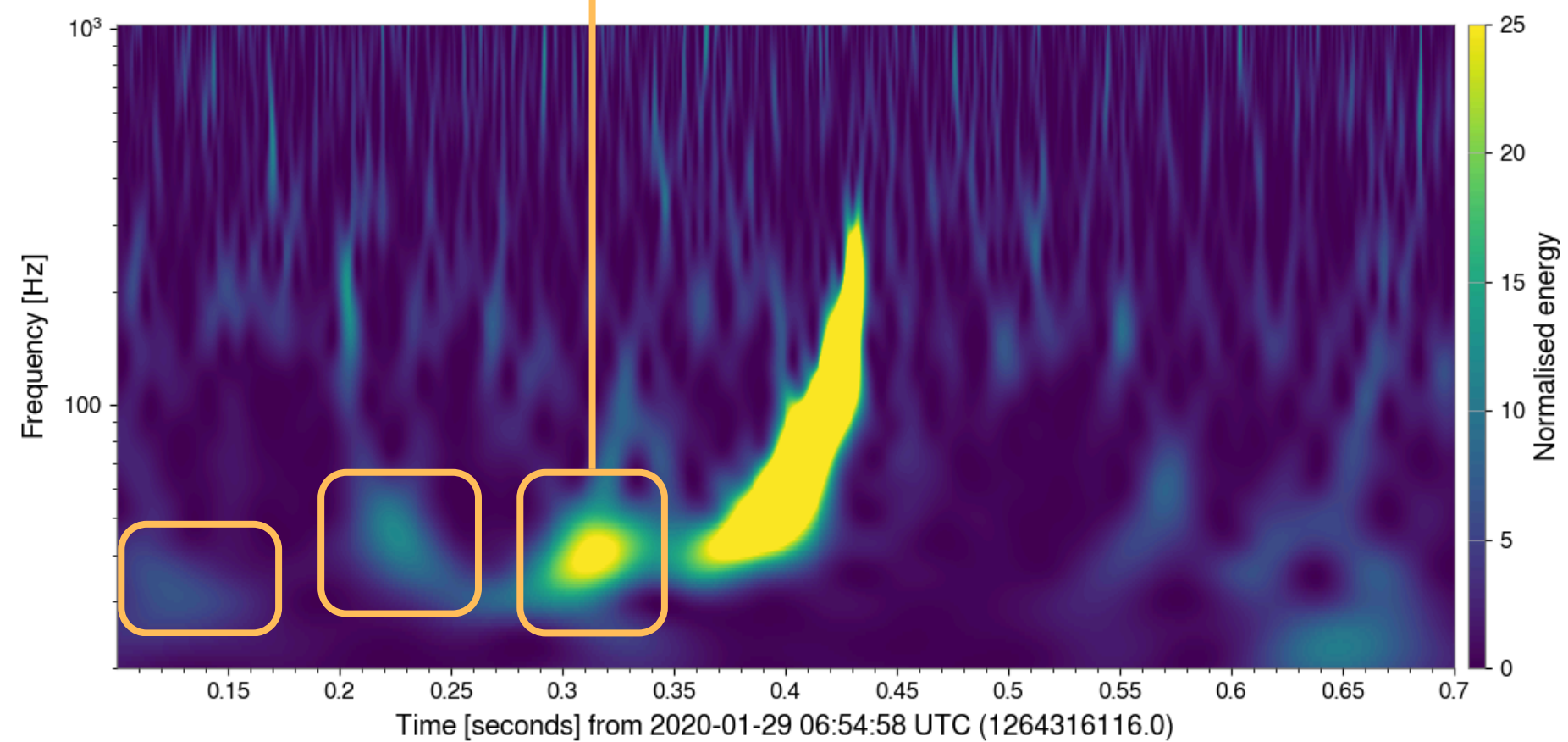
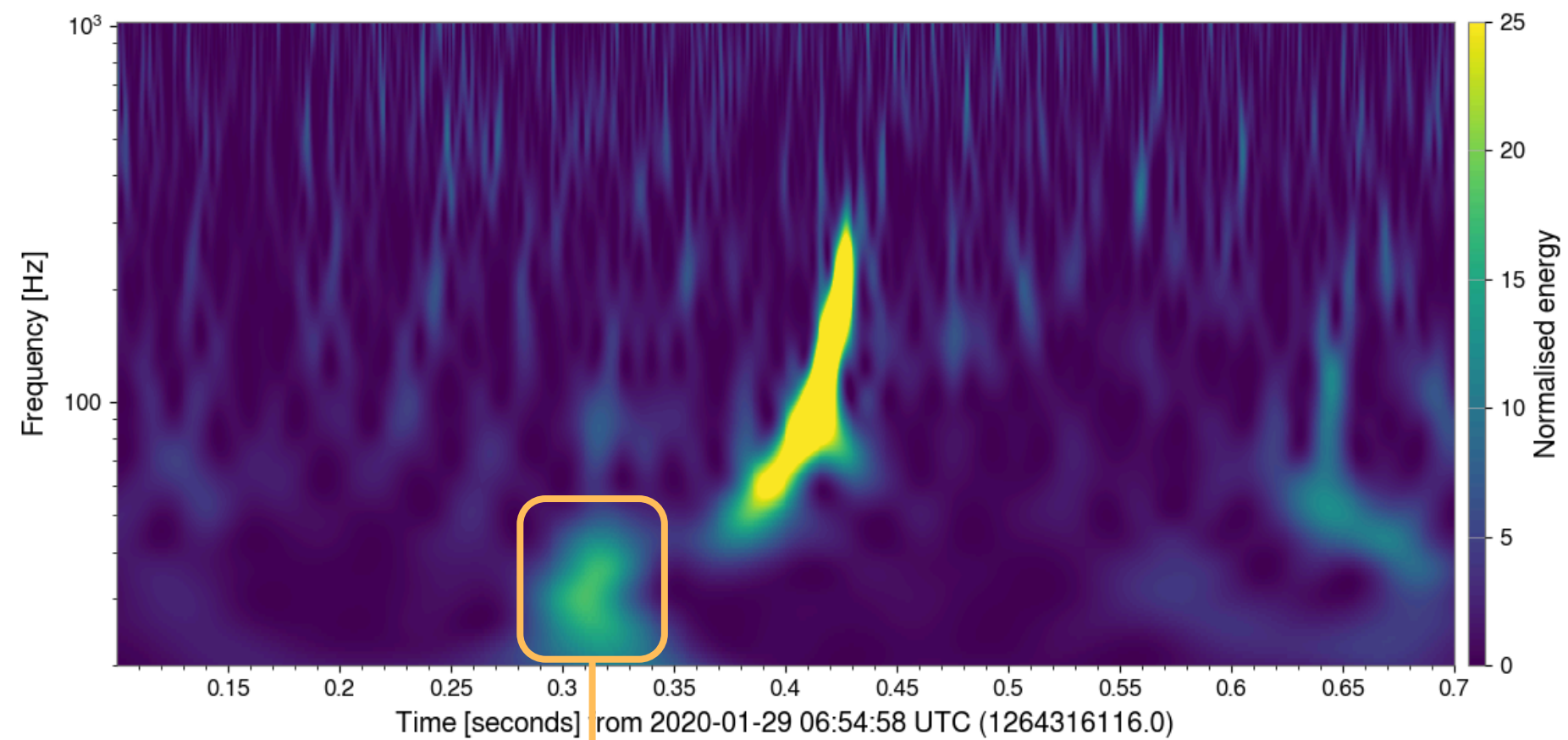
GW200129 — prior: uniform-e

- Gupte+ 2024 (SEOBNRv4EHM, DINGO-IS)
- Planas+ 2025 (IMRPhenomTEHM, bilby)
- Malagon+ 2026 (SEOBNRv5EHM, RIFT)
- Malagon+ 2026 (TEOBResumS-Dali, RIFT)
- Pompili+ 2026 (SEOBNRv6EHM, bilby)



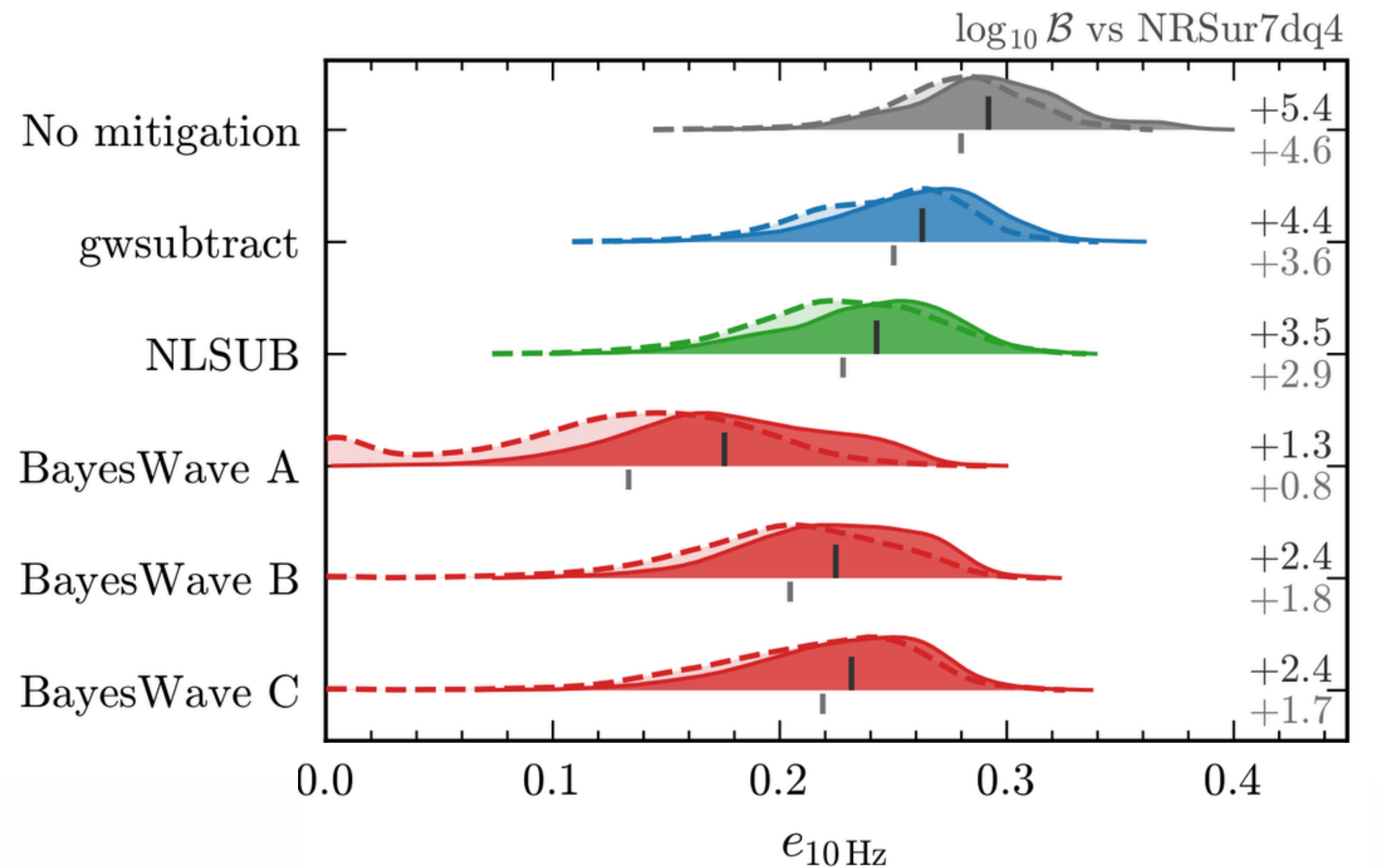




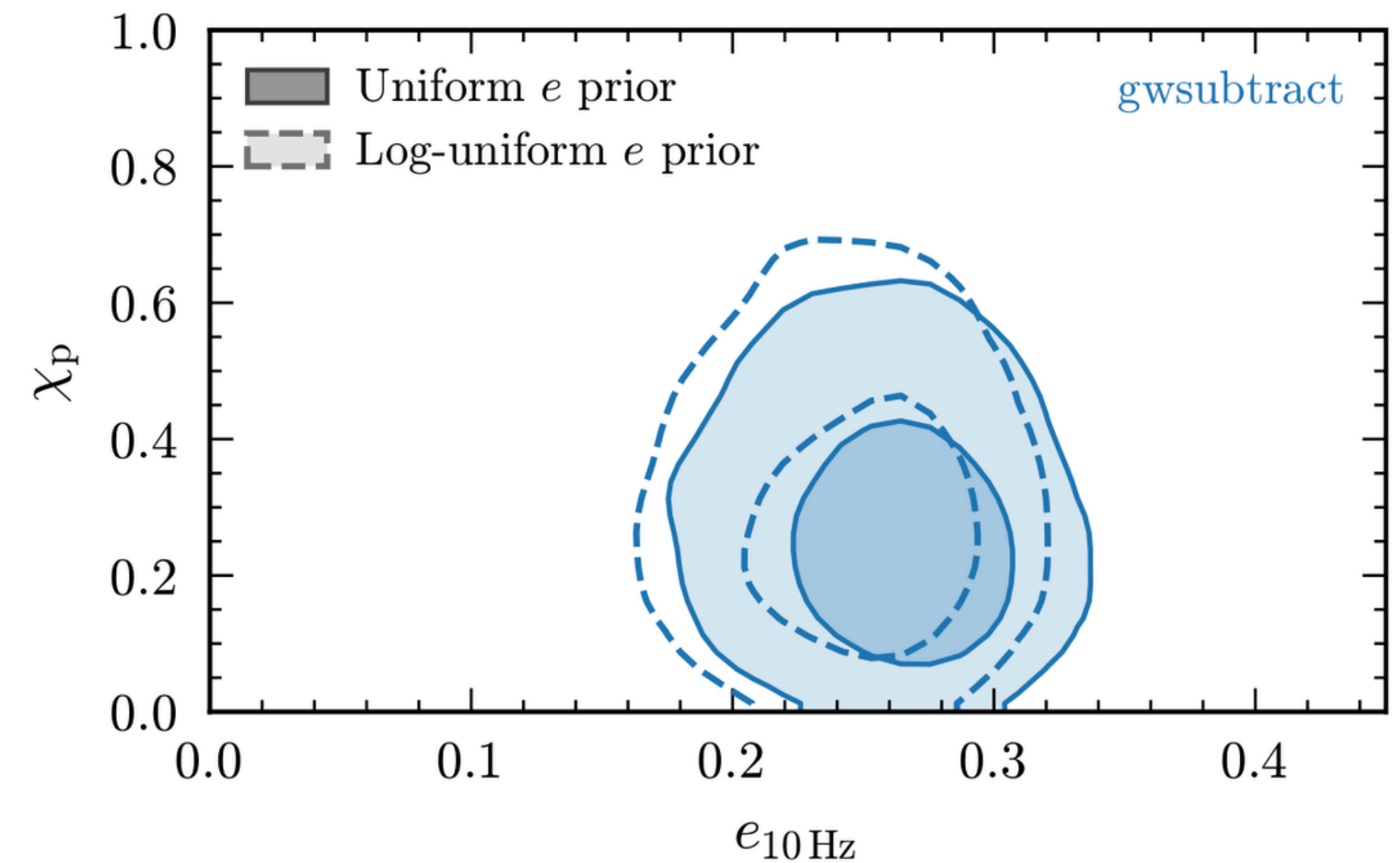


GW200129

In a recent study Gamboa+ in prep. showed that when analyzed with a **eccentric and spin-precessing waveform model** (SEOBNRv6EPHM) GW200129 **gains even more evidence for eccentricity, but does not show evidence for precession.**



Gamboa+ in prep.

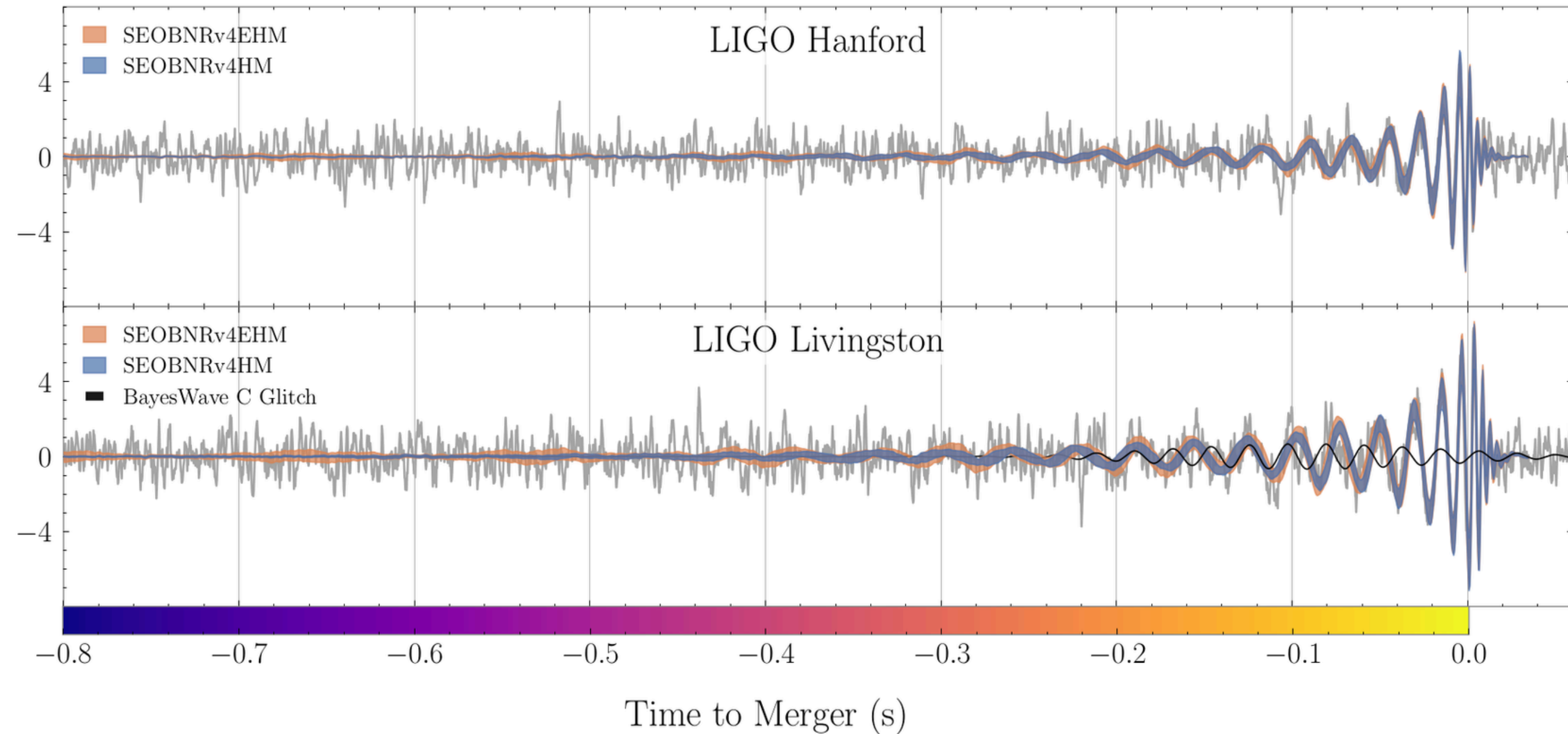


Gamboa+ in prep.

GW190521

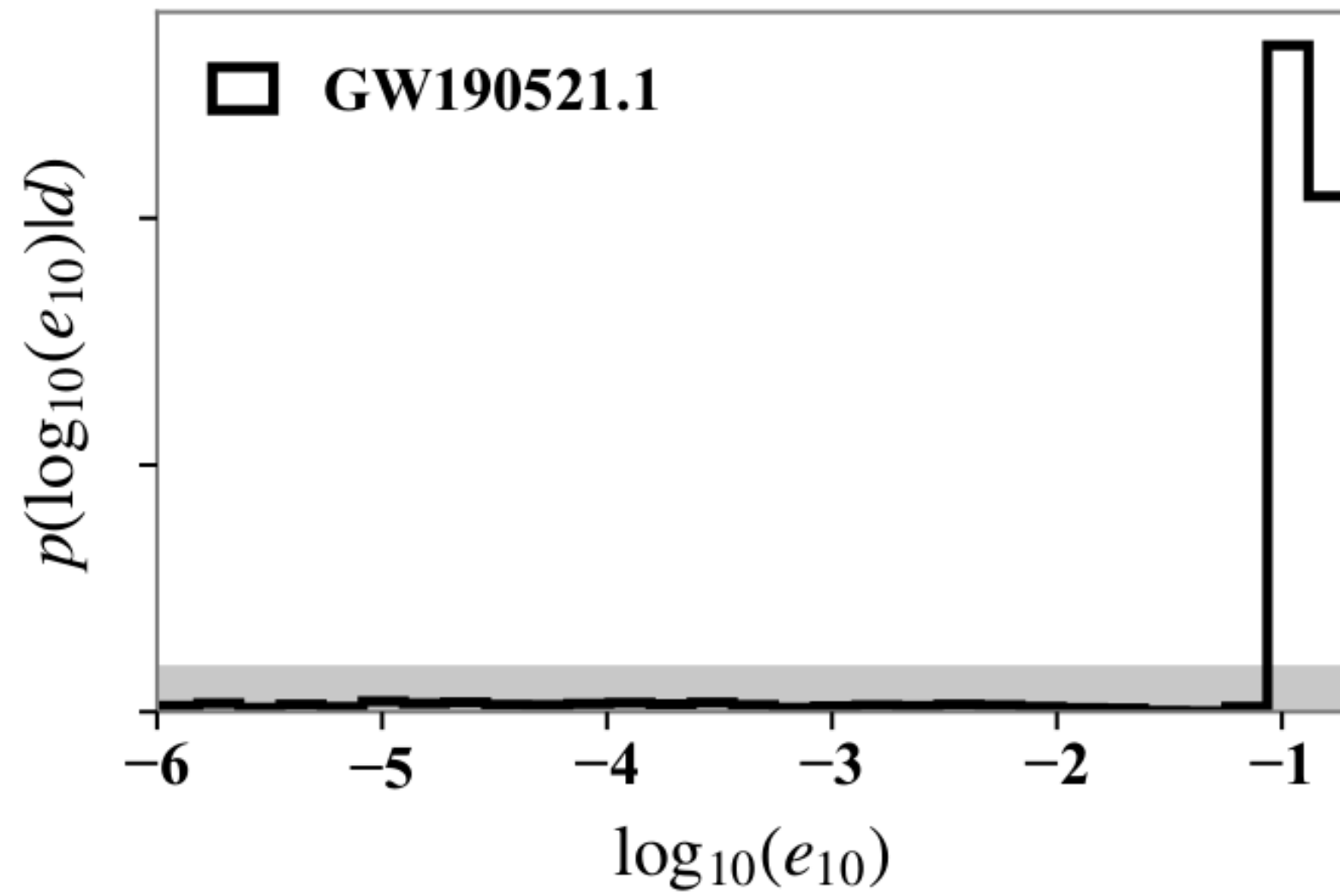
This is a 70 solar mass BBH.

The interesting thing about this event is that there is a glitch. But unlike GW190701, even after marginalizing over the glitch, the evidence for eccentricity remains.



GW190521

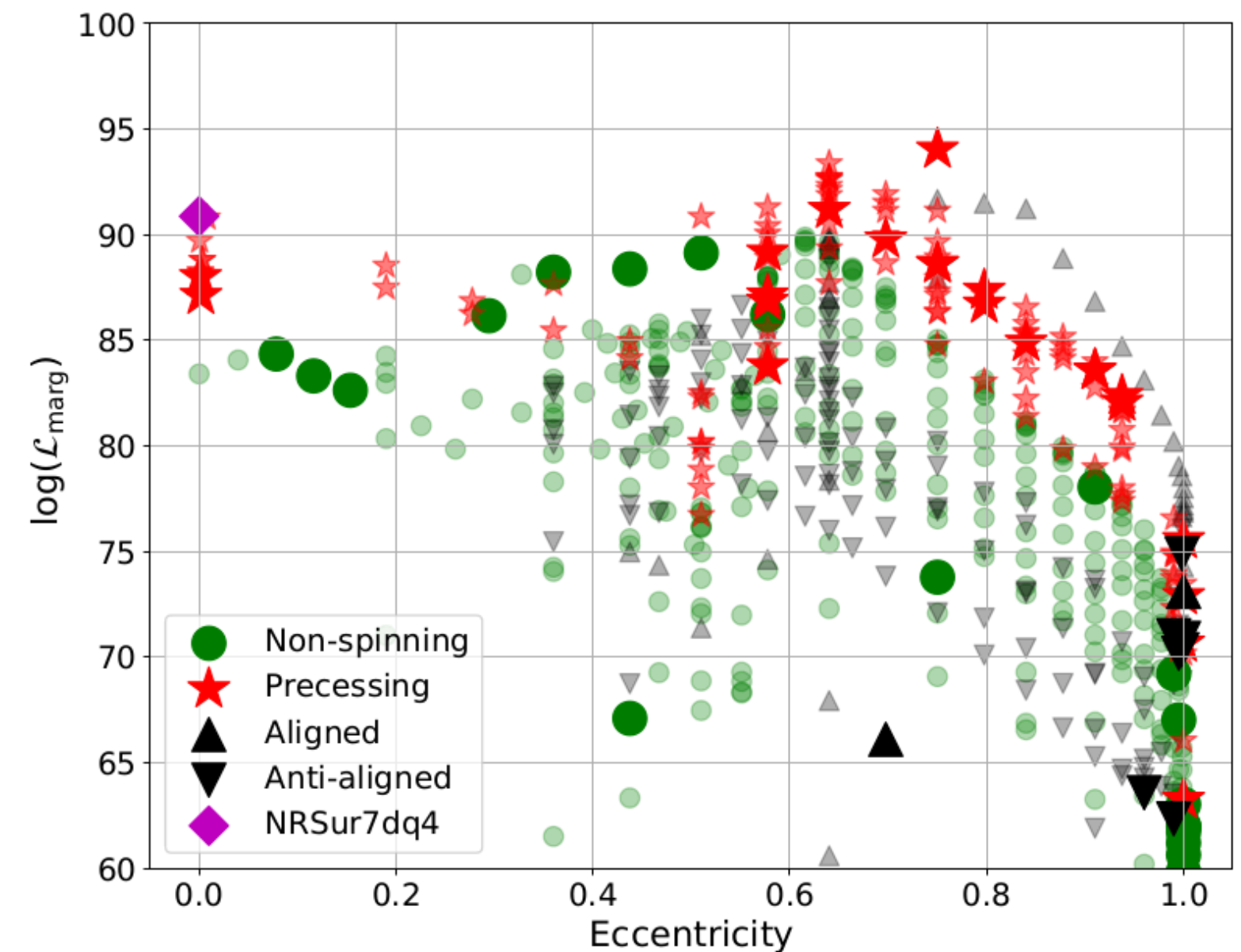
This is a ~146 solar mass BBH. Bayes factors of 21 (log-uniform prior)



Romero-Shaw+ 2020

There were also studies using NR simulations to compare the likelihood only

These include eccentricity near the merger

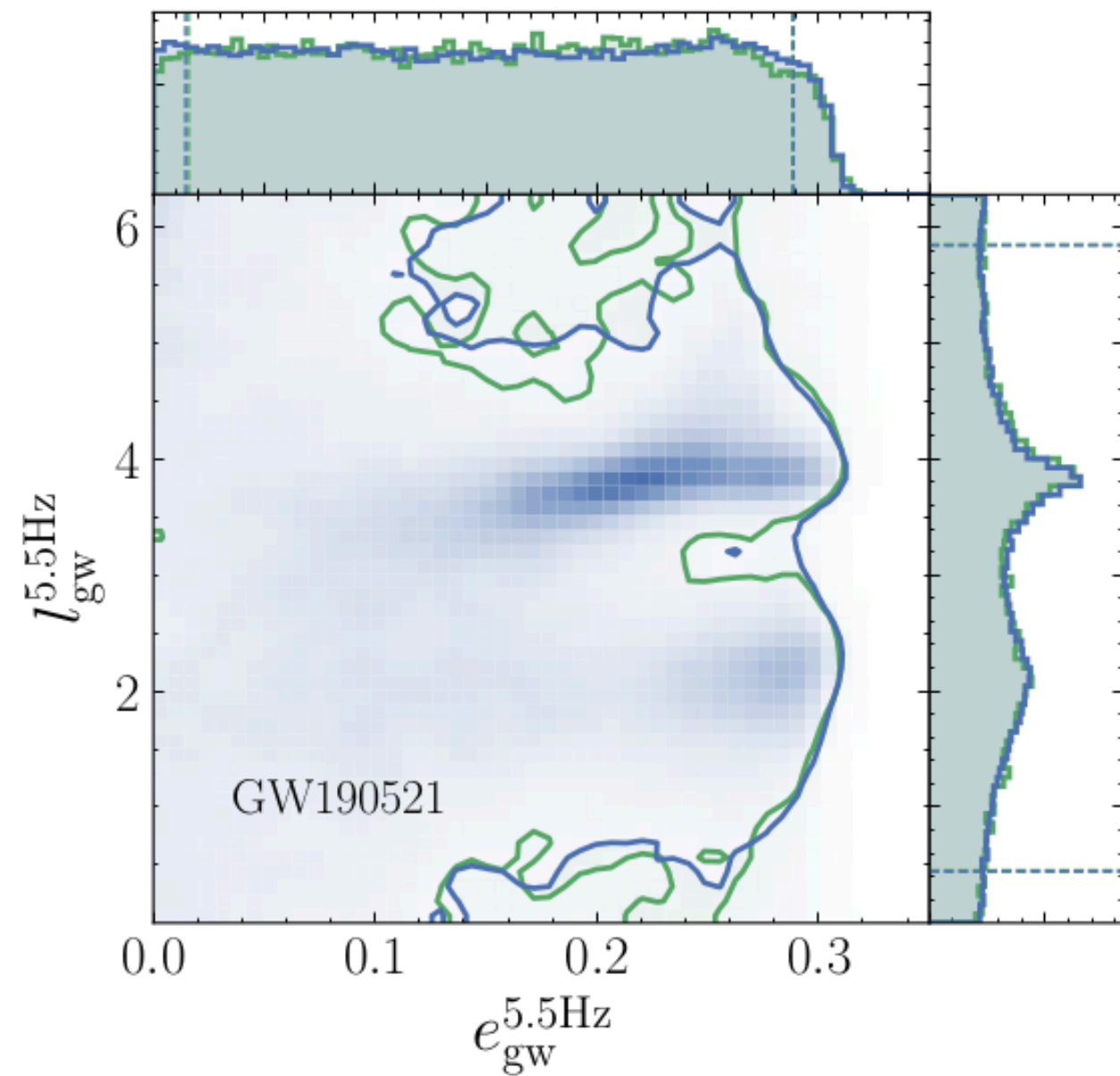


Gayathri+ 2020

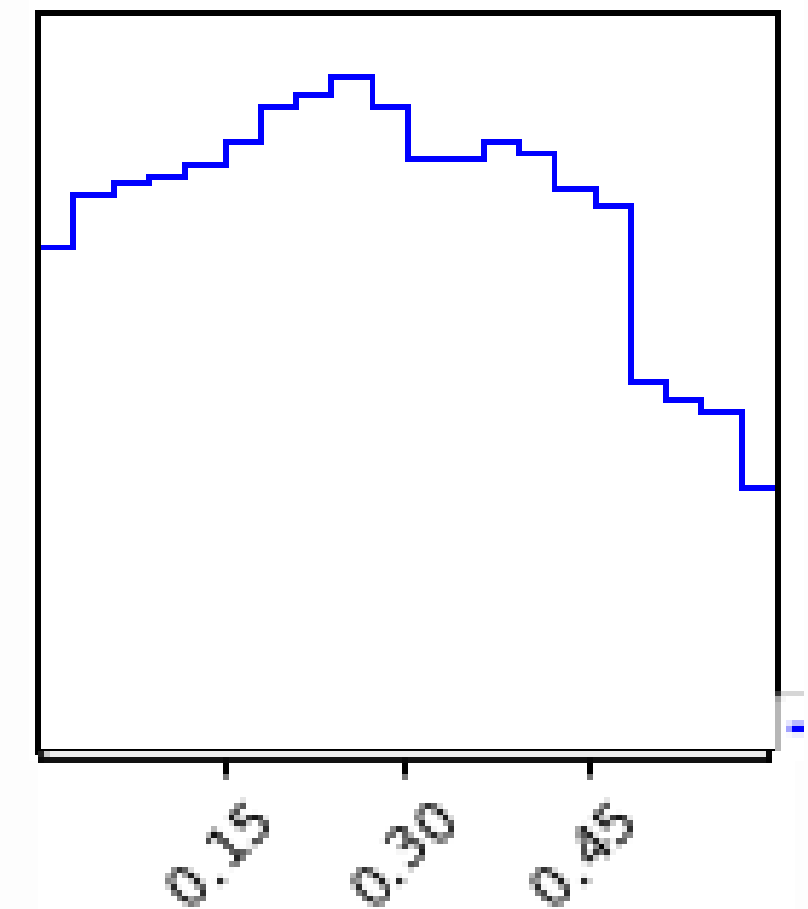
GW190521

Problem is, subsequent papers were did not find eccentricity in GW190521 (Iglesias+ 2023, Ramos-Buades+ 2023, Gupte+2024, Planas+ 2025, Malagon+ 2026).

Note, the waveform models used in these papers circularize towards merger



Ramos-Buades+ 2023

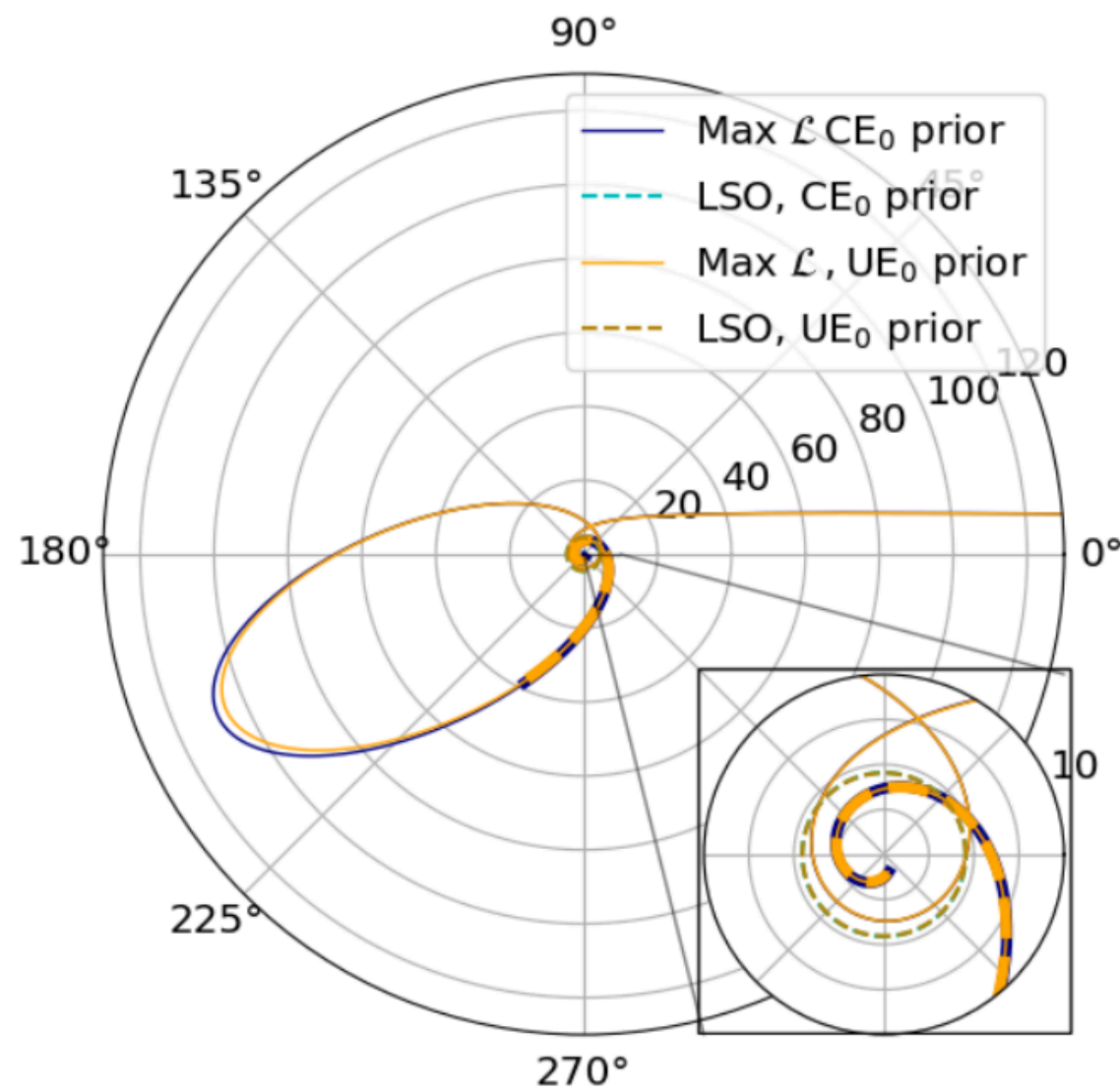


Iglesias+ 2022

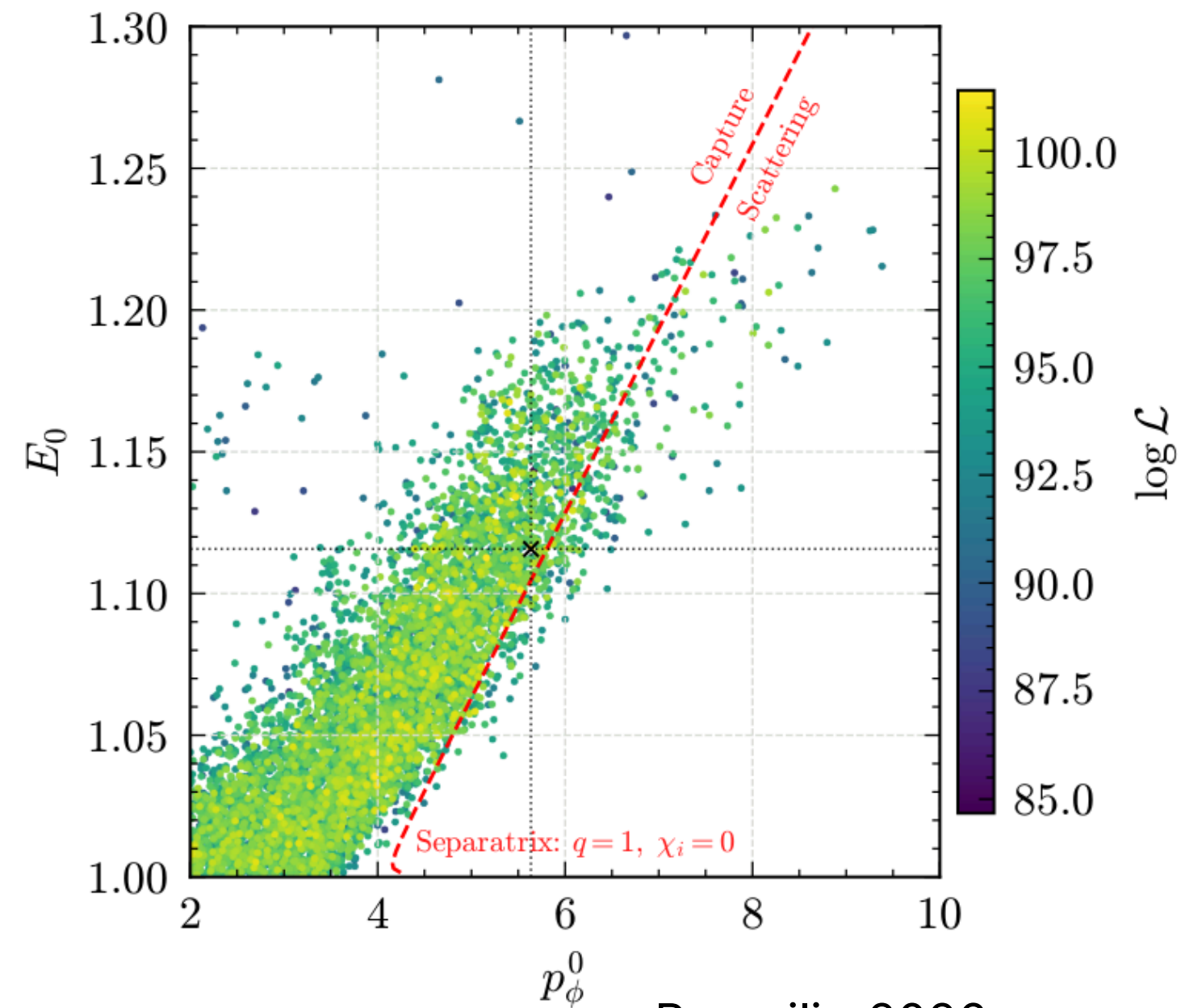
GW190521 - HYPERBOLIC?

Gamba+ 2021 proposed that GW190521 was a dynamical capture of two non-spinning black holes with Bayes factors of 4315

Recent works show that when using updated waveform models the Bayes factors become 4 (Pompili+ 2026 SEOBNRv6EHM) and 7.5 (Lange+ 2026 TEOBResumS-DALI)



Gamba+ 2021

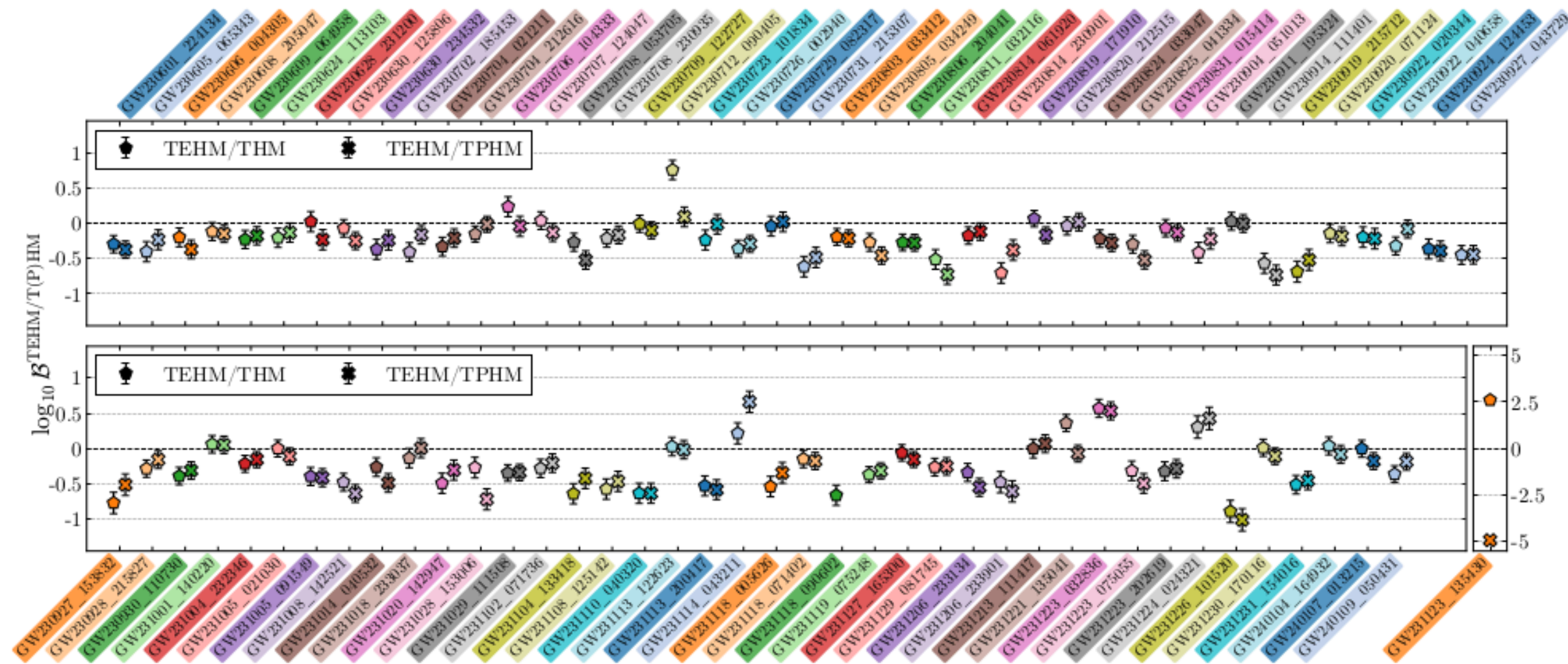


Pompili+ 2026

> These papers sample in the energy and angular momentum at infinity. This allows one to see the posterior compared to a fiducial separatrix

PLENTY OF OTHER EVENTS, THOUGH NONE REACH HIGH SIGNIFICANCE

> Three independent catalogs (Xu+ 2025, Gupte+ 2026, Malagon+ 2026) have confirmed O4a has very little eccentricity



Xu+ 2025

> That's not to say these events are unimportant! They still constrain the prior

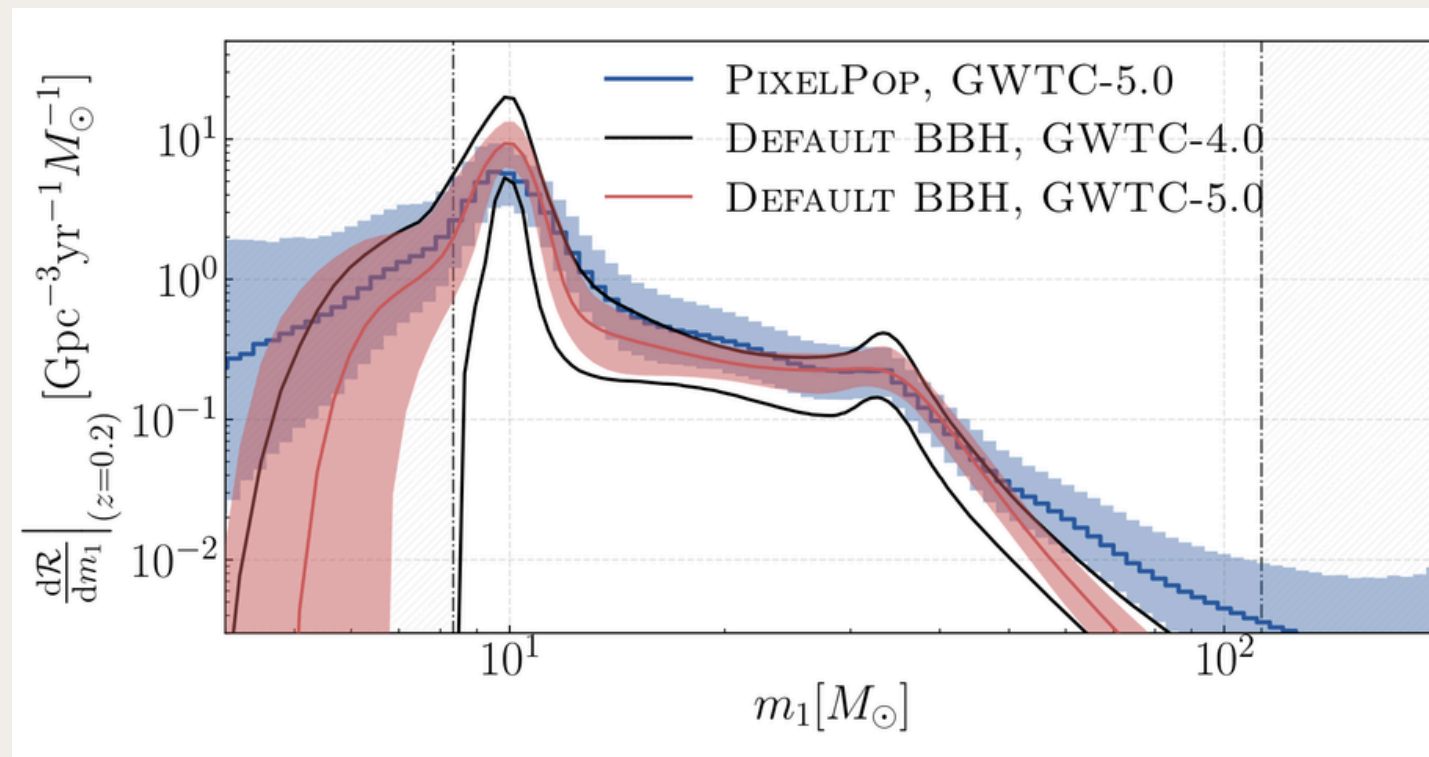
> A few ones with “mild” eccentricity support are GW190620, GW191109, GW231223, GW230706, GW230712.

Event name	Waveform Systematics	Noise Systematics	Sampler/Settings Systematics	Search Significance	Parameter Estimation Significance	Overall Significance
GW190701	High	High	Medium	High	Medium	Low
GW190521		Low		High	Medium	
GW200208_22	Low	Low	Low	Low	Low	Low
GW200129	Low	Medium	Low	High	High	High
GW200105	Low	Low	Low	High	Low	Low

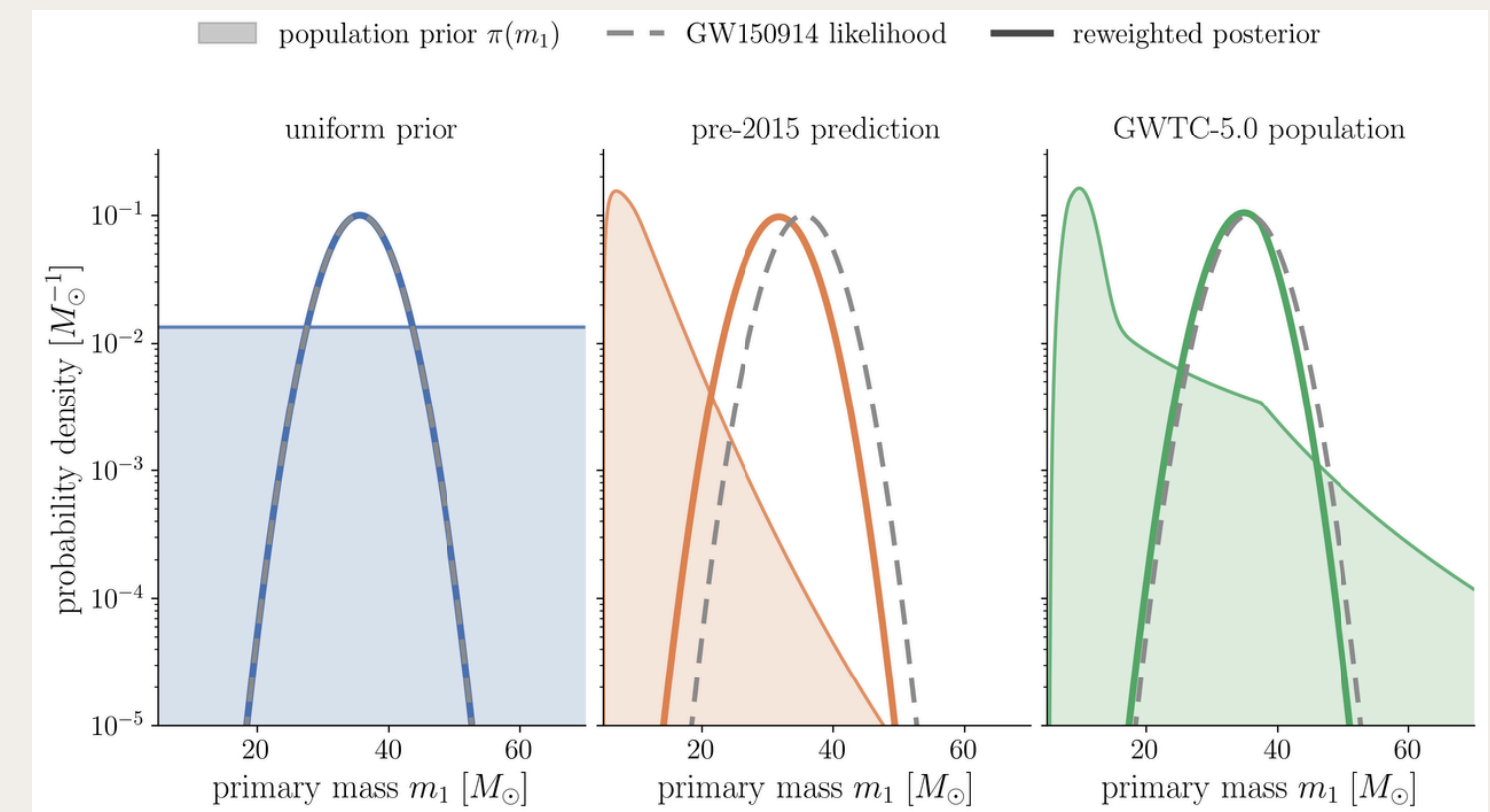
BUT WHAT IF OUR PRIORS ARE ALL WRONG?

Famously, before we started taking GW measurements, the expectation was that the majority of detections would be BNS not BBHs. So we need some way of updating our priors based off the data.

Step 1: Update our priors about the population using all the GW data



Step 2: Update our belief about the individual events



This accounts for the look elsewhere effect. You are looking at a “global” property of the data. We can compare astrophysical predictions directly to this

By using data from other events, we can find

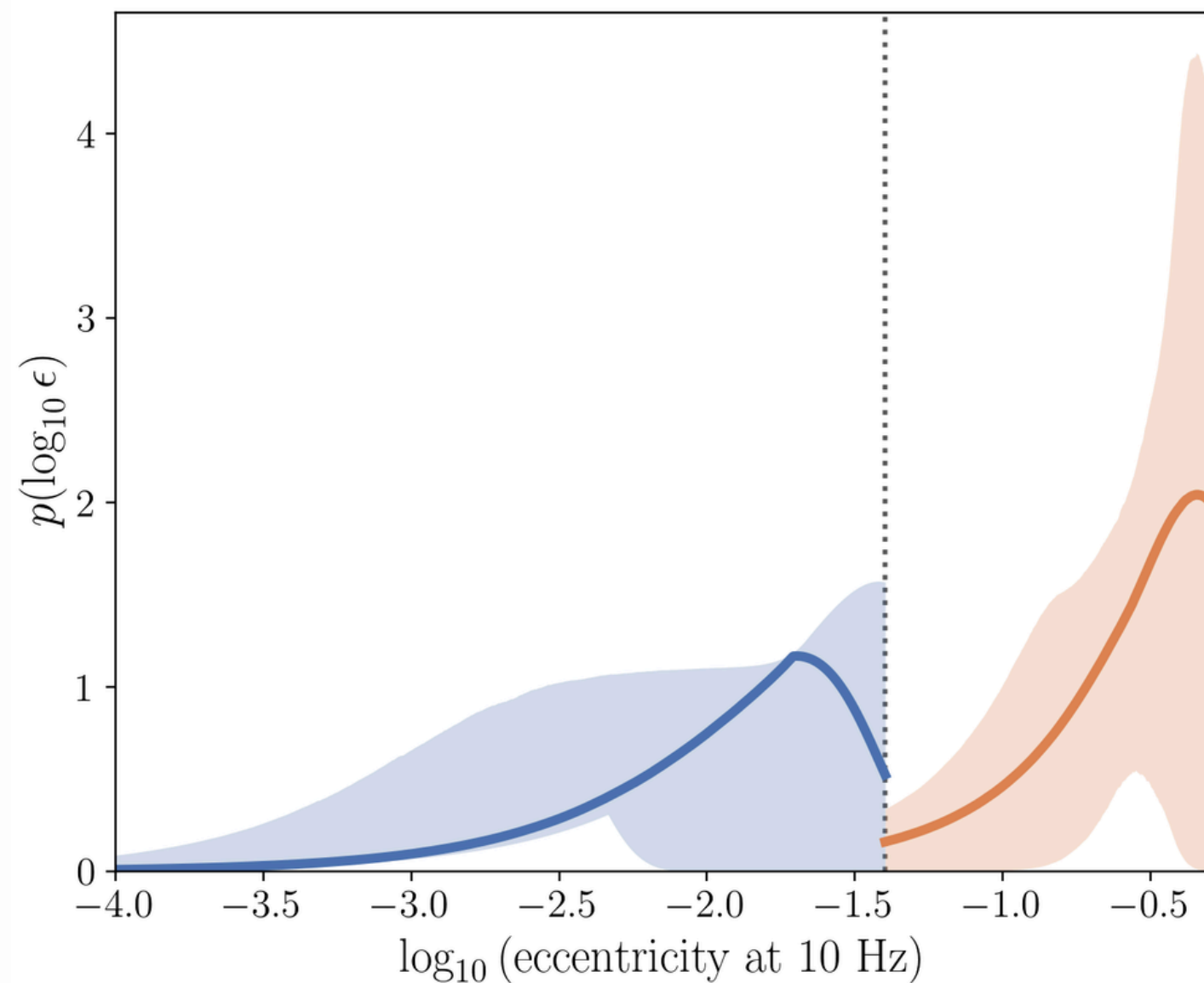
$$p(\text{eccentric} \mid \text{all GW data, our astrophysical priors})$$

This makes mildly eccentric events important as they affect the population!

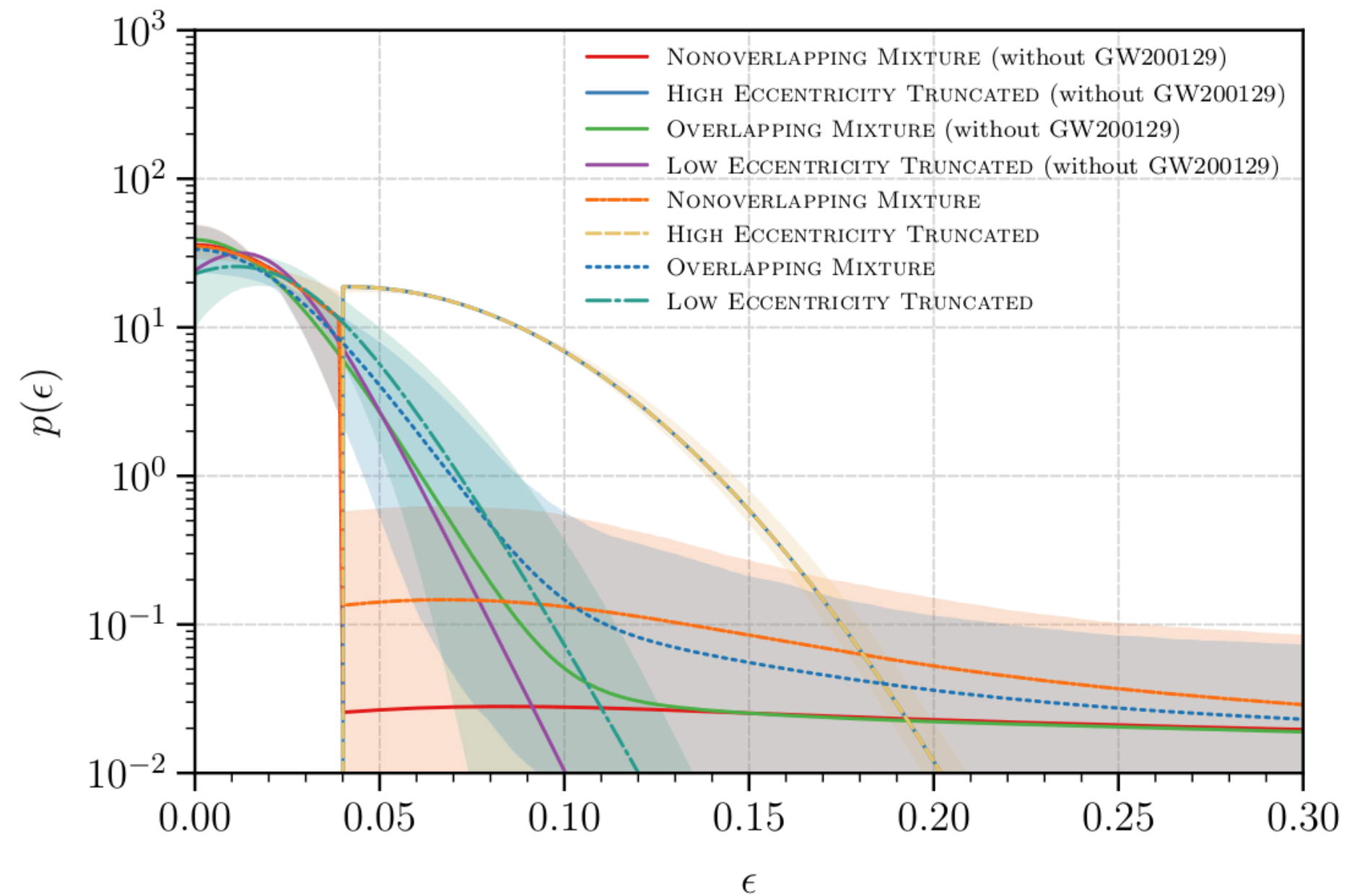
STUDIES LOOKING AT THE POPULATION PROPERTIES

Zeeshan+ 2026 defined a mixture between two Gaussians as a population model.

$$\pi(\epsilon) = (1 - \zeta_\epsilon) \cdot p(\epsilon \mid 0, \sigma_{\text{circ}}) + \zeta_\epsilon \cdot p(\epsilon \mid \mu_\epsilon, \sigma_\epsilon).$$

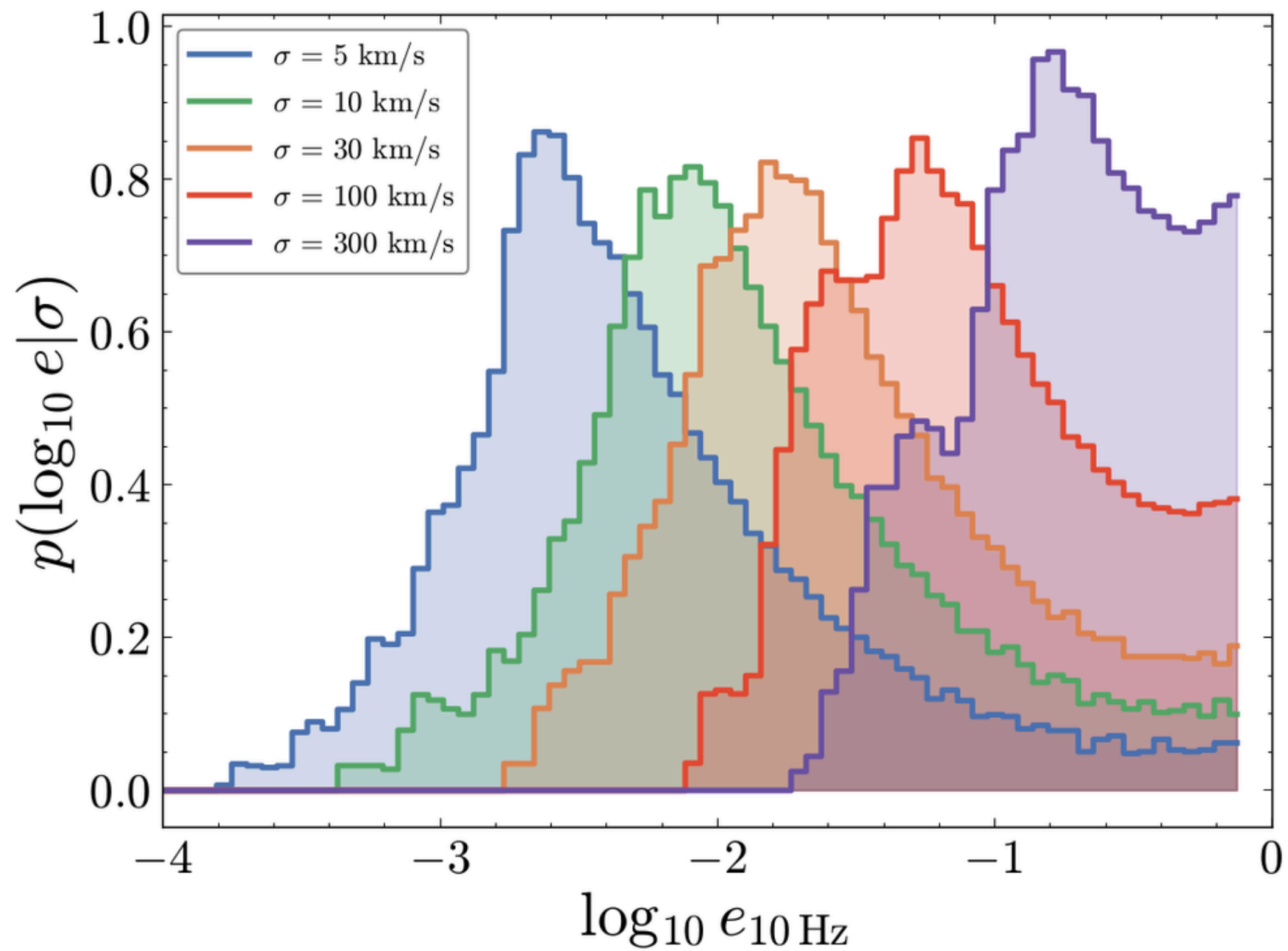


With a FAR cutoff of 1 per year, <5% of binaries have eccentricity > 0.04. When not including GW200129 this goes to <2%.



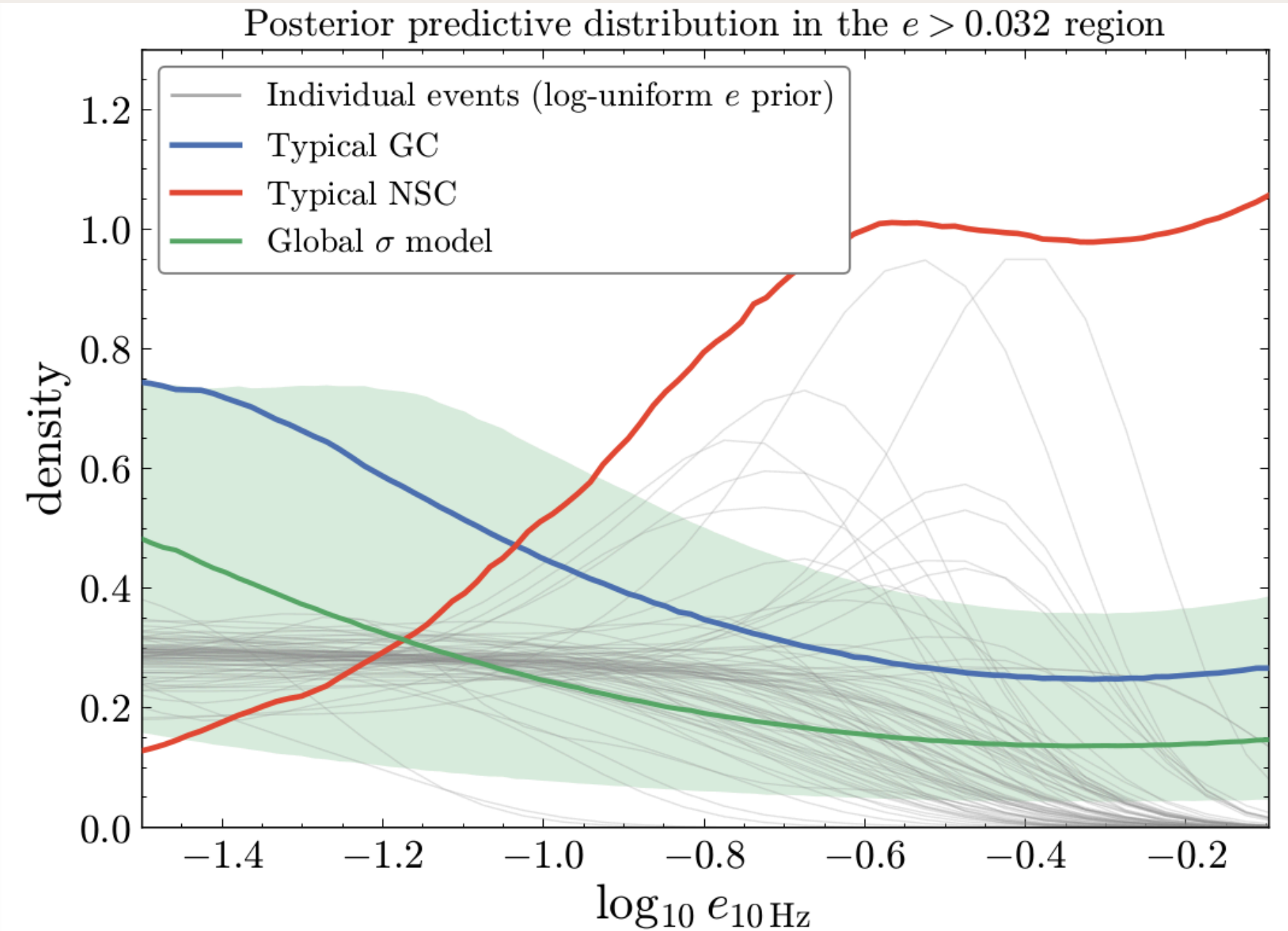
A PHYSICS INFORMED POPULATION MODEL

The distribution of eccentricities expected from dense stellar clusters with a dispersion, σ , due to BH captures



Gupte+ 2026

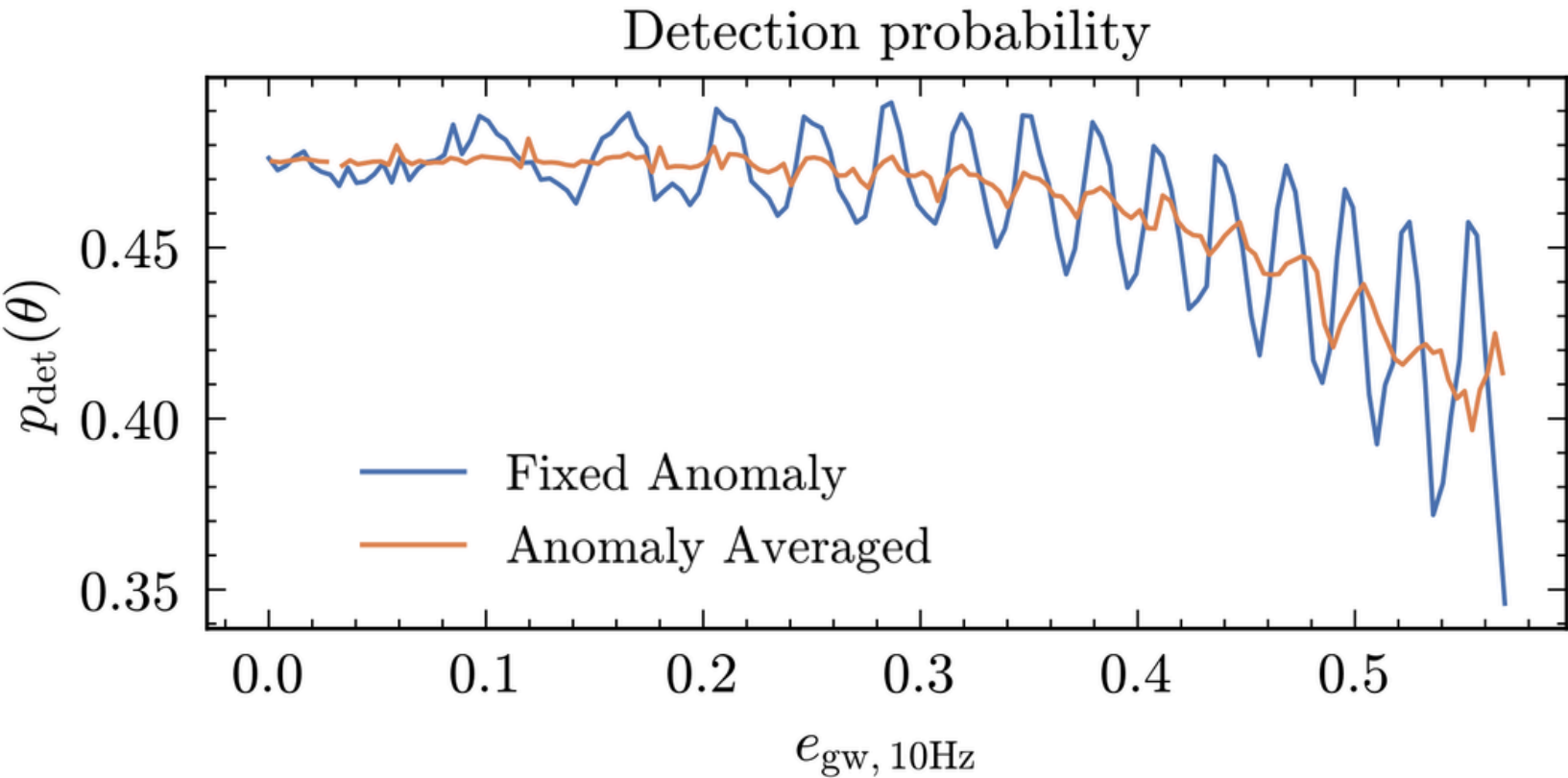
This implies that less than 47% of binaries come from BH captures in nuclear star clusters



Gupte+ 2026

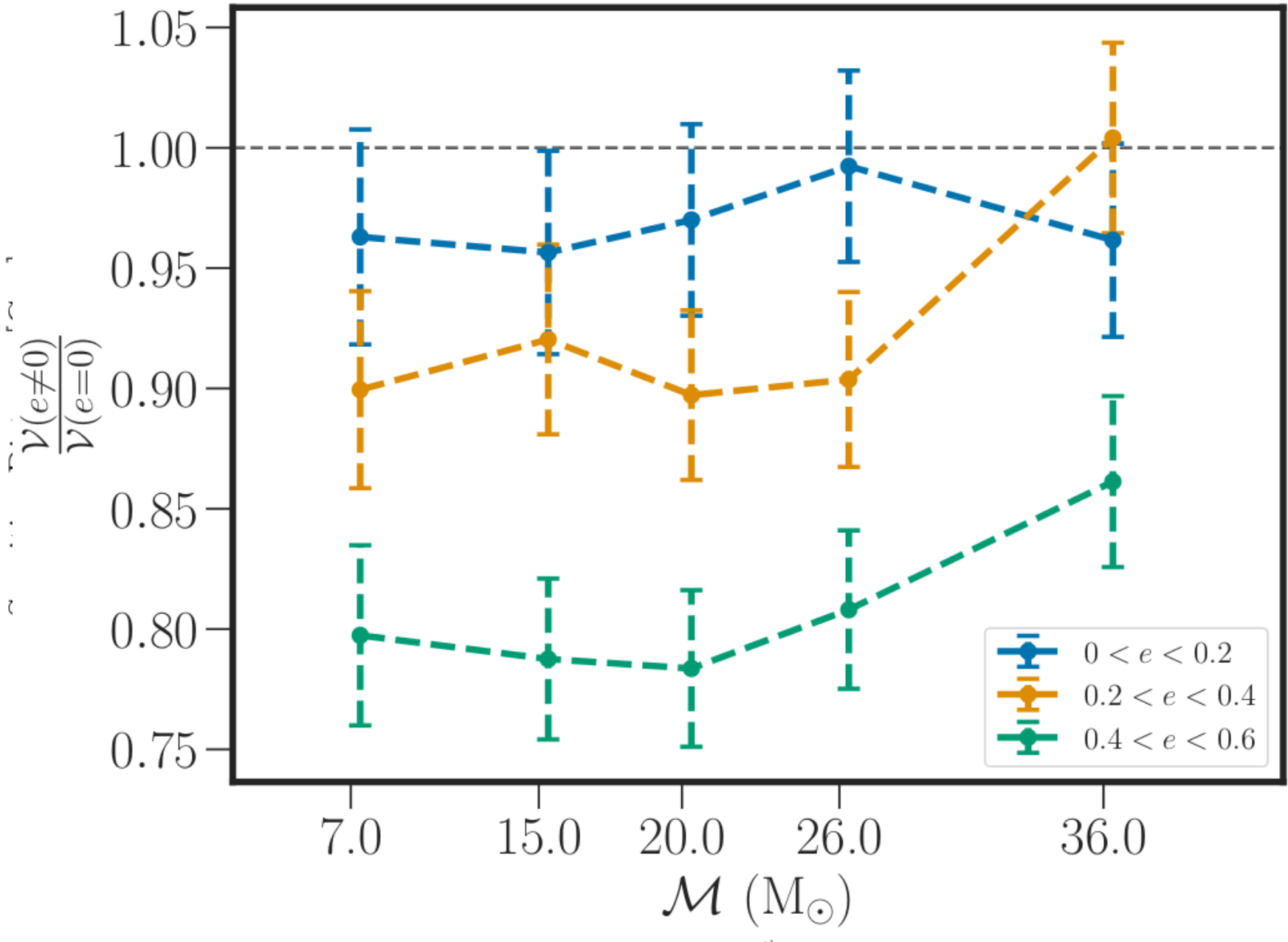
SELECTION EFFECTS DUE TO QUASI-CIRCULAR TEMPLATES

> We do not use eccentric templates when searching for BBHs. This can lead to Malmquist bias.



Gupte+ 2026, Vijaykumar+ 2026

> The above plot was produced with a semi-analytical method that estimates sensitivity without using search pipelines.

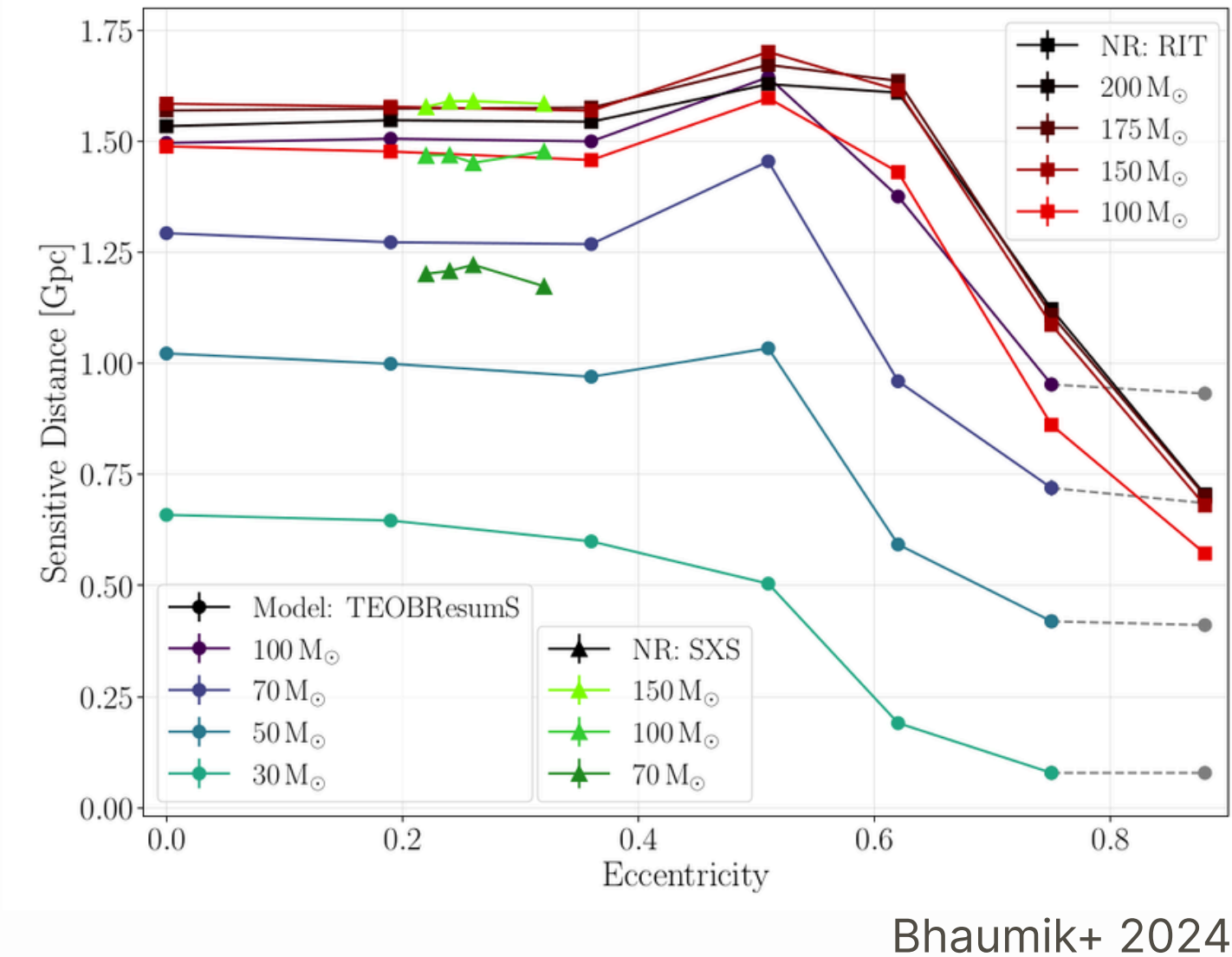
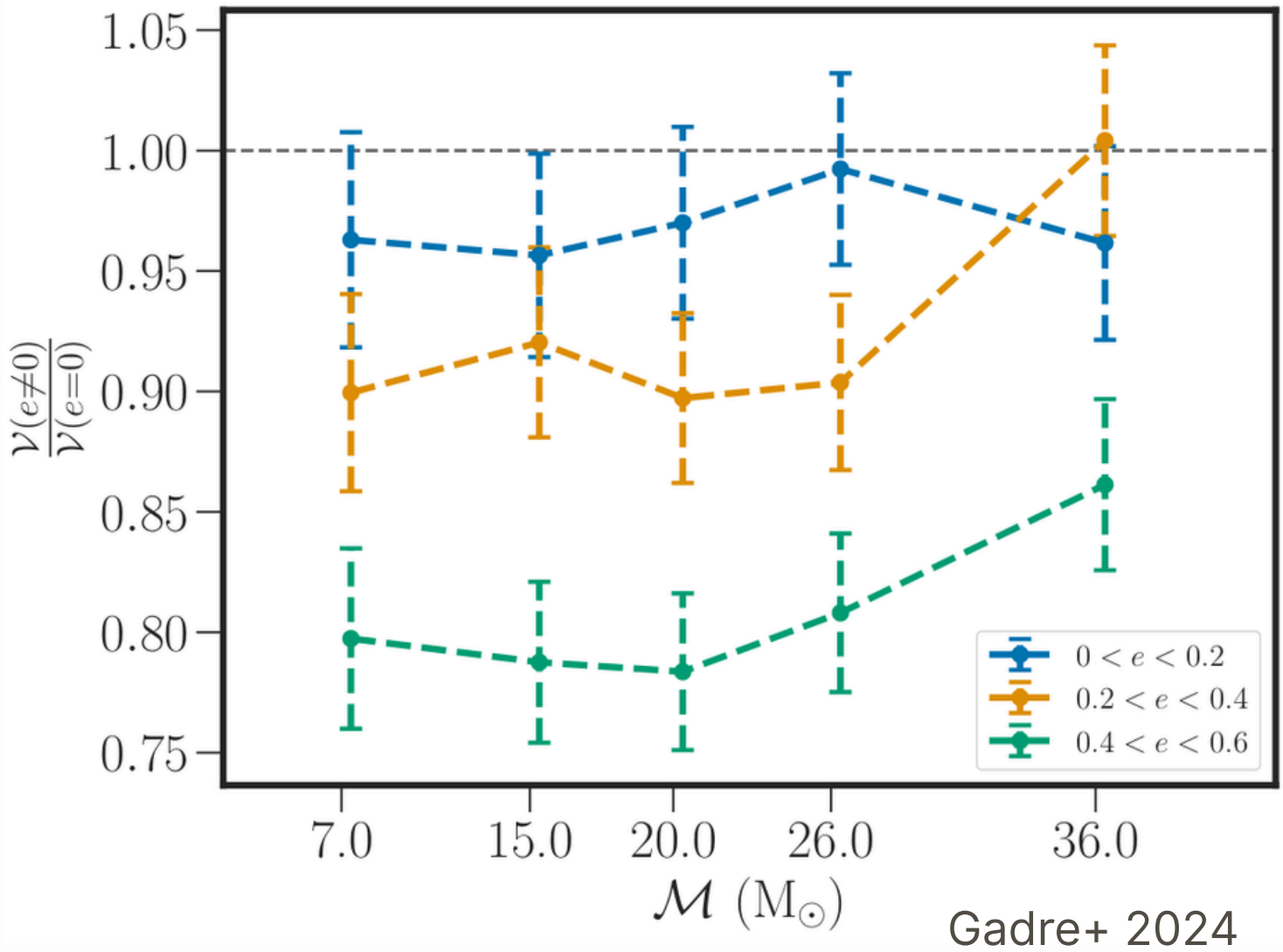


Bhaumik+ 2024

SELECTION EFFECTS DUE TO QUASI-CIRCULAR TEMPLATES

> We do not use eccentric templates when searching for BBHs. This can lead to Malmquist bias.

> There are search papers that quantify our sensitivity to eccentricity



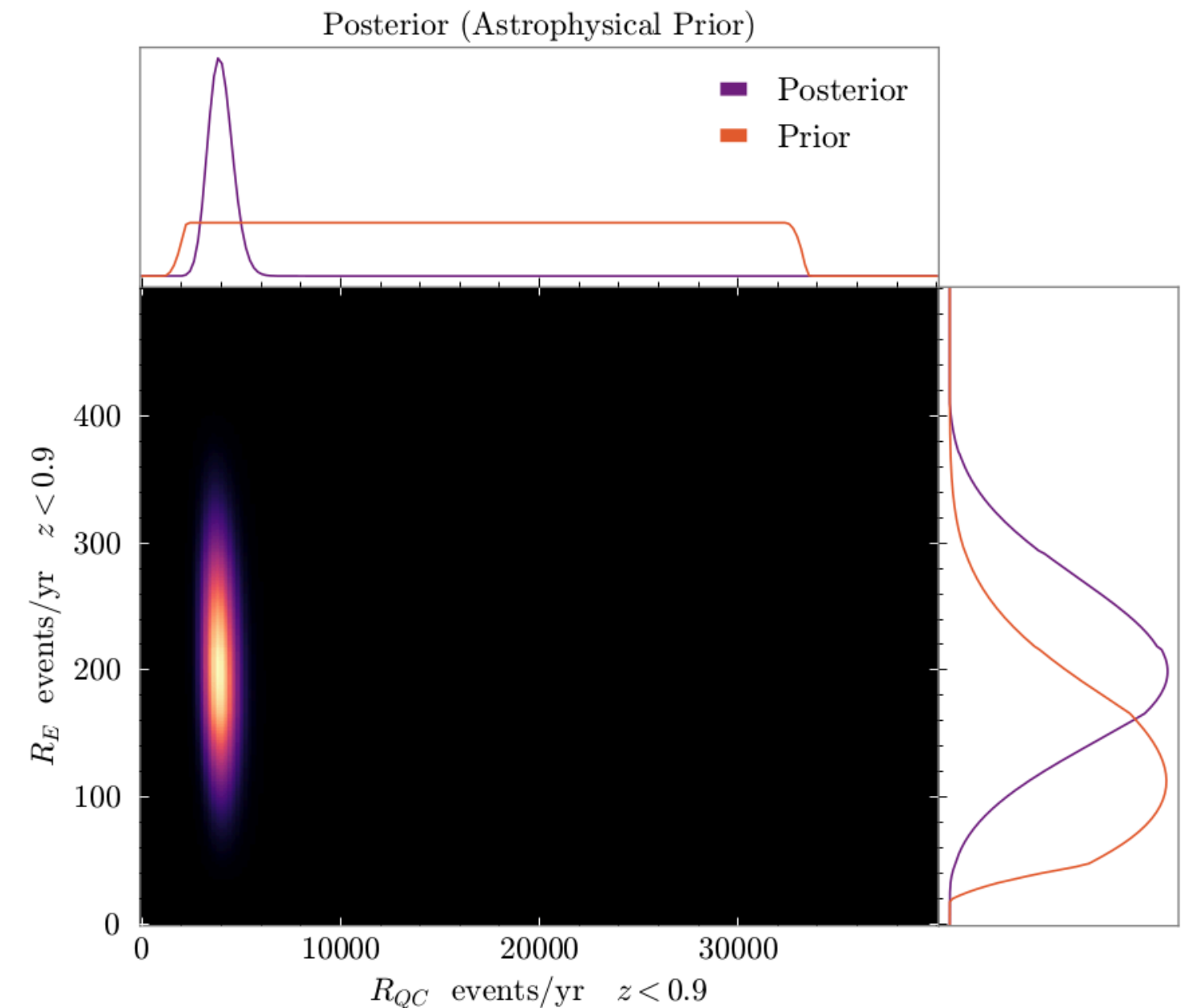
WE COULD ALSO INCORPORATE THE POPULATION INFORMATION VIA PRIOR ODDS

We can “balance” the look elsewhere effect by obtaining a global property of the data. First, we can ask the probability that one event is eccentric.

$$p_{\text{eccentric}} = \frac{\mathcal{O}_{E/QC}}{1 + \mathcal{O}_{E/QC}} = \frac{R_E p(d|M_E)}{R_E p(d|M_E) + R_Q p(d|M_Q)}$$

We can then use all the GW data to obtain a posterior over the rates and marginalize our probability over this.

Finally, we can use the same formalism to ask what the probability is that an event of the 57 is eccentric. This comes out to ~99%.



Gupte+ 2026

CONCLUSIONS

- > There is likely eccentricity in the current LVK data in at least one event
- > Once we have a full eccentric population distribution, it will be interesting to compare to various astrophysical formation channels
- > We need sophisticated methods to be aware of non-Gaussian noise. Simulation-based inference can be very useful here (Raymond+ 2024)
- > Modelling eccentricity effects near merger is important. Just because eccentricity is expected to approach zero near merger does not mean it will for all systems
- > Priors are not scary! We just need to choose a good one (see recent work by Rozner+ 2026)

APPENDIX